Master in Marine Renewable Energies

Offshore resources assessment: Integration of Wind and Wave Energy Resources

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“Difficulties are just things to overcome after all”

Ernest Shackleton

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Abstract

For any power plant to generate electricity, it needs fuel. For a wind power plant, that fuel is the wind. Wind resource assessment is the process of estimating how much fuel will be available for a wind power plant over the course of its useful life. This process is the single most important step for determining how much energy the plant will produce, and ultimately how much money it will earn for its owners. For a wind project to be successful, accurate wind resource assessment is therefore essential.

To be sure, not every aspect of resource assessment is fully mature. There are continuing debates over topics such as remote sensing and numerical wind flow modelling. New challenges are constantly emerging as wind turbines and projects grow in size and more advanced methods of measuring and modelling the wind is introduced. It is necessary to gain enough information and insights to make sound decisions about the tools and methods that should use.

The main outcome of this thesis will be to review the offshore wind resource assessment and to investigate the benefits of combining the effects of wave and wind energies in the electricity system, e.g., which care the special considerations and requirements for offshore resource assessment? How does wave power correlate with wind power generation? What are the potential gains of combining these two sources? In addition, the offshore wind and wave resources were analysed, using *Windographer* software, within the framework of this thesis with a case study, in *Chapter 8*.

Since the two sources are highly correlated in one area, wave power in general is available at the same time as wind power is available, and thus the gain of combining the two sources in this specific area is reduced. In an area with lower correlation between waves and winds, the gains of combining the two sources could be higher. *NEMOS* is an innovative system to generate electricity from ocean waves, based on research findings from 2010, can be implemented at low costs and is particularly suitable for combination with offshore wind turbines.

Wave power also has other benefits compared to wind power, such as higher public acceptance and higher energy density in the resource. The coming years might be the great exploitation of the wave power industry. Experience from the wind and solar industry shows that once things start happening, the growth in installations may be explosive. Prices, investment trends, public acceptance, are some key elements will help to develop the wave power industry.

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PART 1 – Wind Resource

1. Introduction

The main objective is to get clean energies that have the lowest impact on the climate change. Then, the government and private companies even have to assess the cumulative environmental impacts of the process through all its life stages (the whole production line), including an impact assessment to determine their environmental impact as well as social and economic aspects.

Wind farms can be an important clean energy source, and can provide a considerable source of 'home-grown' energy with minimal emissions. A growing share of the world's wind energy development is now occurring offshore. In part, this is because the offshore areas have good wind resource. Even, they offer opportunities for large-scale development. The most of world's offshore wind projects have been built or are being planned in Europe, where the population density is high and there is relatively little land available for large wind projects. Today, the most offshore wind projects have been constructed less than 25km from the shore and less than 40m depth. With improvements in turbine foundation technologies and installation practices as well as O&M activities, the new projects may be sited further from the coast and in deeper waters.

The wind energy potential of a site is defined by *S. Krohn* [**11**] as the energy, which can be extracted over a given duration, normally a year, of wind turbine's operation on the site. Wind energy conversion systems must be adapted to characteristics of the area's energy potential, which are determined before the design phase of the wind farm.

Wind resource assessment is the process of estimating how much fuel will be available for a wind power plant over the course of its useful life. This process is the single most important step for determining how much energy the plant will produce, and ultimately how much money it will earn for its owners as well as selecting a turbine (in recognition of the extreme and fatigue loads) and the optimisation of the wind turbines layout (both further explained in *Chapter 6*). For a wind project to be successful, accurate wind resource assessment is therefore essential.

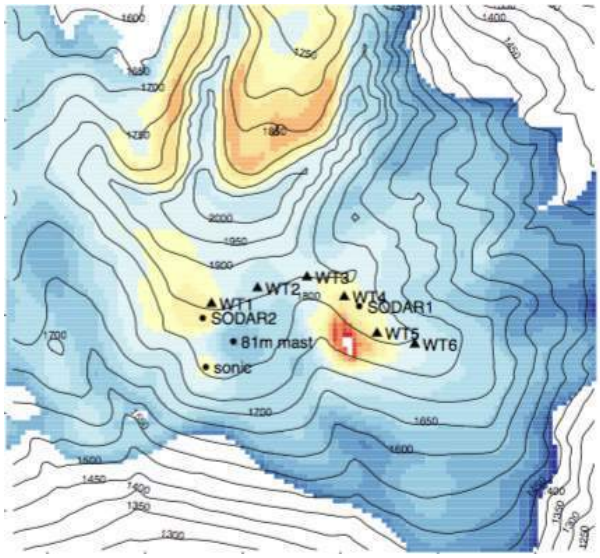


Figure 1 CFD model WinSim simulation of a wind park site in Switzerland

The greater the wind speed, the greater the energy it contains. Offshore wind environment differ from onshore environments in a number of ways. One difference is the surface roughness of open water is much smaller the land surfaces (other differences are explained In *Chapter 5*). Offshore wind speeds are generally higher than on land as there are no complex terrains to disrupt the wind flow. As wind blows over the water surface (generating waves) it loses some energy due to friction. The energy in wind is therefore usually higher further away from the coastline, and increases in strength with increasing height above the water surface.

Even in Europe, where wind plants have operated offshore for over 20 years, the projects have been financed based on a variety of measurement methods, including meteorological masts at close to the hub-height equipped with anemometers, stand-alone or floating LiDARs, and, in some cases, no on-site measurements at all. In *Chapter 3* the basic sources of wind information was introduced to obtain data from direct and indirect measurement techniques.

The offshore wind industry has yet to establish a standard approach to offshore wind resource assessment. For several years the classic sources of wind measurements (weather stations, instruments on board ships) have been being joined by space-borne remote detection: radar

altimeters, radar diffusometers, synthetic aperture radars, radiometers, anemometers, LiDAR… (See in *Chapter 3 and 5).*

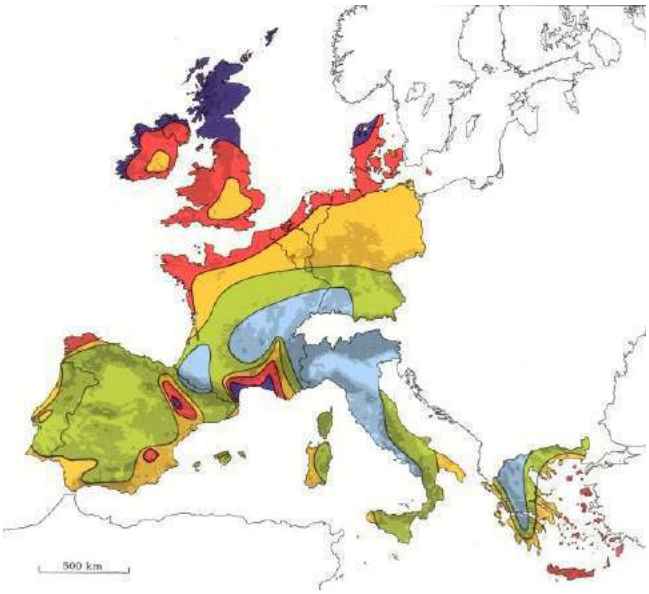


Figure 2 Wind resources at 50 meters above ground level for five different topographic conditions

Figure 3 Offshore Wind Project in the future

1.1 Key Characteristics of the Wind

The annual average wind speed is the main parameter as a way to rank wind project sites. However, the mean wind speed is only a rough measure of the wind resource. To provide the basis for an accurate estimate of energy production, the wind resource must also be characterized by the variations in speed and direction, as well as air density, in time and space [**2**]. The vertical dimension is just as important for projecting wind speed measurements. The variation in speed with height is known as *wind shear*.

Figure 4 Annual Mean Wind Speeds at 92m, 58m and 27m determined with the measurement data of Meteomast IJmuiden, http://www.meteomastijmuiden.nl/en/home/

1.1.1 Origin and properties of the wind

The solar energy received by the earth is partly converted into wind energy. Not surprisingly, all

the energy needed to supply our weather comes from the sun. The uneven heating of the Earth's surface by the sun is the first link in a fascinating chain of events. The movement of atmospheric air at the level of the earth's surface is called *wind*. Wind is simply the movement of air; this kinetic energy which can be harnessed by wind turbines. The atmospheric winds that circle the earth are created because the earth's surface in unevenly heated by the sun. The sun's energy does not heat up the poles and Equator evenly (with the poles receiving less solar energy than the equator), and by the rotation of the planet.

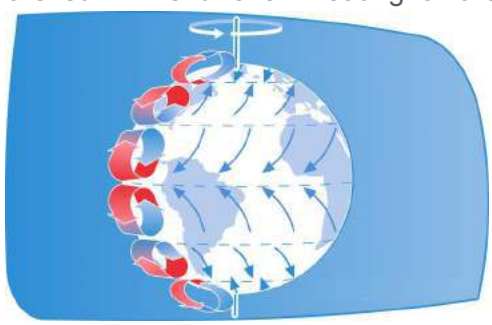
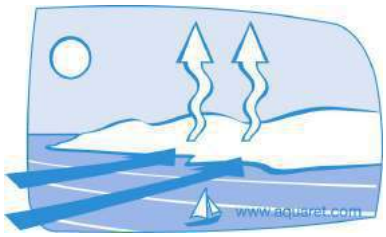


Figure 5 Atmospheric circulation around the world

Warm air is less dense and lighter than cold air, therefore air in the warmer regions rises, and cooler more dense air flows in to take its place, creating winds. Fluctuations in temperature give rise to differences in the density of masses of air, leading them to move from one latitudinal zone to another. This movement is from areas where atmospheric pressure is high to areas where it is low.

Land and water absorb the sun's heat at different rates, the air over land heats up and cools down more quickly than the air over water. During the day, the air over land expends and rises and is replaced by the cooler sea air, crating sea breezes on coastlines, and during the night, these winds are reserved.



1.1.2 Temporal and spatial dimensions

Today, short-term and long-term time series data are used to define the wind resource assessment at a determined site. Fluctuations in wind speed and direction also occur over periods of minutes to hours. Unlike true turbulence, where there is a very short timescales of caused by passing pressure disturbances, or eddies. However, these variations are captured by wind turbines, and resulting in changes in output. It is consequently the needs of a focus of short-term wind energy forecasting.

By the way, on a timescale of 12–24h, the wind variations associated with the daily pattern of solar heating and radiative cooling of the earth's surface. Depending on the height above ground and the nature of the wind climate, wind speeds at a given location typically peak either at mid-afternoon or at night, see an example of diurnal speed variation in *Chapter 8*.

Also, the influence of the seasons begins at timescales of months. In most mid-latitude regions, the better winds usually occur from late fall to spring, while the summer is less windy. Sites experiencing strong warm-weather mesoscale circulations, are often an exception to this rule, and winds there tend to be strongest from spring to early fall, see an example of monthly mean wind speed variation in *Chapter 4*.

The spatial dimension of wind resource assessment is especially important for wind plant design. Most wind power plants have more than one wind turbine. To predict the total power production, it is necessary to understand how the wind resource varies among the turbines. So the spatial scales of interest are related to the size of wind turbines and the dimensions of wind power projects. The rotors of modern, large wind turbines range in diameter from 70 to 140m in onshore and from 120 to 180m. Wind turbines are typically spaced some 5x8D apart (where D is the rotor diameter), and large wind projects can span a region as wide as 10–30 km. Within this overall range, a detailed map of the variations is essential for the optimal placement of wind turbine and accurate estimates of their energy production.

1.1.3 Other characteristic of the wind resource

Although wind speed is the dominant characteristic of the wind resource, there are other important ones, including wind direction, air density, and icing frequency, all of which need to be well characterized to produce an accurate energy production estimate, see *Chapter 4*.

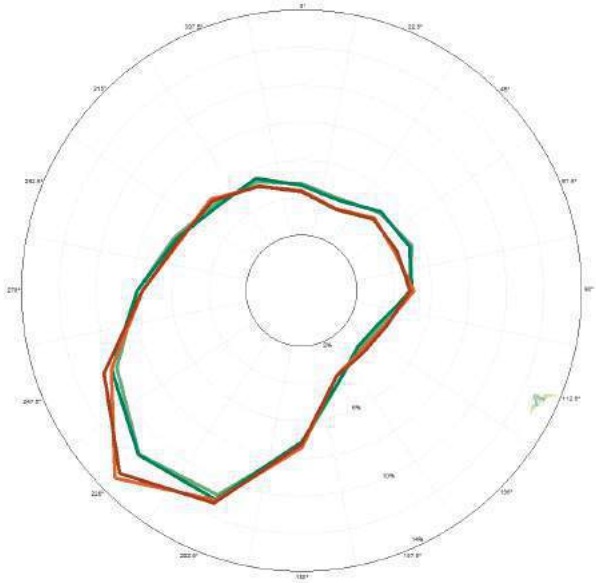


Figure 6 Wind rose to display the wind speed and direction at a hub-height at meteomast IJmuiden

The frequency distribution of wind directions is key for optimizing the layout of wind turbines, which are generally spaced farther apart along the predominant wind directions than along other directions.

Air density depends mainly on temperature and elevation. Air density determines the amount of energy available in the wind at a particular wind speed: the greater the density, the more energy is available and the more electric power a turbine can produce.

In the *Chapter 8*, it shows the wind characteristics of wind resource assessment at an offshore site in the *Dutch North Sea*.

1.1.4 Types of wind

In the wind energy context, types of wind are defined using the different curves, generally called wind characteristic curves, and averages generated at a site, see an example of these curves in the *Chapter 8*.

The wind speed determines the strength of the wind at a given in space. Normally, it is measured in 10min period using anemometers. For large turbines, the International Electro-technical Commission (IEC) defined the type of wind by the norm IEC-61400-1, considering 5 types of turbines based on basic parameters, see more about wind turbines in the *Chapter 6*.

1.2 Wind Power Plants

For any power plant to generate electricity, it needs fuel. The wind is the fuel for a wind power plant.

Conceptually, a wind turbine is a simple machine. The motion of the air is converted by the blades, which generate a torque on a shaft [**2**]. Today, a large offshore wind turbine is an immense, complicated machine ranging from 65 to 120m in height at the hub, with a rotor about 140-180m in diameter, and a power capacity of about 3 to 7 or the new turbines of 8MW, see *Chapter 6 Technology scale and deployment arrangements*.

The key characteristic of a wind turbine from the perspective of wind resource assessment is the turbine power curve. This describes the power output as a function of wind speed measured at the hub.

Figure 7 Wind turbine model nacelle and blades

1.3 Wind Flow Models

The optimal design of a wind project and the accurate prediction of its energy production depend on having an accurate and detailed understanding of the spatial distribution of the wind resource across the project area. The creation of a wind atlas to map the geographical distribution of the wind resource potential over a territory is often the first step for the sustainable development of a wind energy sector. It can be used to identify the best sites for more advanced studies and to optimize economical and territorial planning. In short, we need wind flow modelling, because we do not have measurement at every potential turbine location.

Nowadays, numerical wind flow modelling, combined with on-site meteorological measurement, is used to approach to estimate the wind and wave resources.

1.3.1 Types of wind models

Spatial modelling approaches are classed in the following four categories [**2**]:

- Conceptual models: These models are theories describing how the wind resource is likely to vary across the terrain. They are usually based on a combination of practical experience and a theoretical understanding of boundary layer meteorology. A very simple conceptual model might state that the wind resource at one location (turbine) is the same as that measured at a different location (met mast).

Some examples of conceptual models:

- The influence of elevation on mean wind speed
- Conceptual models
- Winds on upwind slopes versus downwind slopes
- Impact of trees and other vegetation (forest)

As wind projects become larger and are built in ever more varied wind climates, it becomes more and more difficult to implement a purely conceptual approach in a rigorous or repeatable way.

- Experimental models: An experimental model refers to creating a sculpted scale model of a wind project area and testing it in a wind tunnel. The conditions in the wind tunnel, such as the speed and turbulence, must be matched to the scale of the model to replicate real conditions as closely as possible.

Potential benefits and problem with this approach:

- Wind tunnel speed and turbulence must be matched to the scale of the model to replicate real conditions
- Takes time and skill to build such a model
- Difficult to reproduce some conditions, such as thermally stable conditions

In addition, the method has some limitations (such as the difficulty of modelling thermally stable conditions and the challenge of appropriately matching atmospheric parameters to the physical scale).

- Statistical models: These models are based entirely or mostly from on-site wind measurements. The statistical models use several predictive parameters to find those that have the strongest relationship with the wind observations at several masts. Different predictive parameters, such as elevation, slope, exposure, surface roughness, and other indicators might be considered to establish this relationship. Statistical models are grounded in measurement, fairly simple, and transparent, but a problem with statistical models is potentially large errors when results are extrapolated outside the range of conditions used to develop the model. Therefore, it is necessary a validation and determinate its accuracy. Statistical models can be combined with other wind flow modelling approaches, such as numerical flow models.

Suppose one has measured the mean wind speeds at several different towers at different points within a wind resource area. For example, a linear equation (y = mx + b) could be derived and then applied to predict the speed at any other point in the area in function of hub-height. An application of this method can be seen in the *Chapter 8*.

- Numerical wind flow models: The most popular methods of spatial modelling rely mainly on numerical wind flow models. These models attempt to solve at least some of the physical equations governing motions of the atmosphere, with varying degrees of complexity. The models fall into four general categories.

- *Mass-Consistent models:* They solve just one of the physical equations governing fluid flow: Mass conservation. They were the first generation of wind flow models (1970s and 1980s). These models cannot handle thermally driven winds or flow separation in lee of hills. The solution is not unique so constraints must be added, such as a first-guess field or measurements from towers.

- *Jackson–Hunt models:* The next generation of linear wind flow models were developed in the 1980s and 1990s based on theory of Jackson and Hunt (1975). Considering mass conservation and momentum conservation, the model assumes that terrain causes a small perturbation to a constant background flow to solve a linearized form of the Navier-Stokes equations governing fluid flow. The most popular and still the most widely used numerical wind flow model in Europa is *WAsP*, which is a Jackson-Hunt model developed at

- *CFD Models:* These models are considered the next generation of wind flow models. The CFD models solve a more complete form of the equations of motion known as the Reynolds-averaged Navier-Stokes equations. This kind of software simulates the influences of roughness changes and obstacles directly. There is a continuing need to validate CFD results with high quality wind measurements.

- *NWP or mesoscale numerical weather prediction models:* In principle, fully compressible, non-hydrostatic NWP models can simulate and capture a broad range of meteorological phenomena from synoptic to micro scales, but the required computing power is substantial and increases rapidly with decreasing grid spacing [**38**]. NWP models are usually coupled with *linear microscale wind flow models* to achieve a high spatial resolution.

The next level of sophistication is NWP models coupled to large-eddy simulations (LES). LES models have their origin in meteorology and weather prediction and solve the unsteady, non-linear Navier-Stokes equations with the full physics parameterization schemes. The validity of LES depends crucially on the quality of the chosen turbulence closure scheme because of limited grid resolution and thermal stratification effects [**38**].

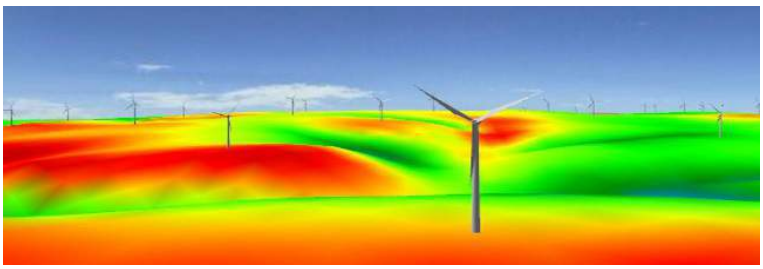


Figure 8 Wind farm modelling with Meteodyn

1.3.2 Examples of different wind flow models:

- *Windographer* - statistical analyses including MCP
- *WindMap/openWind® Enterprise* - a mass-consistent model
- *WAsP* - a linear Jackson-Hunt wind flow model
- *O.F.Wind* - an OpenFOAM@ based CFD tool
- *WindSim or Meteodyn WT* - CFD models or
- *SiteWind®* - a coupled mesoscale NWP-mass-consistent model
- *ARPS* - a coupled mesoscale NWP-LES

In the present study, in *Chapter 8*, aims to characterize the mean wind speed at different heights with a short-term time series data and a long-term time series data (based on a mesoscale model) with the *Windographer* software, which is the industry leading software for analysing wind resource data measured by met tower, SODAR, or LiDAR. It imports virtually every data format commonly encountered in the wind power industry, allows rapid quality control and statistical analyses including MCP, and exports to all wind flow models common to the wind power industry.

2. Overview of Wind Resource Assessment

A wind resources assessment campaign, like other technical projects, requires planning and coordination guided by a clear set of objectives. Tight budgets and schedules are very important and the complexity of project could also be a contributing factor to delay and cost overruns extremely. Its success rests on the quality of the program for the assets, particularly, sound sitting and measurement techniques, high quality equipment, trained staff and data analysis and modelling techniques.

It should be noted that wind resource assessment is the first step in the cycle of a utility-scale wind energy project. A wind assessment campaign could be divided into three main stages [**2**]:

- Site identification
- Assessment of external conditions
- Resource analyses

Wind resource assessment	Permitting	Financing / Due Diligence	Construction	Operation and decommissioning
Site identification	Location	Permits	Site preparation	Performance testing
Assessment of external conditions	Installed capacity	Project design	Turbine installation	Plant operation
Wind resource analysis / energy estimates	Ownership	Energy estimates	Connection / commissioning	Project removal / re-powering

Figure 9 The cycle of a utility-scale wind energy project. Source: AWS Truepower

The design process involves an initial site selection followed by an assessment of external conditions, selection of wind turbine size, subsurface investigation, assessment of geo-hazards, foundation and support structure selection, developing design load cases, and performing geotechnical and structural analyses.

2.1 Site Identification

One or more candidate sites for a wind energy project should be identified. These sites may involve a large region. Knowledge of the wind resource is essential for all wind energy projects, so as to ensure a production of wind energy that fulfils the predicted levels. Besides favourable wind conditions, there are other important considerations of a wind farm site which may include site availability, capacity and transmission access (including proximity to power demand sites and local electricity distribution companies), site infrastructure and access (including impact to existing shipping routes, so visibility and distance from shore) as well as an environmental

impact, neighbourhood and government support and cultural sensitivity.

2.1.1 Assessment of external conditions

Following initial site selection, the developer makes an assessment of external conditions such as the level of existing wind conditions, water depth, currents, tides, wave conditions and ice loading (or *metocean conditions*), the site geology and associated geo-hazards, such as sea-floor mudslides, scour and seismic hazards. Oceanographic surveying is the primary method of obtaining this baseline data. These can provide characterisation of a specific offshore site or a more regional marine environment. The *metocean survey* combines oceanographic with meteorological components.

At sea, the wind is more regular than on land, but the surrounding coasts influence the wind conditions. For a good evaluation of the energy resource it is necessary a detailed modelling of the area of sea in question based on more global meteorological models. Its validation is quantified by comparing them with measurements taken at nearby weather stations, satellite or numerical hind-cast models. A meteorological mast, named *Met Mast*, is often installed in the site and left for an entire year, and will continue to provide information throughout the lifetime, to collect reliable and representative wind data (measures of speed and direction of the wind up to the height of the nacelles on the farm being planned (i.e. 90 m for the baseline wind turbine) to provide more accurate estimates of the energy it could produce.

2.1.2 Foundations

The ground must be able to withstand the mechanical strain and therefore has a very important to play. A 5MW turbine is something subjected to wind speeds of up 50m/s, which translates until the submerged structure. The foundations frequently interact not only with the upper layer of the seabed to a depth of several meters when pile foundations are used. Given a variable in topology in seabed a study of geological history of the site makes it possible to predict the different features likely to be found. Two types of these measurements, *geophysical* and *geotechnical*, are carried out on the site one after the other. A boat will be carry out a detailed survey of the area using ultrasonic sounders and for the geotechnical measurements a special craft test the resistance of the ground to penetration and takes core samples of the sea bed which enables us to specify the nature of the sub-oil and extract the mechanical data needed for the foundations. This seabed study cover a larger area than farm itself does, since the cable's path capable of taking to shore included. All these tests are costly. Wind farms are new infrastructures in the marine space and it is to raise the question of their environmental impact. In one way or another birds, fish and marine mammals are affected [**23**].

2.2 Monitoring and Measurements Campaign

Common to all monitoring programs is the need for a measurement plan. This plan should specify any following elements [**2**]:

- Measurement parameters (e.g., speed, direction, temperature);
- Equipment type, quality, and cost;
- Equipment monitoring heights and boom orientations;
- Number and locations of monitoring masts;
- Minimum desired measurement accuracy, duration, and data recovery;
- Data sampling and recording intervals;
- Parties responsible for equipment installation, maintenance, data validation, and reporting;
- Data transmission, screening, and processing procedures;
- Quality control (QC) measures;
- Data reporting intervals and format.

The data recovery for all measured parameters should be as high as possible, with a target for most tower sensors of at least 90%, with few or no extended data gaps. The rate actually achieved will depend on a number of factors, including the remoteness of the site, weather conditions, the type and redundancy of instruments, and methods of data collection.

2.2.1 Met mast and LiDARs

Usually a mast includes a mono-pile foundation, a lattice tower, equipment, navigational aids and aviation lighting so it can be seen clearly from both the air and sea. About its configuration, the mast is likely to be a robust *free-standing* lattice tower designed to withstand offshore conditions. The face width of the mast will be minimised, within technical constraints, to reduce the influence of the mast on measurements, especially at the lower levels.

Also the same infrastructure allows measurement to taken of the waves. The installation of a measuring mast will involve in the preliminary studies, to obtain the necessary authorization, assembled ships to carry out the works, … In addition, since hydrodynamic forces are very strong, the characteristics of the surrounding sea, including tides, waves, currents, should be measured. Buoys with short masts are being used to measure the parameters [**23**].

In the current market, the exceedingly high costs for conventional (*mono-pile*) offshore met masts make justifying on-site wind measurements challenging. However measurement campaigns can indeed deliver considerable value (its more complete dataset may well compensate for the higher horizontal uncertainty). The cost of the traditional solution for offshore wind measurement, a hub-height lattice met mast on a fixed foundation, has become extremely expensive, due to some factors, like deeper waters, more severe wave climates, vessels able to work under determined conditions and costs for a one-off operation [**54**].

Met Mast remains cost-effective at today's installation prices, and several potentially attractive new technologies are on the horizon, e.g. LiDAR Unit on Lightweight Jacket (LiGA) or Floating LiDAR.

Much attention is currently being focused on floating LiDAR in particular and this assessment quantifies the attraction of such a Wind Measurement Strategy. LiDARs on Light-weight Jacket Structures also appear promising with the advantage of a lower technical risk. LiDAR units on a floating buoy will offer a number of very important practical and economic advantages, especially now offshore, once reliable performance has been demonstrated and the wider industry.

2.3 Wind Resource Analysis

The third stage of the wind resource assessment campaign entails the description of the wind resource at all relevant temporal and spatial scales to support the optimal placement of turbines within the project area and the most accurate possible estimation of energy production, including such as data validation, characterization of the observed resource, numerical wind flow modelling, project design and energy production calculations.

For an accurate estimate of the energy output, the statistics of wind speeds are described, than an overview of methods to calculate the wind speed at hub height, e.g. Annual Mean Wind Speeds profile, is given by the *Chapter 4 and Chapter 8.*

3. Data Analysis and Resource Assessment

For information on the offshore wind climate several sources of data at hand. Generally, a short-term detailed data and long-term data of a good quality and availability is not common. Therefore different types of data will probably have to be combined. *Accuracy, duration and resolution* are three main criteria for each type of data source.

For instance, *to* obtain wind speed data, *data voluntary observing ship*, *light-ship data*, *satellite data*, *data from platforms* and *other offshore* sites are a summary of the typical data sources. Some wind resource monitoring systems was presented in next *Chapter 5.*

Considerable effort is devoted to obtain information about the availability of data. Knowing the existence of databases and of the owner or user is not enough to actually get hold of them. Today, there are various databases, which have established online, can be assessed for wind energy applications. For example, the *Hydra project* is a database in the Netherlands, which can be used by external parties. In the case study, which is presented in *Chapter 8*, the measurement data were obtained from the *meteomast IJmuiden*.

3.1 Data collection and handling

The main objective of the data collection and handling process is to make the meteorological measurements from a wind monitoring campaign available for analysis while protecting them from tampering and loss.

3.1.1 RAW data storage

Data are normally stored in the data logger in a compact, binary file format. This kind of data is commonly referred to as raw data. To ensure high data recovery during the monitoring program, the data logger storage medium should be non-volatile, meaning its data are retained even if the logger loses power.

Once transferred from the logger to a computer, the raw data files be permanently archived and preserved, and it cannot be altered. The access to this kind of data will be carried out for special software.

The data recording interval is typically 10min with a number of sensors between 8 and 12 a 60m tower, since a data recording interval shorter than 10min, such as 1min or 2s, means larger data storage or more frequent data retrieval may be necessary. Filtering of the raw 10-min average data was carried out in order to remove effects caused by turbine and mast wakes, poorer quality data and low wind speeds.

To ensure data are protected while stored in the data logger, proper installation procedures should be followed, including grounding all equipment and using spark gaps.

3.1.2 Uncertainty in wind speed measurement

The uncertainty in measurement accuracy have been assessed for each of the met mast datasets, in terms of instrument accuracy and mounting, as well as data quality and processing in *Chapter 8*.

Normally, the meteorological masts, which acquire data from anemometers, may experience flow distortions due to the platforms and equipment, and the effects of tower shadow at the mast have been quantified, as shows in the Energy research Centre of the Netherlands (ECN) report [**5**]. There is a relatively large uncertainty due to instrument mounting, since the booms are relatively short for the size of the mast, although effects are mitigated by the use of a filtering protocol to select the least disturbed instruments.

In *Chapter 8*, it was analysed a filtered raw data of the wind measurements from the *meteomast IJmuiden*. The uncertainty is lower for the *meteomast IJmuiden*, since the top measurement height is taken as the primary reference, which is relatively unaffected by tower effects, and is newly calibrated. An inter-comparison of the wind speed measurements of the LiDAR at 90 m and the met mast at 92 m will be one way to verify the data quality and the uncertainly due to

mast and instruments. In this study, it was not carried out. As the reference *Site Studies Wind Farm Zone Borssele* [**5**], the comparison of these measurements determined an excellent correlation and a linear relationship and indicated that overall there is no significant issue with either the LIDAR and the top anemometry. This study showed us some data points with significant deviation between the LiDAR and met mast anemometry, but this affects only an estimated 0.1% of available data.

3.1.3 Data handling procedures

Improper data handling increases the risk of data loss. All personnel in contact with the data and storage media should be fully trained and should understand the following [**2**]:

- Data retrieval software and computer operating system. Technicians should be aware of all instances in which data can be accidentally overwritten or erased.
- Good handling practices for all data storage media. Data cards and hard disk drives should be protected from static charge, magnetic fields, and temperature extremes.
- Computer operations and safety practices, including grounding requirements.

To reduce the risk of data loss, the raw logger files should be permanently and safely archived and the working database backed up regularly. Archives and backup copies should be stored in a different location from the main files, and not in the same building. Common data backup methods include the use of CD, DVD, and magnetic tape. Online backup services have recently become popular and are especially secure as well as convenient for frequent backups.

With remote data transfers via e-mail, another very effective data protection strategy is to set up backup e-mail accounts. The e-mailed files can go to different computers in different locations.

3.1.4 Data acquisition for a case study

A filtered raw data (short-time time series 10min) was achieved and used in a simple wind assessment study of the *meteomast IJmuiden* in the Dutch North Sea (see Chapter 8). The data was recorded at the offshore meteorological platform IJmuiden using two different technologies, a classical meteorological mast and a LiDAR. In the *meteomast IJmuiden*, the ECN data-acquisition system in use [**5**]:

"Installation of the distributed ECN data-acquisition system is quite simple. At each physical location where sensors must be installed, a front-end system will be placed, e.g. in the turbine nacelle, at the bottom of the meteorological mast or in the turbine tower base.

For each sensor, a dedicated signal conditioning & analog-to-digital-conversion module will be installed in the front-end. Up to a maximum of 16 sensors can be connected to a front-end. These sensors can be strain gauges, rotary encoders, anemometers, wind vanes, temperature sensors, accelerometers, etc.

A special module has been developed to interface with the ECN telemetric system. With this system it is possible to measure up to 12 signals from the turbine rotor. Optical fibres connect all front-end systems to the host computer, completing the data-acquisition system.

A maximum of 16 front-ends can form one data- acquisition system, each front-end at a maximum of 15 kilometres away from the host computer".

3.2 Validation

Whatever method is used, data validation usually proceeds in two phases: automated screening and in-depth review.

The *automated screening* uses a series of algorithms to flag suspect data records. Suspect records contain values that fall outside the normal range based on either prior knowledge or information from other sensors on the same tower. The algorithms commonly include relational tests, range tests, and trend tests.

The second phase, known as *verification*, involves a case-by-case decision about what to do with the suspect values. This is where *know-how* judgment by an experienced person familiar with the monitoring equipment and local meteorology is most helpful. Information that is not part of the automated screening, such as regional weather data, may also be brought into play.

An example is presented in *Figure 10*, showing us after a period of apparently normal operation, the measured mean speeds from an anemometer dropped to the offset value, while the standard deviation dropped to zero. Later, both appeared to return to their normal behaviour. The reviewer checks the temperature and finds it hovered near freezing before the event and rose above freezing at the end. The conclusion is that this was a likely icing event and these measurements should be excluded.

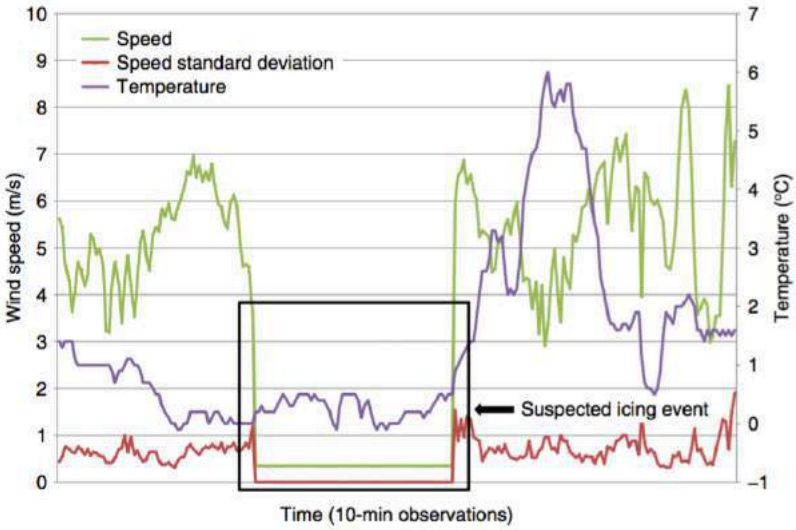


*Figure 10 Example of a plot 10min average wind speed, standard deviation of speed, and temperature [**2**], showing a suspected icing event, source: AWS Truepower*

In the following *Sections*, some aspects to consider in the data validation are presented.

3.2.1 Tower effects

The presence of the tower mast can increase or decrease the observed wind speed compared to the true free-stream speed, see *Figure 11*. This effect depends on direction, sensor distance from the tower, and the width of tower.

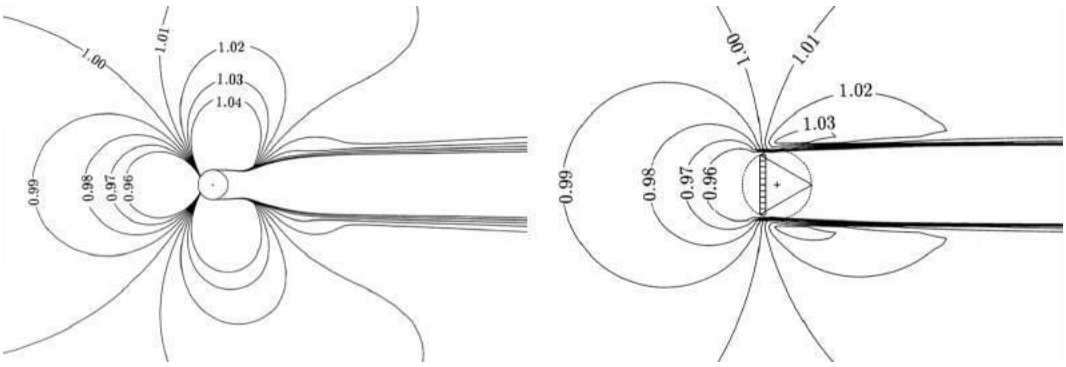


*Figure 11 Illustration of the flows round a solid mast of cylindrical cross section (left) and round a triangular lattice mast (right), with local speed normalised by free-field wind speed [**55**]*

First, to avoid these effects, the mast configuration is likely to be a *free-standing* lattice tower, and the face width of the mast will be minimised, within technical constraints, to reduce the influence of the mast on measurements, especially at the lower levels, see *Section 3.1.2*.

Some effects can be observed, in the case of directly upwind, a tower impedes the wind, reducing the speed; over certain angles on either side of the tower, the tower causes the wind flow to accelerate, producing an increase in the observed speed.

3.2.2 Turbulence

Cup anemometers are known to overestimate the wind speed in turbulent flow conditions because of the tendency of the anemometers to respond more quickly to abrupt increases in speed than to rapid slowdowns. The magnitude of the over-speeding depends on the sensor type and degree of turbulence [**52**].

In contrast to cup anemometers, prop-vane anemometers tend to underestimate the wind speed as a result of turbulence. This is because the wind direction changes so quickly that the vane cannot keep the propeller aligned perfectly into the wind. A propeller anemometer only measures the component of the wind speed that is parallel to its rotation axis; the observed speed is reduced by a factor equal to the cosine of the angle of deviation. Since greater turbulence produces larger direction shifts, the magnitude of prop-vane under-speeding typically increases with increasing turbulence intensity [**2**].

Sonic anemometers, lacking moving parts, are insensitive to turbulence. To bring their measurements in line with those of Class I sensors, however, an adjustment for turbulence may nonetheless be called for, although it is usually small.

3.2.3 Inclined flow

Horizontal axis wind turbines generate power from the component of the wind that is perpendicular to the turbine rotor's plane of rotation. To support an accurate energy production estimate, anemometers should ideally measure only that component. However, cup anemometers, in particular, are sensitive to varying degrees of off-horizontal winds depending on the geometry of the cups and instrument. The instruments with a direct measurement, like a SODAR, a LIDAR, or a vertical anemometer mounted on the mast, can measure the flow angle at the tower.

The flow angle can be estimated from the terrain slope and from wind flow modeling. Inclined flow can also occur in flatter terrain for brief periods under low wind and strong surface heating, but this effect is usually small and thus requires no correction.

3.3 Characterizing the resources

The data can be analysed to produce a variety of wind resource statistics and informative reports. This analysis provides a useful summary of the wind resource observed with a monitoring program. An example of monthly statistic is showed in the *Figure 12*.

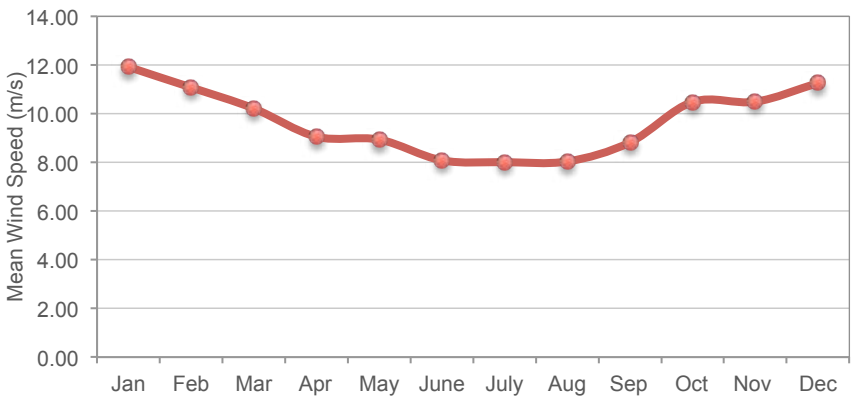


Figure 12 Sample monthly mean wind speeds at a height of 100m

The following *Table 1* shows us some these wind resource statistics. An explanation some statistic parameters are presented in the *Chapter 4*.

	Units
Data recovery	%
Mean wind speed	m/s
Annual mean wind speed	m/s
Mean wind power density (WPD)	W/m2
Wind shear	Non-dimensional exponent
Turbulence intensity (TI)	%
Mean air temperature	°C
Mean air density	kg/m3
Speed frequency distribution	graph
Weibull A and k parameters	m/s, non-dimensional
Wind rose	graph
Daily and hourly speed distributions	graph

*Table 1 Some wind resource report statistics [**2**]*

In the *Chapter 8*, it is showing some statistic parameter calculated from a long-term time series of wind speed and a short-term time series. Some date checks were performed with Know-how of my tutor and *Windographer* software.

4. Estimating the Wind Resource

Wind resource can be defined by mean velocity, described by speed and direction, and turbulence intensity. The wind energy potential of a site is defined by *S. Krohn* [**11**] as the energy, which can be extracted over a given duration, normally a year, of wind turbine's operation on the site. Wind energy conversion systems must be adapted to characteristics of the area's energy potential, which are determined before the design phase of the wind farm.

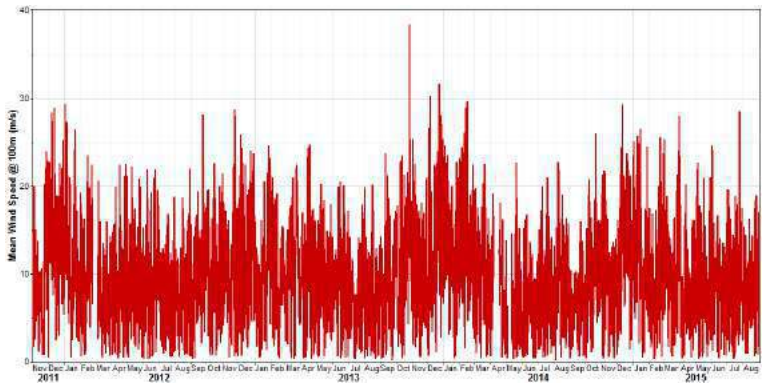


Figure 13 Short-term time series of 10min. means (4 years) at meteomast IJmuiden

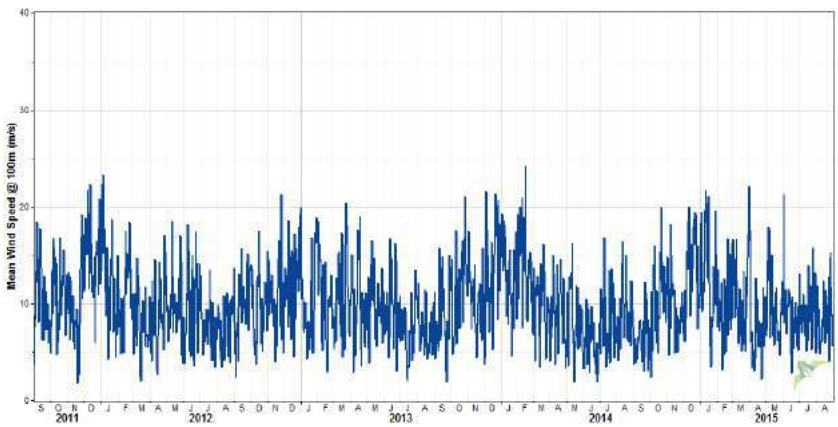


Figure 14 Long-term time series of 4hr. means (33 years) from VORTEX data

Wind is extremely variable, and it fluctuates continuously in speed and direction. This variability extends over an enormous range of time-scale. *Figure 13 and Figure 14* represent Time series of 10min and 4hours, which have been determined with the measurement data at meteomast IJmuiden and VORTEX mesoscale data respectively, see *Estimation of Wind Energy Source, in Chapter 8*.

4.1 Statistical characteristic curves of the wind speed

The speed frequency distribution is a critical piece of information. It is used directly in estimating the power output of a wind turbine. This frequency distribution represents the number of times in the period of record that the observed speed falls within particular ranges. The measurements collected for wind speed define the characteristic curve of the distribution of wind speed *p(v)* and that of cumulative frequency *P(v)*.

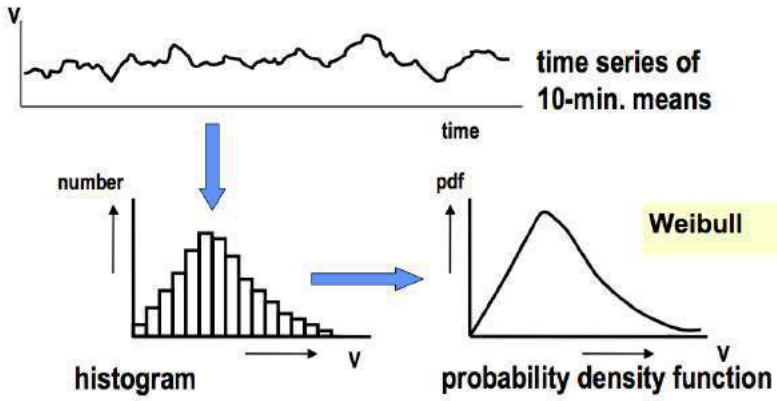


Figure 15 Statistical characteristic curves of the wind speed

While year-to year variation in annual mean winds speeds could be too difficult to predict, wind variations during a year can be well characterized in terms of a probability distribution. Among the distributions used in statistical models, the *Weibull distribution* has proved to be appropriate for describing the statistical properties of the wind [**16**]. Therefore, the *Weibull distribution* is normally used for wind energy estimation at a site and its probability density function is defined as follows [**8**]:

$$p(v) = \frac{k}{A}\left(\frac{v}{A}\right)^{k-1} exp\left[-\left(\frac{v}{A}\right)^{k}\right] for\ v \geq 0, A > 0, k > 0$$

And the *Weibull cumulative distribution function* is:

$$P(v) = 1 - exp\left[-\left(\frac{v}{A}\right)^{k}\right]$$

There are two parameters *A* is the scale parameter, in m/s, which is related closely to the mean wind speed and characterizes the peakedness factor of the distribution, and *k* is non-dimensional shape parameter, which is a measure of the central tendency. *v* is the time series of wind speed observations. The A and k parameters are known, using the diagram of cumulative measured frequencies and the linear regression method, and the moments and percentiles of the wind speed distribution may be computed, e.g. *Abramowitz and Stegun*.

$$\bar{v} = A\Gamma\left(1 + \frac{1}{k}\right)$$

Also once A and k are determined the expected energy density (or *power in the wind – the wind resource that may be harnessed using wind turbine*) is given by:

$$E = \frac{1}{2}\rho A^3 \Gamma\left(1 + \frac{3}{k}\right)$$

The *Weibull* distribution is a good approximation of the true wind speed frequency distribution in the mid- and high–latitudes of the Northern Hemisphere as well as in offshore wind (see *Figure 16*), while the real speed distributions at any sites does not fit a *Weibull distribution* well. Also the use of a *Weibull distr.* is very good fit in offshore wind sites, as known the case study referred in *Chapter 8*.

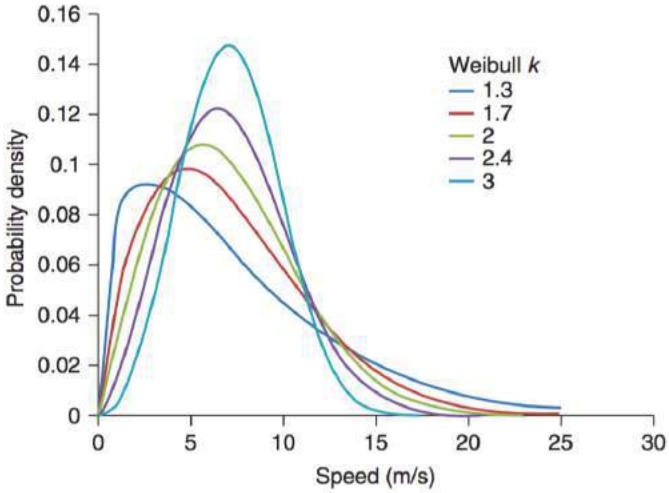


Figure 16 Weibull probability density curves for a range of values of k. All curves have the same A 8.0 m/s. Source: AWS Truepower [**2**].

The calculation of the offshore wind resource is carried out along well-established methodologies using all types of available data: in-situ and remote sensed data and the results of meteorological modelling.

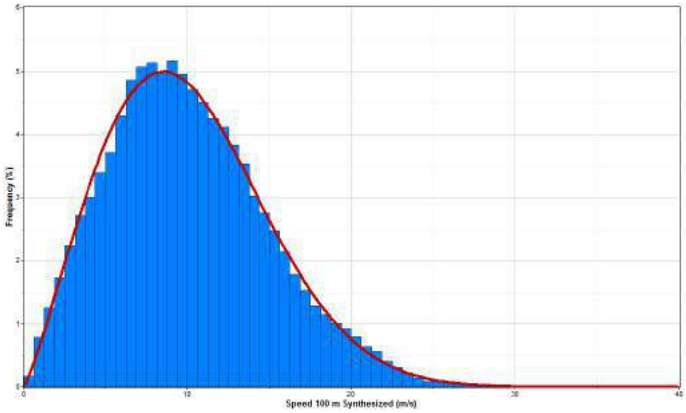


Figure 17 Frequency distribution plotted against a fitted Weibull curve (k = 2.17,c = 11.34 m/s) of the measurement data at mesomast IJmuiden

The observed distribution in this case is modal, that is, it has only one peak, and does follow the *Weibull curve* very well. The goodness of fit allows us to check the energy density (W/m^2) from the measurement data.

Windographer software has three methods (*Maximum likelihood, Least squares and WAsP*) for fitting the *Weibull distribution* to measured wind speed data at *meteomast IJmuiden*. In the worst-case the deviation is around 0.3%. The *WAsP* fitting is typically used.

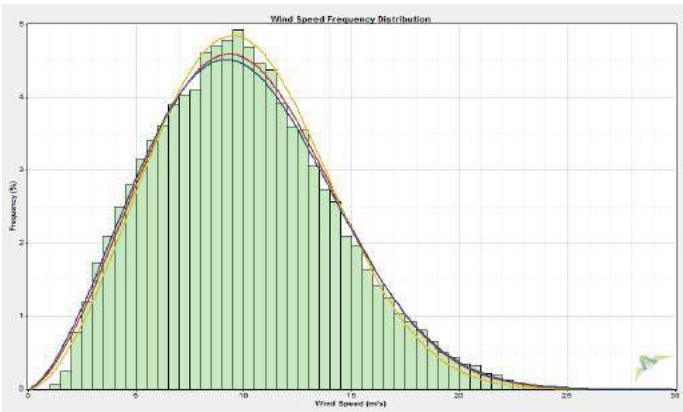


Figure 18 Wind Speed Frequency Distribution and three algorithms for Weibull fitting

By straightforward combination of the *Weibull distribution* and the *power curve of the wind turbine* (power as function of the wind speed at hub height; both averaged over 10min) the annual energy yield can be determined.

The statistical curve is used to evaluate the significance of wind speed samples for energy production. Depending on the nature of the turbine, the energy potential defined will be more or less fully exploited according to the initial physical constraints, the electromechanical system and the action of loads on the structure. Because the wind is so variable, also from year to year (annual average wind speed from one year can easily differ until 20 % from another year) one needs to measure for many years to obtain a reliable average [**2**]. As a time-consuming solution, the so-called Measure Correlate Predict methods (MCP) have been developed, which correlate the short-term measurements (even one year is short term) with simultaneous measurements at a site where long-term statistics and wind atlas methods are available. It is of course also useful to combine both methods and use the short-term correlation to validate/calibrate theoretical

methods. An example of different MCP methods are analysed in the case study in *Chapter 8*.

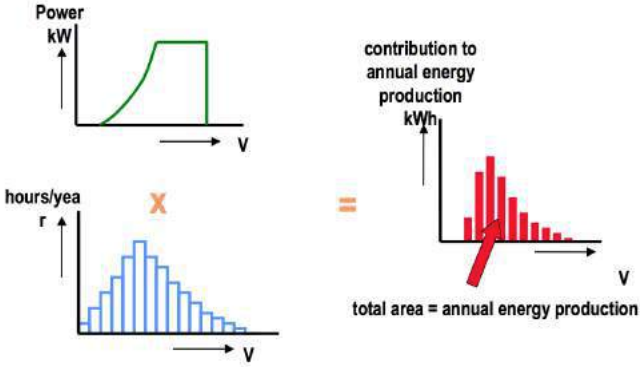


Figure 19 Annual energy yield production [**6**]

4.1.1 Speed-duration characteristic curve

The speed-duration curve is used to predict the periods of a wind turbine's operation at a site. The *Weibull model* is used to trace this curve, which shows the percentage of time during which the wind speeds observed are above a certain value.

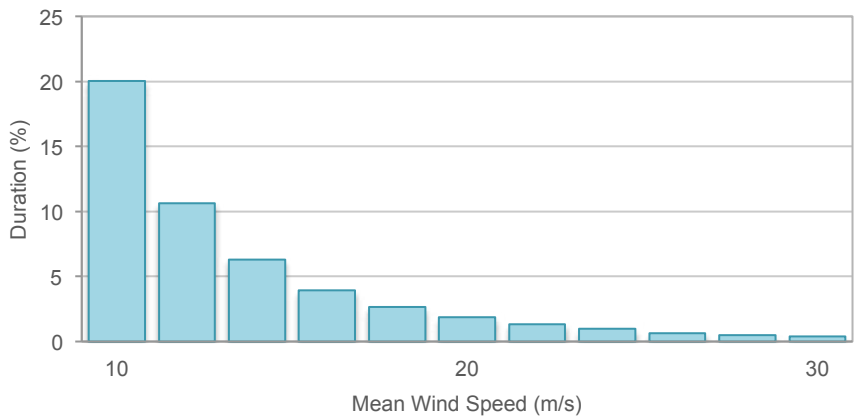


Figure 20 Speed / duration characteristic curve

4.2 Wind speed

The *mean wind speed* is the average of the valid speed values for the period at a given location:

$$\bar{v} = \frac{1}{N_{valid}} \sum_{i=1}^{N_{valid}} v_i$$

, where N_{valid} is defined as the number of valid data records against the total possible number of records N for a reporting period. The mean wind speed can sometimes be a misleading parameter of the wind resource. If the data span a period much shorter than a full year, the mean will not reflect the full seasonal cycle of the wind variations.

The *annualized mean wind speed* attempts to correct for gaps in a record or an overestimate the true annual average speed. This parameter can be estimated, first, the mean for each calendar month in the record, and second, the weight by the number of days in each month [**2**]:

$$\bar{v}_{annual} = \frac{1}{356.25} \sum_{m=1}^{12} D_m \bar{v}_m = \frac{1}{356.25} \sum_{m=1}^{12} \left(\frac{1}{N_m} \sum_{i=1}^{N_m} v_{im} \right)$$

, where the 12 calendar months with D_m being the average number of days in month m. The inner sum is over the speeds that include within a particular calendar month.

The *Figure 21* represents the monthly wind speed of a short-term time series 10min wind speed measurement (4 years) and a long-term time series hourly dataset (33 years). The most wind turbines start up at wind speeds of 3 or 4 m/s, it can see in the *Figure 21* that the exploitation of

wind energy is probably favourable on this site, since the monthly average wind speeds are always greater than 8 m/s.

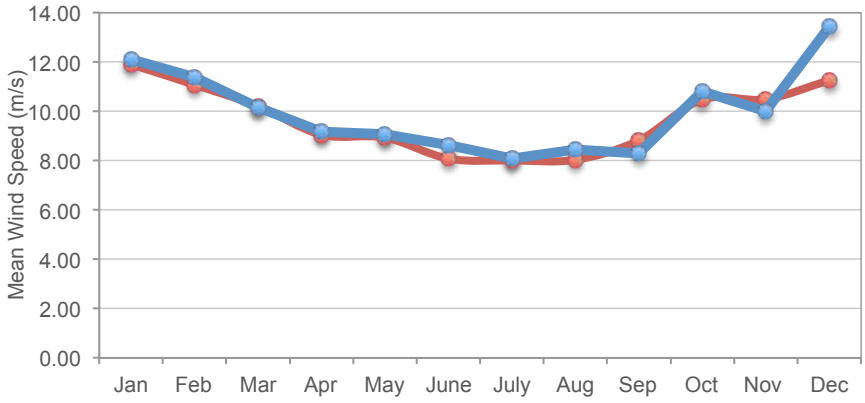


Figure 21 Monthly mean wind speeds at 100m based on Meteomast IJmuiden measurement data (blue) and VORTEX Data Long-term mean wind speeds series (red)

4.2.1 Wind shear

Vertical wind shear is a change in wind speed or direction at different altitudes. Wind speed increases with altitude, and this dependency is expressed for the assessment of the wind resource by power law equation relates the wind speeds, v_1 and v_2, at two different heights, h_1 and h_2 respectively. The wind shear can be characterised by the α power law exponent in the power law equation:

$$\frac{v_2}{v_1} = \left(\frac{h_2}{h_1}\right)^{\alpha}$$

The rate of change in horizontal wind speed with height is typically expressed as a dimensionless power law exponent as know α. Vertical mean speed profile is used to extrapolate the wind speed to the wind turbines hub-height.

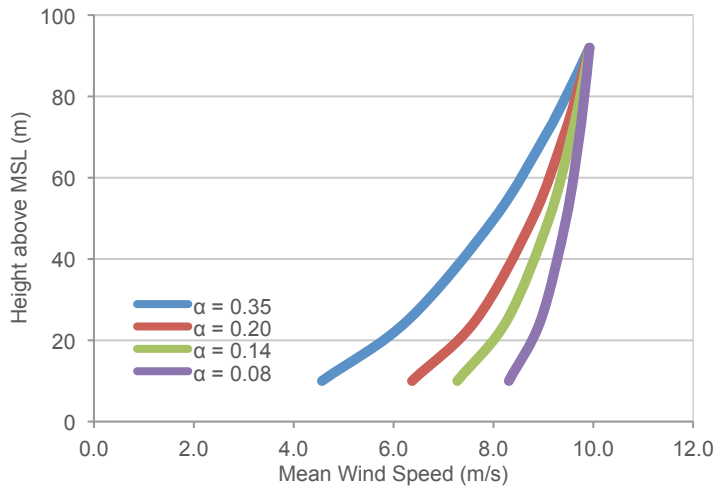


Figure 22 Theoretical profiles of wind speed with height for a range of values of the exponent α (0.08, 0.14, 0.20, and 0.35). All curves assume a speed of 10 m/s at 92 m height

For onshore wind turbines, the exponent α depends on the layout the terrain and varies between 0.1 and 0.4. For irregular terrain, α takes the values of the upper limit. In *Table 2* it shows some representative α values. These estimates may not be valid for specific sites; measurement is required.

Just one average shear value for each pair of heights is usually provided in wind resource reports. This shear is calculated per sector as noted above, by averaging the concurrent speeds from two anemometers at two different heights, then taking the ratio and calculating the exponent. Some analysts choose to exclude speeds below 3 or 4 m/s in this calculation, as shear tends to be more variable in light winds, and low speeds do not contribute significantly to energy production [**2**]. Negative or very high values may be caused by tower or shielding effect.

A first step towards a physical description of the wind profile for a thermally stable atmosphere is

the *logarithmic law*. It incorporates the roughness length z_0, and gives therefore more information, in this case about the underlying surface [**6**]:

$$v(z_1) = \frac{v_*}{k}\ln\left(\frac{z_1}{z_0}\right) \rightarrow \frac{v(z_1)}{v(z_2)} = \frac{\ln\left(\frac{z_1}{z_0}\right)}{\ln\left(\frac{z_2}{z_0}\right)}$$

Terrain type	Land cover	Approximate range of annual mean wind shear exponent
Flat or rolling	Low to moderate vegetation	0.12-0.25
Flat or rolling	Patchy woods or forest	0.25-0.40
Complex, valley (sheltered)	Varied	0.25-0.60
Complex, valley (gap or thermal flow)	Varied	0.10-0.20
Complex, ridgeline	Low to moderate vegetation	0.15-0.25
Complex, ridgeline	Forest	0.20-0.35
Offshore, temperate	Water	0.10-0.15
Offshore, tropical	Water	0.07-0.10

Table 2 Typical shear exponents for different site conditions

The value of the Von Kármán constant *k* is 0.4, while *u** is the friction velocity, a measure for the effectiveness of vertical exchange of momentum. In this expression it is assumed that there is no heat exchange (adiabatic conditions), and the wind profile is completely determined by mechanical turbulence. This assumption is certainly true for high wind speeds, and appears to be true on average for offshore conditions.

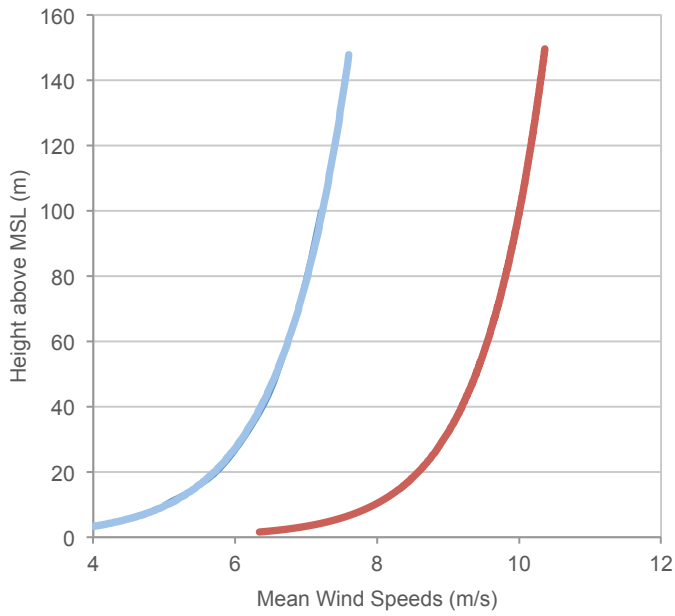


Figure 23 Example of a Mean Wind Speeds profile – onshore (blue) and offshore (red)

While onshore the value depends on the terrain conditions, offshore conditions the value of z_0 does depend on the wind speed, see *Figure 23*. Due to the generation of waves by the wind there is a strong correlation between the wind speed and the roughness length. In a simple form was formulated by *Charnock* and related between z_0 and the friction speed as follows:

$$z_0 = 0.0144\frac{v^2}{g}$$

, where g is the gravitational constant. In practice, what is used is either a logarithmic law or an exponential-type law, taking 10m as the reference height. Most regulations call for the exponential-type law.

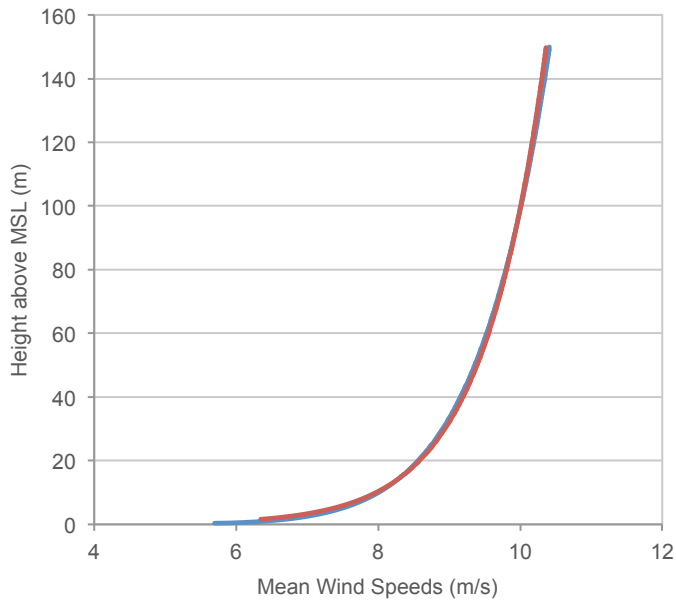


Figure 24 Mean Wind Speeds profile – Power law fit (blue) and Logarithmic law fit (red)

4.2.4 Wind direction

Determining *wind direction* plays an important role when installing wind turbines, since there must be as few obstacles as possible in the prevailing direction [**1**]. It is desirable to space turbines much further apart along the principle wind direction than perpendicular to it to minimise wake interference between the turbines. For this reason, the directional frequency distribution is a key characteristic of the wind resources.

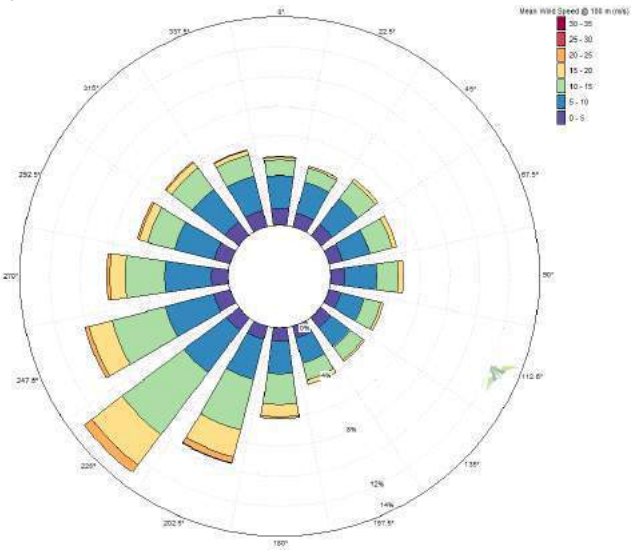


Figure 25 Frequency of Mean Wind Speed vs. Wind direction at 100m

A *wind rose* is a polar plot displaying the frequency of occurrence and direction over a period of measurements (for one year, one month, or one hour of the day). Three elements are displayed [**1**]:

- Often display the percentage of total time, or the frequency of occurrence f_i, during which the wind blows on a given direction in certain speed ranges dividing each segment of the plot into different colour sectors.

$$f_i = 100\frac{N_i}{N}$$

- Known as *energy rose*, displays the percentage of total energy, E_i, in wind coming from each direction, where *WPD* is wind power density (see below).

$$E_i = 100\frac{N_i \cdot WPD_i}{N \cdot WPD}$$

- The average intensity of turbulence is represented on each sector.

Wind roses are spatio-temporal diagrams produced and used by professional to qualitatively evaluate the site potential. This resource will allow us to work out the optimal relative positioning of the turbines within a farm.

4.3 Other parameters

4.3.1 Air density

The air density depends on temperature and pressure, thus as altitude. Seasonally this parameter can vary between 10 – 15%. Considering an ideal gas law, the air density can be calculated by [**2**]:

$$\rho = \frac{P}{RT}\ (kg/m^3)$$

, where *P* is the site air pressure, in Pa or N/m^2, *R* is the specific gas constant for dry air, and *T* is the air temperature in Kelvin. It the site pressure is not available, the air density can be estimated as a function of the site elevation and temperature, as follows:

$$\rho = \frac{P_0}{RT}\ e^{\left(-\frac{gz}{RT}\right)} = \frac{353.05}{RT}\ e^{\left(-0.03417\frac{Z}{T}\right)}$$

, where P_0 is the standard sea-level atmospheric pressure, *g* is the gravitational constant, and *z* is the elevation of the temperature sensor above mean sea level in meter. The error of this equation increases with increasing elevation because the air pressure does not follow the exponential function exactly.

The Wind speed-power curve, which is provided by wind turbine manufacture, is very dependent on the air density. This important parameter was not analysed in the case study in *Chapter 8*.

4.3.2 Turbulence intensity

The planetary boundary layer is always characterized with some degree of turbulence. The fluctuations in wind speed and direction, which have a significant impact on turbine performance and loading, are known as *wind turbulence*.

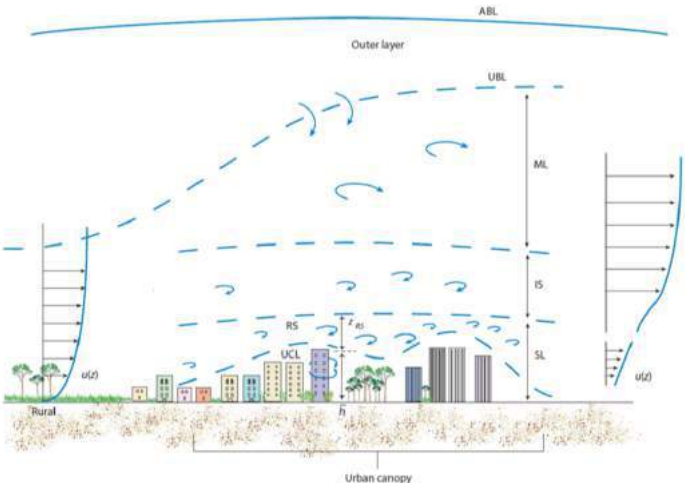


*Figure 26 Illustration of the different layers of the planetary boundary layer above an urban area. The urban boundary layer (UBL) is divided into a mixed layer (ML), inertial sub-layer (IS) and surface layer (SL), which in turn is divided into a roughness sub-layer (RS) and an urban canopy layer (UCL). Also illustrated are the vertical wind speed profiles, u(z), and its difference between rural and urban areas (Fernando, 2010) [**15**].*

In wind energy, the turbulence is evaluated by the *turbulence intensity*, calculated by dividing the *standard deviation of* 10 minutes wind speed series by its mean wind speed according to the following equation [**14**], as known the following equation:

$$TI_v = \frac{\sigma_v}{\bar{v}_{mean}}$$

This parameter will be normally assessed for each wind speed bin (about 1m/s). The turbulence intensity in the neutral atmosphere clearly depends on the surface roughness. For instance, *TI* values above 10 m/s typically range from less than 0.10 in relatively *flat terrain with few trees* or other obstacles to more than 0.25 in *forested, steep terrain*. Along with the *mean wind speed*

and *air density*, the *TI* at 15 m/s enables a preliminary determination of the suitability of a turbine model for the project site.

The final decision to select a turbine should be taken into account the frequency distribution of turbulence as well as turbulence generated by upstream turbines.

4.3.3 Wind power density

Wind power density or WPD is defined as the flux of kinetic energy in the wind per unit cross-sectional area. This parameter provides an indication of the wind energy production potential of a determined site. It is calculated in the following equation:

$$WPD = \frac{1}{2N}\sum_{i=1}^{N} \rho_i \, v_i^3$$

, where *N* is the number of records in the reference study, ρ_i is the air density in kg/m^3, and v_i is the wind speed for record *i* in m/s. The air density in this equation must be calculated, as described in the above section.

By integration in relation to the time of the density of the wind's kinetic power (proportional to the cube of the wind and expressed in kW/m^2) it is possible to determine the density of gross density, expressed in kWh/m^2, of the surface intercepted.

The real WPD is generally a few per cent greater than that calculated from this formula. This is usually not important for wind resource assessment, since WPD is not used directly in calculating energy production [**2**].

5. Offshore Resource Assessment

This Chapter focuses on the special considerations and requirements for offshore resource assessment.

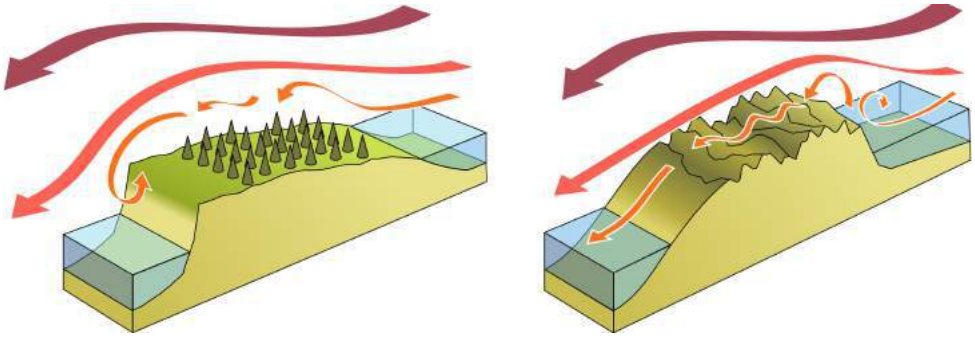
5.1 Nature of the offshore wind environment

The greater the wind speed, the greater the energy it contains. Offshore wind environment differ from onshore environments in a number of ways. One difference is the surface roughness of open water is much smaller the land surfaces. Offshore wind speeds are generally higher than on land as there are no complex terrains to disrupt the wind flow. As wind blows over the water surface (generating waves) it loses some energy due to friction. The energy in wind is therefore usually higher further away from the coastline, and increases in strength with increasing height above the water surface.

Onshore winds are generally slower than those at sea because of friction caused by obstacles such as trees, cliffs and mountains.

Another difference is that the daily cycle of surface temperature variation is usually attenuated offshore because the seawater heat capacity is much greater than soil and maintains a more constant temperature throughout. This particularity produces smaller variations in atmospheric stability and wind profile [**2**].

Because of the lack of terrain, winds and other meteorological conditions tend to be more spatially uniform offshore, especially farther than around 5 km from the land. This is fortunate for wind project development, as it means that fewer measurement stations are generally required to characterize the resource accurately within a project area.



Even so, surprisingly complex wind phenomena can occur. In the offshore zone, close to the coast (5km from the coast), several factors complicate the assessment of wind profile and wind resource. It can be expected that stability effects are important; very low roughness and large differences in temperature structures, both over the day and over the seasons.

Three different cases are for instance:

1. In the summer sea breezes can develop during the day and with onshore wind the air is in equilibrium with water surface temperature.
2. The return flows at night results in relatively warm air flowing out over colder water.
3. In the winter the water is usually warmer than the land, so when the wind flows from land to sea, cold air flows over warm water.

Looking at the general situation onshore compared to that offshore the following characteristics can be identified, see *Table 3*.

	onshore	offshore
diurnal pattern	daily maximum	uniform
seasonal pattern	less pronounced	more pronounced
stability	diurnal pattern	seasonal pattern
wind profile	unstable on average	neutral on average
mean wind speed	decreasing inland	higher than on land

*Table 3 Comparison of atmospheric characteristics onshore and offshore [**6**]*

5.2 Evaluation of the offshore wind resource

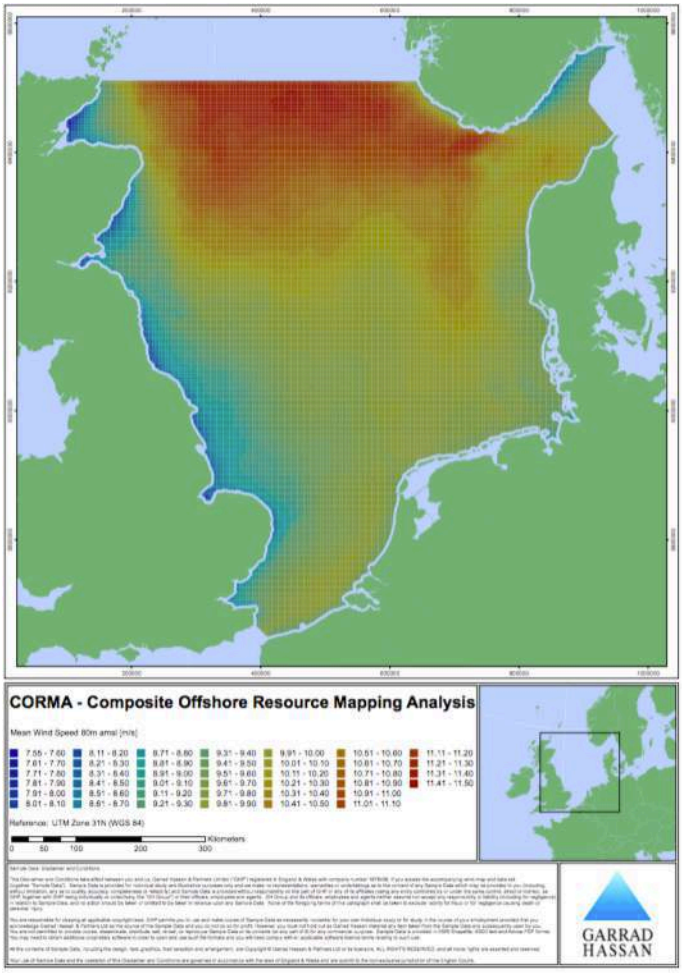
Evaluation of the wind energy resource is done by a general estimation of the average energy contained in the wind over an extended area.

Offshore wind resource estimation has a number of special issues. The thermal stability of the atmosphere has an important effect on the vertical wind profile and on estimates of the wind resource at a particular height. Therefore, a correction to the logarithmic profile must be applied to take it into account for the complexity, *Gryning et al. 2007* [**17**].

In offshore, atmospheric stability depends on the temperature difference between sea and air. To attribute the correct roughness to the sea surface is also a problem (see Ch.4 Wind shear), this roughness depends on the sea state, which is also a function of the depth of the sea. However, it has been shown that varying sea surface roughness has a minor impact on wind speed profiles, *Barthelmieg et al. 2001* [**18**].

Specific measurements for offshore wind farms have been mostly undertaken below 100m in height, but the growth in size of wind turbines, which are developing now, gives rise to higher altitudes.

Since there are evidences from wind speed profiles measured above 50m suggest that the use



of atmospheric stability corrections based on logarithmic profile (or *Monin-Obukhov*) theories may not be adequate for offshore wind speed profile, different measurements programmes have been made at prospective offshore wind farm sites using purpose built meteorological masts at a number of sites in northern Europe.

For information, one of the main issues of extrapolating vertical wind speed and turbulence profiles to turbine hub-heights without the use of a tall meteorological mast has been addressed using both SODAR (*Coelingh et al.,* and *Barthelmie at al., 2003*), and more recently LIDAR (*Antoniu et al., 2006*) [**8**].

Figure 27 Offshore wind resource distribution in the Dutch North Sea, Atlas CORMA (DNV GL)

5.2.1 Variation of wind speed with altitude

With reference to previous *Chapter 4*, it is accepted that the average speed follows the following empirical logarithmic law [**19**]:

$$v(z) = \frac{v_*}{k} \ln\left(\frac{z}{z_0}\right)$$

, where v is the friction speed, K is Karman's constant and z_0 a reference height depending on the sea state, and therefore on the wind speed. The variation of the wind speeds with altitude, what is used is either a logarithmic law or an exponential law, taking normally 10m as the reference height.

As mentioned in *Chapter 4 (Section 2)*, the most regulations call for the exponential law and a α coefficient between 0.1 and 0.2. If we consider at a height of 200m, the difference in wind speed can be up to 50% with an effect on the distribution of the loads on the components of a turbine.

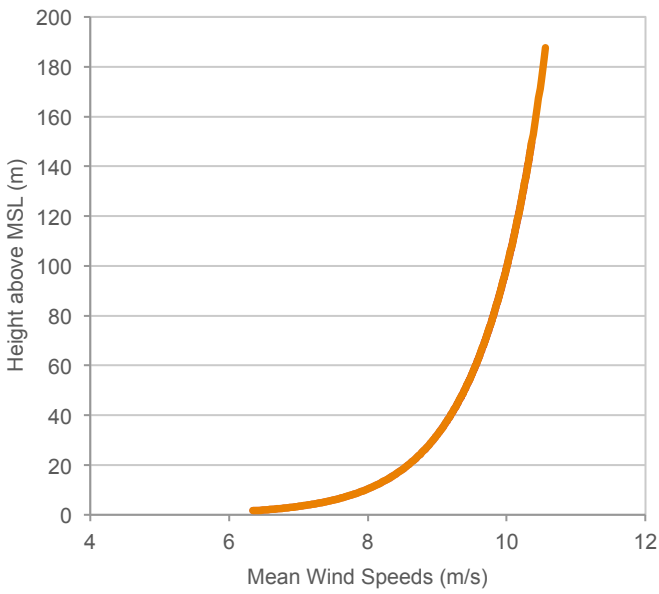


Figure 28 Offshore Mean Wind Speeds profile at a location in North Dutch sea

For offshore wind projects, IEC 61400-1 defines extreme wind speed, V_{ref}, based on a 10-min. average. The 50-year extreme gust *V50(z)* is defined to have duration of 3 seconds based on the following equation:

$$V50(z) = 1.4V_{ref}\left(\frac{z}{z_{hub}}\right)^{0.11}$$

, where *z* is the measurement height. The empirically derived exponent, 0.11, in this formula is a coefficient that defines the change in wind speed with height, typically referred to as the wind shear profile. Normally, the extreme wind speeds are defined based on a long-term on-site measurement, to which the turbine might be subjected. In *Chapter 8*, the extreme wind speed are analysed in a specific study.

5.2.2 Mapping offshore wind energy potential

The installation of offshore wind farms enables us to benefit from stronger, more regular wind speed than on land. This represents a higher and more regular energy production.

No two offshore sites are the same, and this can make choosing the right site for the offshore wind farm quite a challenge. There are many different aspects to consider when choosing the location for an offshore wind farm, e.g. wind and wave resources, bathymetry, environmental impact, access to ports and building facilities, …

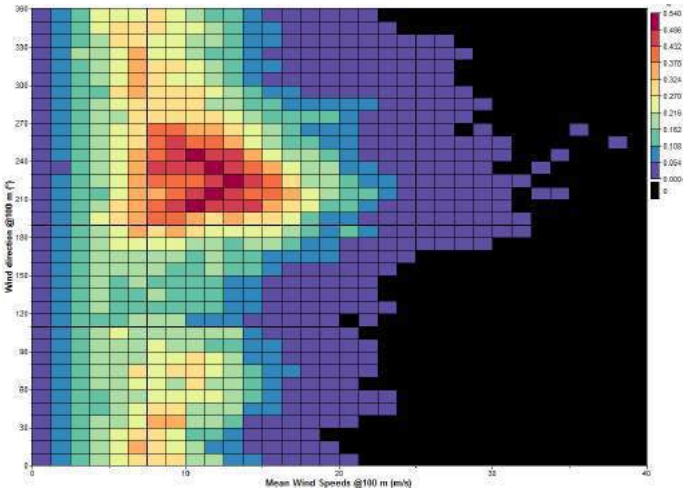


Figure 29 Representation of the wind resource at a location in the Dutch North Sea: Mean Wind Speed vs. Wind Direction @100m

The climatology of the wind at a determined site is one of the most important factors, since it conditions the energy resource, and therefore, the electricity produced. This is represented by the frequency or occurrence of different wind direction and the wind speed. At the stage of choosing the installation site, it is necessary a precise map of the wind resource with the evaluating of its spatial-distribution. In the next Section, it is explained the some wind monitoring systems, which are applied in the offshore wind industry.

Other characteristics aspects are described below, are summarised of the article *How to choose the right location for an offshore wind farm?* [**35**], it is based on the experience of the largest wind turbine manufacturer, like *Vestas*.

- **Minimising the risk:** High-quality site data is the best tool for minimising the risk for all parties involved in the project. Therefore it recommend starting the survey process as early as possible to make sure that the results of the Offshore Design Basis are complete and accurate. A lower risk means a higher investment.
- **Begin with a simple desk study:** It is recommended carrying out a simple desk study, before investing in expensive equipment and surveys. A fair amount of information about the different sites under consideration might already be available from local sources. If the information reveals shallow waters, good winds and stable seabed conditions, however, it might be recommendable to go ahead with a full survey. Manufacturers can evaluate and verify the site data they receive, but the responsibility for collecting and sharing this data rests solely with the developer.
- **Waves, weather and geotechnics:** In addition to measuring wind and wave conditions, the developer will need to execute a *Geotechnical Campaign*, one

geotechnical and two geophysical surveys in order to obtain the full assessment needed, in order to have the solid picture of the seabed changes that occur over time.

- Demands on the harbour: It is important to make sure that a nearby harbour can accommodate the installation of an offshore wind farm. It is necessary that the harbour can handle the tonnage involved in the construction phase, and can meet the logistic demands.

5.3 Wind Resource Monitoring Systems

The offshore wind industry has yet to establish a standard approach to offshore wind resource assessment. For several years the classic sources of wind measurements (weather stations, instruments on board ships) have been being joined by space-borne remote detection: radar altimeters, radar diffusometers, synthetic aperture radars, radiometers, …

Even in Europe, where wind plants have operated offshore for over 20 years, the projects have been financed based on a variety of measurement methods, including hub-height masts equipped with anemometers, stand-alone LIDARs, and, in some cases, no on-site measurements at all. In *Chapter 3* the basic sources of wind information was introduced to obtain data from direct and indirect measurement techniques.

The equipment used for gathering offshore wind and wave data includes a meteorological mast and a wave meter (or buoy). The two measuring devices must collect data at the site for at least one full year to ensure that all seasonal weather variations are recorded. Today, the measurement data are only representative of a very limited area around the point and limited on the time. Therefore, we must turn to other measurement data, or to mesoscale models, …

The most precise means of measuring wind speeds and directions is to install a Met mast (or two) at the point where the resources needs to be evaluated, but there is a financial point of view, it is unrealistic to install a tower and wait several years to obtain a long-term time series data.

5.3.1 Meteorological mast

The equipment used for gathering offshore wind and wave data includes a meteorological mast and a wave meter (or buoy). The two measuring devices must collect data at the site for at least one full year to ensure that all seasonal weather variations are recorded.

Offshore wind monitoring towers are typically equipped with instruments very similar to those deployed on land, including anemometers, direction vanes, and air temperature and pressure sensors, although marine-grade models may be called for. In addition, the towers and their platforms may have motion sensors to detect deflections caused by wind and waves, which can affect speed measurements, and a variety of instruments to measure ocean currents, wave heights, water temperature, and other parameters-even bird activity. These other elements can transform an offshore mast into a complete *ocean-atmosphere-wildlife monitoring system*, providing data critical to the design and permitting of a wind project [**2**].

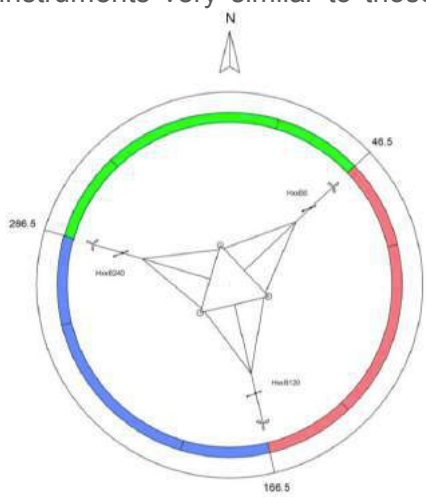


Figure 30 Position of cup anemometers, Met mast IJmuiden [**25**]

Generally, the anemometer configuration on offshore mast differs from the land-based mast. Three or more anemometers are installed at each monitoring height. This allows for the wind to be measured by at least two sensors every direction, and to reduce the influence of the tower on the wind measurements, minimizing the effect of flow distortion on the measurements (IEC

guidelines recommend some specifications). Specific measurements for offshore wind farm have been mostly undertaken below 100m in height. Here it will introduce a meteorological mast, *Meteomast IJmuiden*, whose measures are used in the case study, see *Chapter 8*.

Meteomast IJmuiden is a met mast, which built in 2011, approximately 82 km west of the coast of IJmuiden N52°50.89' / E3°26.14', and water depth on-site is about 28 meters. The Unit Wind Energy of the Dutch research institute ECN is carrying out a four-year meteorological measurement campaign on the Dutch part of the North Sea, requested by the Ministry of Economic Affairs, Agriculture and Innovation. The goal is to collected better wind data that can be used in the preparations of new offshore wind farm in the area. And to design safe and cost efficient foundations for future offshore wind turbines the campaign also includes measurements on sea current and wave data making use of a wave buoy.

Figure 31 Meteomast IJmuiden and floating LiDAR buoy

ECN verifies the data quality in several ways. Subsequently the data is checked manually. Only valid data is kept in the provided raw data files. Missing values are indicated with blanks. All data is stored in a database that are made publically available for the public use on the ECN's website. Then, a high quality of measurement data is available from the *Meteomast IJmuiden*.

An additional filter was implemented for periods with high turbulence intensity, as measured by the met mast, since significant deviations were found between the met mast and LiDAR wind speeds during those periods, but affecting only a small amount of data.

At various heights, at 21m, 58m, 85m and 92m LAT, sensors are mounted observing and recording wind speed, wind direction, temperature and pressure differences. The design of the met mast and data processing techniques ensure high data quality. Flow distortion due to the tower is minimised by installing anemometers on three different booms at each height (two at the mast top). Each boom is pointing in another wind direction, so that data can be selected only from relatively undisturbed sensors.

Also a ZephIR ZP300 LIDAR system is also installed on the mast platform and is placed that measures wind speed and direction at heights above the mast, up to 300 m LAT, by using laser technology. To the study's accuracy in January 2014 a Fugro's SEAWATCH wind LiDAR buoy was deployed 75 kilometres off the coast of *IJmuiden* in The Netherlands where wind data will be compared with data from three levels on a met mast at the site, together with data from a LiDAR mounted on the mast, and provide oceanographic parameters such as ocean waves and current profiles.

Figure 32 Floating LIDAR buoy

While the oceanographic parameters (during the period 02.11.2011 - 31.12.2015) were directly obtained from the *Meteomast IJmuiden* Website, http://www.meteomastIJmuiden.nl/, the raw data of the wind measurement (during the period 01.11.2011 - 31.08.2015) for the wind resource study was provided by my tutor Javier Rodriguez Ruiz of *ScottishPower Renewable Offshore*.

5.3.2 Other monitoring systems

Mainly, Lidar and Sodar are a useful complement to traditional met mast measurement

campaigns offshore. They are smaller and lighter than tall towers, and can usually be deployed at a lower cost. It may be possible to take advantage of existing structures to support offshore monitoring systems without interfering in its operation.

Often, an important challenge of using a support existing structure is that it may influence the wind measurements, which can cause substantial flow distortions all around them, either reducing or increasing the observed speeds depending on their shape, the wind direction, and the placement of the wind measurement system. Therefore, the accuracy of measurements usually suffers.

Another device are weather buoys, which can serve as the primary on-site data-gathering station for ocean parameters, including waves, currents, water levels, temperatures, salinity, and others, relieving the need to measure these parameters on a fixed platform. The buoy's basic atmospheric monitoring package should include wind speed and direction, barometric pressure, and air and water surface temperature.

5.4 Satellite data and mapping

Given the costs and inconveniences of surface-based wind monitoring systems, it is natural that there should be a strong interest in other potential sources of offshore wind resource measurements. Sensors, altimeter, SAR and scatterometer have the clear advantage that they penetrate clouds and are not dependent on sun of remotely sensed objects, but they need a pre-feasibility studies or in combination with classical offshore measurements and modelling results.

The last 10 years, the use of satellite spatial and temporal information has been a valid support for wind and wave energy assessment, especially as an instrument to validate modelling approaches.

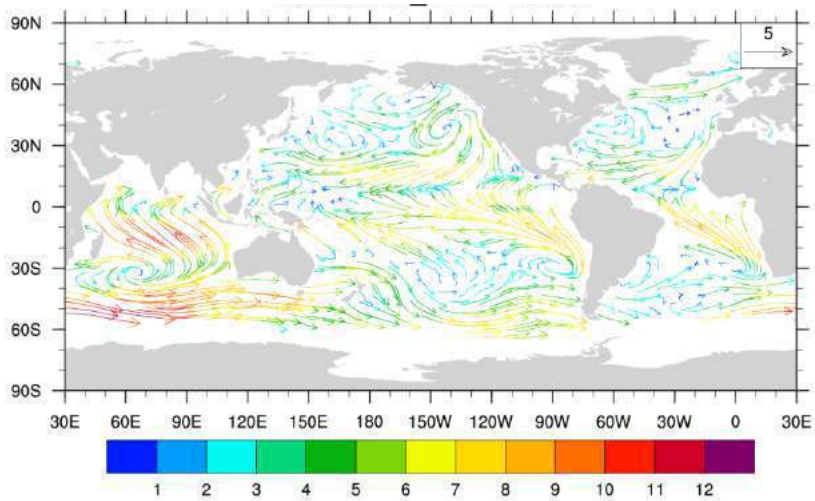


Figure 33 Near surface wind vectors colored by wind magnitude with Quikscat scatterometer (D. Shea, Climate Data Guide, Sep.2003)

5.5 Offshore wind climatology from non In situ offshore data sets

The long-term measurements at nearby land sites are compared with short-term records offshore. An overview of the different methods used to estimate the wind climatology offshore is presented by the reference [**8**].

5.5.1 Standard Measure Correlate Predict (MCP) method

It assumes a linear relationship between wind speed at paired sites where one site with a long-term record acts as *predicto*r and the wind speed at short-term measurement as the *predictand* [**8**]. A regression equation lets to condition the measurements overlap period, the regression parameters can be used to approach an extended data record for the site of interest. This method was used in the case study in *Chapter 8*.

5.5.2 Application of WAsP to offshore wind energy

It calculates the wind climatology at one site from the wind climatology of long-term representative stations. The model WAsP® by Risø is commonly used for calculating energy productions in many countries. Several parameters used in the underlying model are tuned to yield correct results on average during average conditions. In principle WAsP® can be used for offshore purposes, but the tuning of the parameters is not adapted for offshore applications yet, especially near-shore due to the gradients of wind speed. However, several publications have shown encouraging results in certain particular conditions. Some distance away from the coast, like the *meteomast IJmuiden*, the wind resource will be entirely homogenous. It means a reasonable approximation of the reality.

Another modelling product on the market can be used for offshore purposes, such as *VORTEX* is a cutting-edge metocean modelling tool for long-term statistics within an area plus high-resolution temperature, mean wind, extreme wind (*Vref*) and turbulence maps to detect local effects at less than a kilometre.

5.6 Wind resource assessment using numerical models

Created by winds blowing over the sea surface, ocean waves can be described by numerical wind-wave models that solve the equation of transport of the energy density distribution taking as input surface wind fields computed by atmospheric models. Solving the energy density balance equation the spectrum is modified locally through a function that represents the energy input from wind.

5.6.1 Global atmospheric models

The resolution of global atmospheric models is in general too coarse for wind speeds and wave results to be used directly in the majority of applications; they can be downscaled using either regional models or statistical approaches.

5.6.2 Mesoscale models

Another interesting trend is the increasing use of so-called mesoscale models, such VORTEX referred above, which resolve the local and regional circulation patterns and the boundary layer in contrast to the current methodologies. Mesoscale modelling offers a number of advantages for wind resource assessment, such as the ability to simulate, with reasonable accuracy, complex wind flow in areas where surface measurements are scant or non-existent [**37**].

6. Energy Production Estimation

6.1 Setting up to the project

The methods and challenges of designing a wind power plant and estimating its energy production could easily fill their own book. The purpose of this chapter is to round out the discussion of wind resource assessment by providing an overview of this subject. The reader is introduced to the underlying principles and methods of energy production estimation and taken through the main steps of selecting an appropriate turbine model, importing or creating a wind resource grid (WRG), wake modelling, designing and optimizing a layout, and estimating losses.

One of the first steps in designing and evaluating a wind energy project is selecting the plant design software. Before choosing a program, the resource analyst should investigate its features, design limits, user interface, required operating system and computer platform,

compatibility with other file types, and, of course, price and technical support options. Today, DNV GL's *WindFarmer,* ReSoft's *WindFarm*, EMD International's *WindPro*, and AWS Truepower's *openWind* are several software packages available on the market. These programs all present a number of features and capabilities in common:

- Import wind frequency distributions from one or more monitoring towers
- Create or import the results of a wind flow simulation
- Characterize turbines according to a number of parameters including hub height, rotor diameter, and power output and thrust over a range of speeds
- Lay-out the turbines within the project area
- Estimate gross energy production (w/o or with losses)
- Estimation of Wake losses and apply other losses to arrive at the net energy production.

These features constitute the basic elements of an energy production simulation. Most software packages also have the ability to adjust the turbine positions automatically to maximize the net energy production while meeting certain constraints (such as respecting property boundaries and setbacks). This process is known as *optimization*. It generally involves striking a balance between placing turbines close together in the windiest locations and spacing them far enough apart to keep wake losses.

During the project setup, the user must define the boundaries of where turbines can be placed. Depending on the software, these methods may require a GIS.

6.2 Wind Resource data

Once the project has been set up, the next step is usually to import or create the wind resource information required for the simulation.

Generally, two main types of wind resource information are used in wind plant design software:

- Wind resource measurements from a point location, which are normally provided wind-monitoring mast
- The output of a numerical wind flow model describing the variation of the wind resource across the determined area (*modelled Wind Resource Grids or WRG*)

I would like to mention some words about the second key piece of resource information (WRG). The conventional WRG format (made in ACSII nearly universal by the *WAsP software*) is a resource grid file contains the coordinates, height above ground level, the estimated overall wind climate and estimated power density or power production for each of a number of sites in a resource grid, which is simply a regular grid of sites. Although the WRGs are widely used, it has significant limitations, including limited data precision and an inability to incorporate wind resource data other than the *Weibull* factors. In most cases, both types of information are required for an accurate simulation, and they are used together.

6.3 Technology types: Wind turbines

6.3.1 Selecting a turbine

Once the project is set up and the wind resource information has been completed and analysed, the analyst is ready to design the turbine layout with the standard IEC turbine classifications.

The turbine suitability should be addressed provisionally early in the process to avoid wasting time assessing unsuitable models or developing unrealistic expectations of the performance of the project. The final decision on turbine suitability is up to the manufacturer, who must decide whether the turbine will be warranted for the site, and under what operating conditions.

Wind turbines are designed to extract the kinetic energy in wind, which will convert into useful electric power. This is achieved by allowing wind to blow past rotor blades, causing them to rotate and power an electric generator.

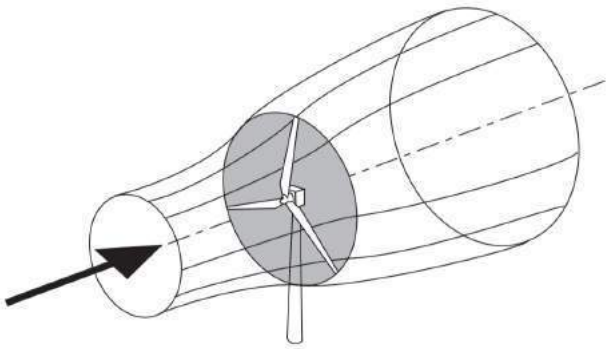


Figure 34 The energy extracting stream-tube of a horizontal axis wind turbine [**3**]

There are two basic types of Wind turbines:

Horizontal axis turbines with three-blade design are by far the most common form of onshore and offshore wind turbine used around the world today, but modern horizontal axis wind turbines also include single, twin, and multi-bladed designs.

Vertical axis wind turbines can be further differentiated into drag type devices, lift-type devices, and a combination of both.

Figure 35 Siemens Offshore Wind Turbine

Offshore wind turbines are subjected to oceanographic, meteorological and electrical conditions, which may affect their loading, durability and operation. To ensure the appropriate level of safety and reliability, the environmental, electrical and soil parameters as well as other relevant parameters shall be taken into account in the design, and shall be explicitly stated in the design documentation [**31**].

6.3.2 Wind turbine classes

For an offshore wind turbine the definition of wind turbine classes in terms of wind speed and turbulence parameters remains appropriate as the basis for design of the topsides structure. The values of wind speed and turbulence intensity parameters are intended to represent the characteristic values of many different sites and do not give a precise representation of any specific site. The *Table 4* specifies the basic parameters, which define wind turbine classes.

Wind turbine class	I	II	III	IV	S
Vref (m/s)	50	42.5	37.5	30	Site Specific
Vmean (m/s)	10	8.5	7.5	6	
V50 (m/s)	70	59.5	52.5	42	
I15 (-) A	0.18				
α (-)	2				
I15 (-) B	0.16				
α (-)	3				
I15 (-) C	0.145				
α (-)	3				

Table 4 Basic parameters of a wind turbine class (International norm IEC-61400-1)

, where *Vref* is the reference wind speed, which correspond to the 10min mean of the extreme wind speed with a recurrence period of 50 years at hub height, and Vmean is the annual mean wind speed over many years at hub-height. A, B, C are the categories of the turbulence intensity values (for higher, medium and lower).

I15 (or turbulence intensity at the site with the influence of the wind turbines) is characteristic value of the turbulence intensity at 15m/s (10min average) and α slope parameter for turbulence characteristics.

6.3.3 Technology scale and deployment arrangements

Across the market, there is a clear tendency toward using ever more powerful wind turbines, see *Figure 36*. Today, there are turbines in existence, which have a capacity of 8MW.

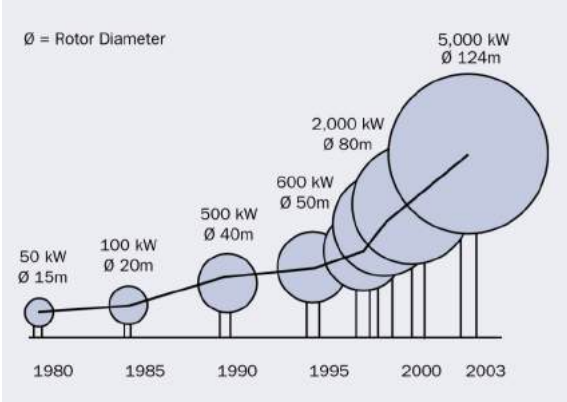


Figure 36 Growth in size of commercial wind turbine designs until 2003 [**22**]

The offshore wind power industry has promised to deliver significant cost savings. One way to do so is by boosting the economy of scale. As overall offshore capacities continue to grow, the performance of the wind turbines themselves also increases. Nowadays, offshore wind turbines in the range of 5 to 8 MW is feasible, bearing in mind that installing bigger turbines can imply a higher investment effort but reduce the total CAPEX per MW significantly.

For instance, Siemens' flagship offshore wind turbine, the *SWT-6.0-154*, has already set new standards in gearless turbine design. In early 2015, Siemens raised the bar yet again by introducing a generator upgrade to 7 megawatts. The Siemens *SWT-7.0-154* delivers nearly 10% more energy than its predecessor while retaining its proven reliability. An example, considering the *Neart na Gaoithe project*, the transition from 6MW to 7MW would decrease the number of turbines from 75 units to 64 units and thus lowering the project cost significantly.

Wind turbines can be installed as single devices or in arrays of multiple devices. Therefore, projects may have capacities ranging from a few megawatts to several giga-watts in larger offshore wind farms.

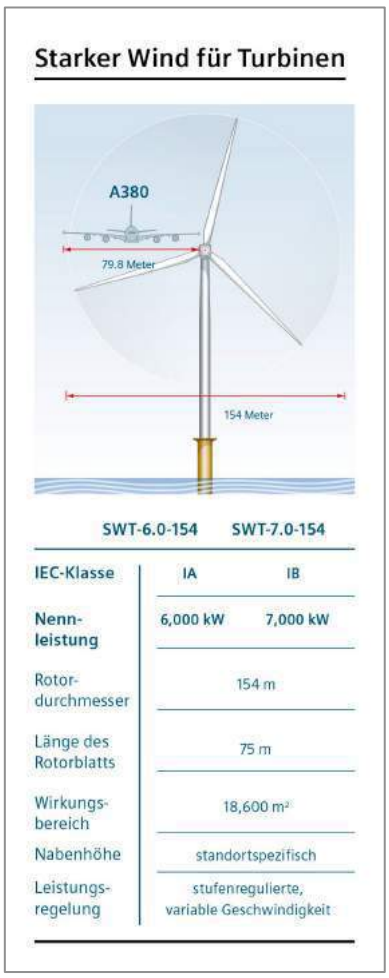


Figure 37 Comparison of the two biggest offshore wind turbine from Siemens [**6**]

6.3.4 Statistical characteristic curves

The key characteristic of a wind turbine from the perspective of wind resource assessment is the turbine power curve in *Figure 38*. This describes the power output as a function of wind speed measured at the hub.

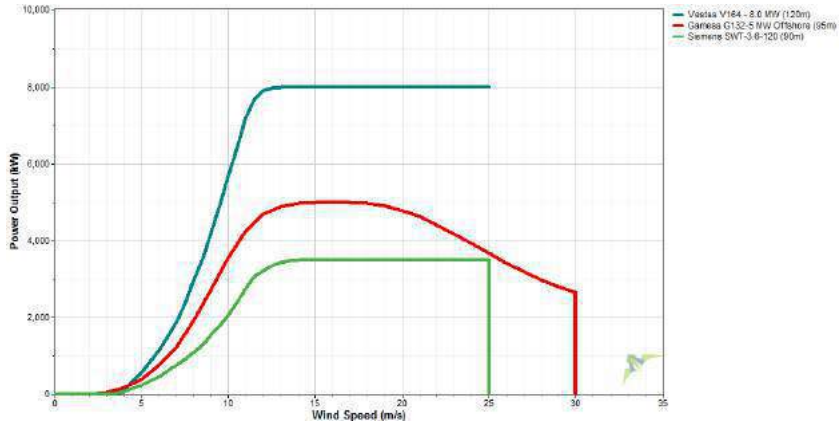


Figure 38 Comparison of three commercial offshore wind turbines (Windographer database)

The performance of a wind turbine can be characterized by the manner in which the three main indicators - *power*, *torque* and *thrust* - vary with wind speed. The power determines the amount of energy captured by the rotor, the torque developed determines the size of the gearbox and must be adjusted by the generator is being driven by the rotor.

In order to express a wind turbine's power characteristics, the power output from a wind turbine is given by the well-know expression:

$$P = \frac{1}{2} C_p \rho A U^3 = \frac{1}{2} C_p \rho \pi R^2 v^3$$

, where ρ is the density of air, 1.225kg/m^3, *A* is the rotor swept area, *U* is the wind speed, and C_p is the power coefficient, which is a function of the tip speed ratio λ, and its maximum achievable value is know as the Betz limit.

$$\lambda = \frac{\Omega R}{v}$$

The torque on the shaft is given by:

$$\Gamma = \frac{\mathrm{P}}{\Omega} \rightarrow \Gamma = \frac{1}{2} C_p \rho \pi R^3 v^2$$

, where C_Γ is the torque coefficient, linked to the power coefficient by the relation:

$$C_p(\lambda) = \lambda C_\Gamma(\lambda)$$

The typical shapes of the characteristic curves $C_p(\lambda)$ for a turbine with variable pitch angle is known in the *Figure 39*.

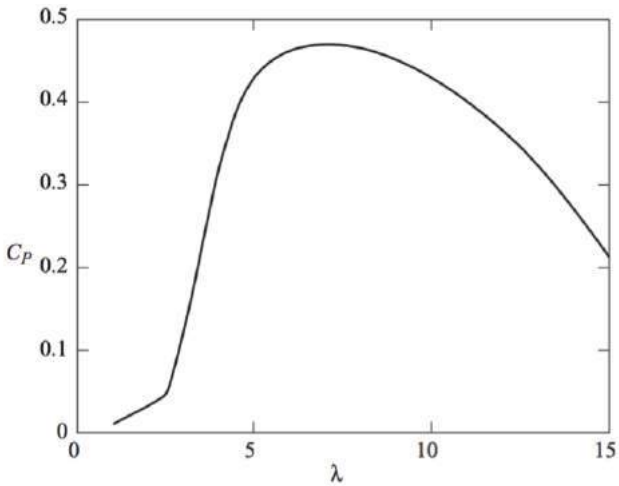


Figure 39 Performance curve for a modern three-blade turbine ***[3]***

The usual method of presenting power performance is the power coefficient, while the torque coefficient does not give any additional information about the turbine's performance, but is used for toque assessment purposes when rotor is connected to a gearbox and generator.

The characteristic curves of power as function of the rotation speed on the shaft of the turbine may be given by the manufacturer or obtained by modelling.

On the next characteristic curve of the operation of a wind turbine, we can see that there are four main function areas [**1**]. A turbine operating at variable speed (constant tip speed ratio) until maximum power is reached and thence at constant speed, constant power and pitch control would capture the maximum possible amount of energy in a given time. And then, if the wind speed could be greater than the maximum speed, which the turbine can withstand, the turbine is stopped by the braking system to avoid damage to the mechanical components of the turbine.

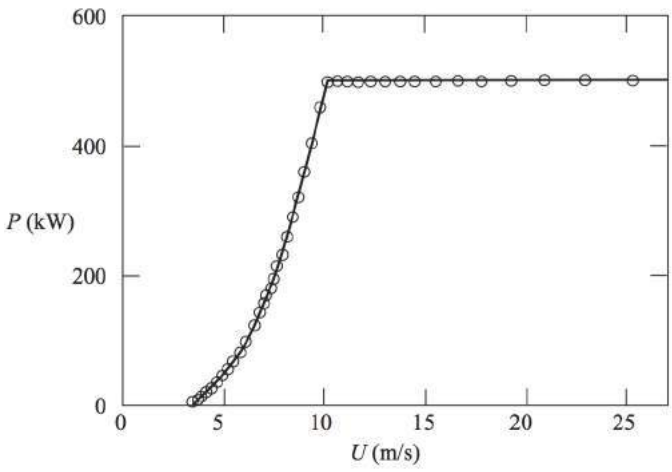


Figure 40 Power curve for variable speed turbine [3]

6.3.5 Optimisation of the wind turbines layout

The layout of wind turbines can have an impact on the power production of a wind farm. Design variables that define the layout of wind turbines within a wind farm include: orientation of the rows with respect to the prevailing wind direction, size and shape of the wind farm, spacing between turbines, and alignment of the turbines (i.e., whether in-line or staggered with one another). There are no universal layout recommendations for offshore wind farms, partly because isolating the contribution of each individual design variable is impossible at existing offshore wind farms, where multiple effects overlap non-linearly on one another, and partly because analysing the sensitivity to design variables requires sophisticated and computer-intensive numerical codes, such as large-eddy simulations (LES), that can simulate the small-scale turbulent features of turbine wakes [**41**].

Wind farm design parameters include separation distance from and layout (spacing, nominal power density) of the wind farm, and hub height and rotor diameter of the wind turbine. The impact of nominal power density is studied by changing the turbine separation distance in the hypothetical wind farm between 5 and 14 times the rotor diameter and keeping the geostrophic velocity at a constant value such that the upstream hub-height velocity is halfway cut-in and nominal.

Figure 41 Aerial photo of the Horns Rev wind farm off the Western coast of Denmark, http://www.nrel.gov/news/features/2012/1995

Once permits are granted, that means the restrictions are fully defined, there is an excellent opportunity to optimise the wind farm layout. The layout is fine-tuned to minimize wake losses and ensure compliance with permits and wind turbine certification limits.

The layout optimisation began with an updated wind resource assessment and an evaluation of the permits to find the key constraints (navigation routes, environment constrains, …), which is typically based on simple grid spacing, but and new industry best practice regarding wake losses and turbulence intensity levels may dictate different spacing.

Offshore wind farm, or OWF, design is often a balance of various factors. For instance, an

offshore wind farm layout depends on several factors [**48**].

First, it is referred to the conditions within the permits for the offshore wind farm (site boundary and place limitations, environmental impact assessment (EIA), tip height, rotor diameter, foundation types, installation methods, other impact assessments). All these conditions will be considered to ensure a fully compliant layout.

A second key factor in the design of wind farms is wake effects. Wake losses can be reduced by the increase of the spacing between wind turbines, but it is necessary to find a balance of technical and financial considerations.

A third constraint is the turbulence intensity and extreme winds speeds specially. For example, the IEC defines the design levels for maximum conditions of the wind turbines, which are divided into different classes. While ambient turbulence intensity levels are generally low offshore, it is possible to reduce the wake losses, which are produced, by increasing the spacing between wind turbines. Thus, the reduction of wake losses is a balance of technical and financial considerations, since a mayor spacing grid means fewer wind turbines within boundary and increased costs fro array cables.

A last factor is the cost. For example, foundations can be more expensive for larger wind turbines, but it is possible to get an overall savings of fewer foundations are needed. The wind farm design can also affect the cost of electrical infrastructure, as already mentioned in the previous paragraph.

Then, the final layout is important to achieve a balance between constraints, site conditions, cost and yield.

The *cost of energy* is a key driver for the optimization of an OWF. It represents the costs of the wind farm spread over the amount of production [**48**].

$$Cost\ of\ energy = \frac{(CAPEX \cdot CRF) + OPEX}{Annual\ energy\ yield}$$

, where CAPEX is the total upfront investment (capital expenditures), OPEX are the operating expenditures, and CRF is the capital recovery factor

Today, there are tools or models, such as *ecofys*, which calculate the annual operating expenses based on the wind farm layout, and then, the annual energy yield can be calculated for various layouts for each layout using industry standard software, e.g. *Windofarmer* may offer solutions to minimise infrastructure cost at the price of acceptable performance costs. [**49**]. This software calculates the wake development by applying a parabolic solution to the RANS equations based on a modification of the Eddy Viscosity model.

The total annual energy yield increases also with larger wind farm capacity. However, with each additional wind turbine, the wake losses increase. Therefore, increasing the wind farm capacity leads to progressively smaller improvements in annual energy yield.

In order to determine the optimum capacity in practice, an *iterative approach* can be employed. The Cost of Energy is calculated for many different layouts, covering a large range of wind farm capacities. The following example demonstrates this process for a fictional wind farm site. These approaches are generated using variations in orientation and wind turbine spacing in order to cover a wide range of wind farm capacities, e.g. reducing the wake-added turbulence intensity.

When the OWF capacity is fixed, the next step, it is to identify the optimum spacing and orientation, through the focus on minimising wake losses.

Finally, once the optimum orientation and spacing is determined, some positions may still be fine- tuned. Spreading the positions over the entire area will often mean the lowest wake losses, and ensures clear wind farm boundaries, which may be important for navigation safety.

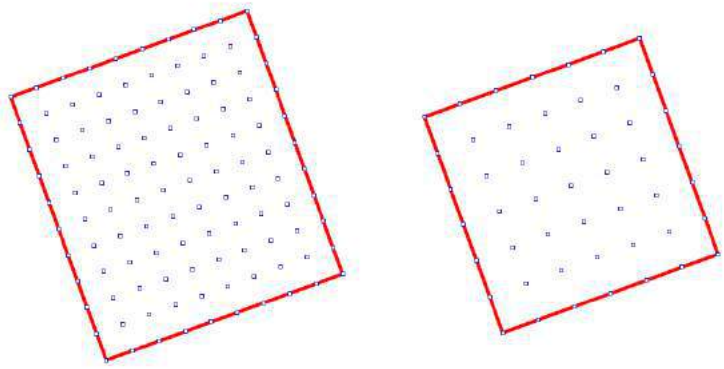


Figure 42 Two interactions of the simple layout for an example wind farm of about 400MW: 112 units of 3.6MW WTGs (Siemens model, left) and 49 Units of 8MW WTGs (Vestas model, right)

In *Chapter 8* it is presented a small layout optimisation of the wake effect with two interactions of the layout for OWF in the Dutch North Sea around the *meteomast IJmuiden*.

6.3.6 Estimating the power developed by a wind turbine

In general, modern wind turbines are designed to reach their nominal power output at wind speeds between 12m - 15m/s and are automatically shut-off at around 25m/s.

Due to the non-linear variation of power with steady wind speed, the mean power obtained over time in a variable wind with a mean velocity is not the same as the power obtained in a steady wind of the same speed.

A wind turbine generates a maximum power on the shaft called nominal power, P_n according to which it was designed (see *Figure 38*). The power P_n is obtained for a wind speed v_n called the nominal speed. For $v > v_n$, the wind turbine must adapts its parameter so that the power generates remains practically equal to P_n. The choice of the nominal speed is based mainly on financial constraints.

$$P_{mean} = \int_0^\infty P(v)f(v)dv$$

To express power as a function of speed in the turbine's function. Normally, it is used the a non-linear model of the form:

$$P = a + bv^k$$

, where k is the form factor of the Weibull distribution which characterizes the statistical properties of the wind. The other parameters are deduced from the conditions:

$$P(v_D) = 0; P(v_n) = 0$$

For a qualitative study of the dependence of the average power P_{mean}, as a function of the parameters v_D and v_n. Then, it is possible to obtain the set of characteristics,

$$\frac{P_{mean}}{P_n}\left(\frac{v_n}{v_{mean}}\right)$$

for different values of the relationship v_n/v_D. This last relationship has a significant effect on the efficiency of a wind turbine. Since the nominal speed is determined by the cost, a better relationship can only be obtained by reducing the cut-in speed v_D.

Then, the construction of the turbine must take in account the following [**41**]:

- The turbine should cut in at as low a wind speed v_D as possible.

- For $v_n < v < v_M$, the speed on the shaft will be constant.
- For $v > v_M$, the turbine will be stopped for operating safety reasons.

6.4 Gross and Net energy production

The gross energy production is the annual output of the wind plant without wake or other losses. It is calculated, for each turbine in the layout, from the following equation [**2**]:

$$E_k = 8766 \sum_{i=1}^{N_d} \sum_{j=1}^{N_v} F_{ijk} P_{ijk}$$

, where the average number of hours in a year (taking into account that one in every 4 years is a leap year, which has 24 extra hours) is the factor 8766. The sum is over the number of direction steps N_d and speed bins N_v. F_{ijk} is the frequency of occurrence (expressed as a fraction) and P_{ijk} is the power output for direction sector i, speed bin j, and turbine k. The units of energy are either kilowatt-hours or megawatt- hours, depending on whether the turbine's power output is expressed in kilowatts or megawatts.

The net energy production equals the gross production minus losses. A good estimate of plant losses is essential for accurately determining the long-term financial performance of a wind project.

The mutual distance between the wind turbines has to meet the requirements of the manufacturers. When siting the wind turbines too close this may result in a lowering of the electricity production. Another, more serious, consequence may be the damaging of primary structural parts caused by the wake of upwind sited wind turbines. The minimum distance depends on the siting with regard to the prevailing wind direction. When siting perpendicular to the prevailing wind direction the distance has to be at least four and otherwise minimal five times the rotor diameter. The gross energy yield of the wind farm is dominated by the local wind distribution and the siting of the wind turbines [**12**]. To calculate the net energy yield it is needed to determine the anticipated losses. The gross annual energy yield can be adjusted for seven loss terms [**40**]:

- Wake effects losses
- Electrical losses
- Turbine performance losses
- Environmental losses
- Curtailment losses
- Other losses
- Availability

6.4.1 Wake effect losses

The wind speed downstream the rotor, as known *wake*, of the turbine is lower compared to the undisturbed wind speed resulting in a somewhat reduced performance of downstream sited wind turbines. The wake is characterized by extra turbulence.

Losses due to rotor generated turbulences and shadow effect between wind turbines arrays reduce overall income wind speeds

The wake effect products losses resulting from aerodynamic interference between wind turbines in a wind farm. For a single wind turbine, the wake losses would be zero. For a wind farm, the wake losses could range from a few per cent to more than 10% (in case of a dense or poorly designed arrangement).

It is common practice to estimate the wind farm wake losses in the range of 3 to 4% of the gross energy yield *(see Chapter 7 and Chapter 8).*

6.4.2 Electrical losses

Electrical or grid losses are defined as the electrical losses between wind turbine switchgear and public grid connection, i.e. all losses to the point of revenue metering. Depending on the layout the electrical losses are in the range of 2 to 3% of the gross energy yield.

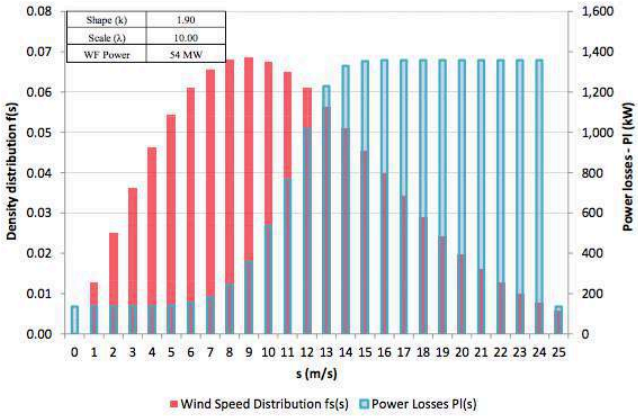


*Figure 43 Histograms for the wind density distribution and the power losses [**39**]*

6.4.3 Turbine performance losses

These losses are due to the turbine not producing to its reference power curve within test specifications (for which the turbines typically perform most favourably)".

6.4.4 Environmental losses

These losses are caused by a wide range of environmental factors, i.e. accumulation of dirt or ice on the turbine blades, degradation of the blades over time, environmental events, high and low temperature events. Such losses could range from near zero to several per cent, depending principally on the frequency of icing events.

6.4.5 Curtailment losses

These losses are resulted from the need to shutdown operation due to external factors, it do not include component failure or repair or environmental events), e.g. reducing physical load on the turbine, limitations of the external power grid, operations to reduce noise or bird impacts, …

6.4.6 Other losses

These losses do not fit into the above categories.

6.4.7 Availability

The availability of a wind turbine is defined as the time the wind turbine is in operation or ready for operation with external conditional, for instance too low wind or grid loss, preventing the system from energy generation. When a turbine is no available is due to maintenance operations or failure of a component. The technical availability of the turbine is 97% or higher [**40**]. This *Figure 44* is based on data of modern operational wind farms.

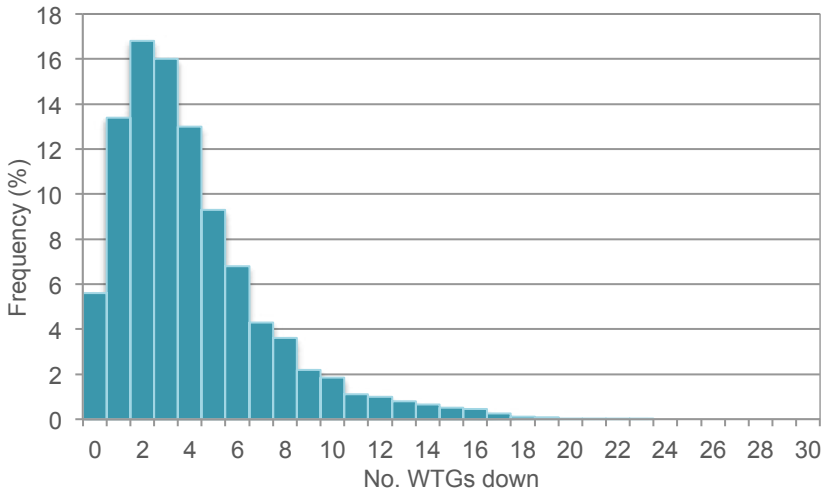


Figure 44 Availability distribution profile taking account of scheduled and unscheduled maintenance activities [57]

6.4.8 Combining losses

The total loss is given by the equation:

$$L_{total} = 100\% - (100\% - L_{wake}) \cdot (100\% - L_{elec}) \cdot (100\% - L_{perf}) \cdot (100\% - L_{enviro}) \cdot (100\% - L_{curtail}) \cdot (100\% - L_{other})$$

In summary, the following *Table 5* provides a breakdown of the main loss categories, each with a range of values, as a percentage of energy production, encountered in practical application.

	low	typical	high
Wake effects %	3	6.7	15
Electrical %	2	2.1	3
Turbine performance %	0	2.5	5
Environmental %	1	2.6	6
Curtailments %	1	0	5
Availability %	2	6	10
TOTAL losses %	**7.8**	**18.5**	**37**

Table 5 The ranges apply in mature operation losses,and availability. Source: AWS Truepower

6.5 Annual Energy Production

With the information provided in the heading related to energy production it is possible to start building some calculations for the expected *Annual Energy Production (AEP).* In order to properly estimate this value some additional assumptions to should be made regarding the wind turbines:

- **Capacity factor:** Is the ratio between the average delivered power and theoretical maximum power. In this case the *capacity factor (CF)* is taken as 40%. This value can also be expressed as the Net equivalent hours, which reflects the total amount of annual effective hours that a turbine is generating at maximum power:

Taking into account all losses described above, and that energy production will not happen until the wind farm is running the estimated AEP is calculated as:

$$AEP = E_k \cdot L_{total} \cdot CF$$

7. Wind Flow Modelling

Measuring and modelling the environmental wind field are the first steps in the process of planning a wind park. If the wind field is known, it is possible to design an optimal park layout in terms of number and positioning of the turbines. Lower wind speed and enhanced turbulence intensity in the wake downstream of the wind turbines mean lower total energy yield of the wind

park, and high fatigue loads which can affect turbine lifetime or failures rate and therefore energy output. In the *Chapter 1* was presented the different wind models. Computational Fluid Dynamics (CFD) models are often used to model the dynamic atmospheric flow at specific points where wind measurements are not available.

Wind field modelling is required for spatial extrapolation of the wind data measured at specific points (meteorological masts) to the position and hub heights of the prospected wind turbines. There are various methods and models available to do so, based on different approaches. Additionally, it should be borne in mind that regardless of the employed model/method, certain requirements must be met, namely (*source: Iberdrola Renewables*):

- Model validation: the model to be used must be capable of describing the most relevant wind field effects and yield physically consistent and plausible results.
- Simplifications of the intended model: the appropriateness of simplified or limited models must be considered and assessed for each case.
- Model input data: the model shall make use of input data that significantly influence wind conditions such as orography, roughness length, vegetation type, obstacles and atmospheric stratification.
- Wind field resolution: the resolution intended for wind field calculations must be selected appropriately and with regards to the scale of the effects expected or known to be relevant for the site.
- Model verification30: the results of the model must be verified against the available real data in each case so as to thus verify the applicability and accuracy of their results. This must be done in all cases by cross-predictions among the towers at the site. The resulting deviations shall be taken into account

In addition, if the situation of the meteorological mast is extremely distant from wind turbine positions or in a clearly different environment, even applying CFD techniques uncertainties in the results will continue to remain high. As such cases arise, it will have to assess whether the wind field modelling will be made with a linear model, since the use of more sophisticated CFD models does not always mean more accurate results.

7.1 The Physics of CFD Models

CFD models are considered the *next generation* of wind flow models for wind energy applications. The CFD models solve the mass and momentum conservation components of the non-linear *Navier-Stokes equations*. The solution is reached using a constant inlet wind profile. For idealized cases, steady-state CFD models perform well and give a high level of detail on the turbulence characteristics of the flow.

Turbulence is an important parameter to evaluate the suitability of the designed Wind Turbine Generator (WTG) through the fatigue loads calculations: the wind speed at any one time oscillates around the average value of a longer period and these fluctuations are transferred by the force of the wind into fluctuating loads on the WTG. The turbulence can be quantified by the *turbulence kinetic energy*, which is calculated from a closure scheme in the CFD model. Today, there are a huge variety of CFD models with different boundary conditions, numerical schemes, turbulence and wake models.

7.2 Numerical models

Predicting the overall impact of large offshore wind turbine array is a complex problem involving dynamic interactions between the turbines and various properties of the atmosphere, including vertical and horizontal gradients of temperature, pressure, wind speed and turbulence.

It has become clear that new models are needed and required to simulate wakes with reasonable accuracy. For instance, some recommended CFD-models, which are well established:

- *Windsim* is recommended for large wind farms spread out over a vast space or having some important orographic characteristic around the site that could disturb the flow in general.
- *MeteodynWT* for small and medium wind farms having modest spatial representativeness.

7.2.1 Wake modelling

Wind turbines operating in a wind farm will be exposed to the wake influence of neighbouring turbines. Wake behind a wind turbine leads to a reduction of the mean wind speed and an increase of turbulence level. Therefore in order to design an optimum layout for a wind farm, the evaluation of the wind turbine wake effect is essential because it leads to high fatigue, which reduces significantly the turbine life-time, and decreases the energy extraction for the down-stream wind turbines.

The wake effect from upstream wind turbines is a hot topic recently and many researches on wake modelling have been carried out in the past few years. There are different approaches to model the wind turbine wakes, from analytical models to three-dimensional CFD rotor modelling. Today, most of the site assessment CFD simulations are performed using commercial packages or an alternative open source CFD toolbox, *OpenFOAM* [**47**].

Therefore, as main guidelines and recommendations a wake model to be used in an offshore wind resource assessment must be able to capture as much as possible the following effects:

- Properties of individual wakes and combination of wakes in low turbulence environment,
- Effect of large wind farms as a disturbing element in the boundary atmospheric layer (apparent roughness respect to sea surface),
- Taking into account various low atmosphere stability situations.

Two early-generation models, the *Park and Modified Park Models* and *Eddy Viscosity Model*, are currently in wide use, and a third model, known as deep-array models, designed for large projects [**2**].

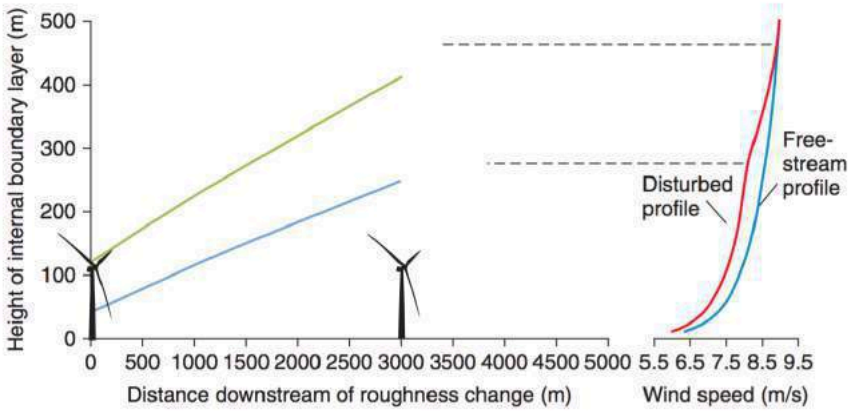


Figure 45 Illustration of the growth of two Internal Boundary Layers created by the wind turbine at left, which is modelled as a surface roughness change (source: AWS Truepower). The two curves on the right illustrate the effect of this turbine on the free-stream wind speed profile at the fourth turbine [**2**].

8. Case Study: Estimation of Wind Energy Source

8.1 Introduction

This study describes a preliminary wind climate assessment for an offshore wind or wave farm zone in the Dutch North Sea. This assessment is based on the combined use of offshore wind and wave measurements records, which be provided by the *Meteomast IJmuiden*, and *mesocale model data*. The software used on this study is the *Windographer* [**24**].

There are four offshore wind measurement masts in the Dutch North Sea, which are suitable for this wind resource assessment. There are two wind-energy-specific offshore meteorological masts (met masts): *Offshore Windpark Egmond aan Zee* and *Meteomast IJmuiden*. Closer to the Borssele zone, there are also two offshore platforms with met masts: *Europlatform* and *Lichteiland Goeree*. There are also co-located LiDARs at two of the locations.

The *Meteomast IJmuiden* offshore mast data forms the primary basis of this wind resource assessment, based on the overall low uncertainty of the wind measurements. For 4 years ECN will do measurements on the meteorological mast *IJmuiden*. All the data is stored in a database and will be made publically available on this website. In previous *Chapter 5* this meteorological mast was introduced.

Since there are no on-site wind measurements, a mesocale model will be used to quantify the gradient between the measurement locations and the wind farm site. As mesocale model accuracy varies, a number of models and input datasets are compared.

Figure 46 Offshore meteorological station, Meteomast Ijmuiden, coordinates N52°50.89' E3°26.14'

The following approach has been followed [**5**]:

- A detailed analysis of datasets from the one offshore measurement locations identifies the highest quality data sources.
- Two mesocale models are compared and the most accurate models and input source for this assignment is selected (Data sources: *MERRA Data* provided by *AWS Truepower* and VORTEX Data provided by *Tutor*).
- The wind climate is calculated based on a combination of good available data sources, presenting the long-term wind farm wind climate across the wind farm zone. The analysis includes variations with height, as well as an assumed an uncertainty assessment.

Summary

Section 2 presents the wind measurement datasets and general information from the offshore *Meteomast IJmuiden* and compares relevant trends. The wind climate calculation is described in *Section 4*, with a detailed assessment of the wind speed with one mesoscale data model, and the correlation of the wind measurements. *Section 5* describes trends in the wind climate at the wind zone. Some conclusions and an estimation of energy resource are presented in *Section 6*.

8.2 Wind measurements

These measurements are used together with *Mesocale model data* to characterise the wind climate for the Dutch North zone. The measured datasets are described in detail, including all data processing, and their suitability is assessed as a primary data. The measurement campaigns are summarised in following *Table 6*. Considering a good data quality, the datasets were thoroughly analysed. Further details are provided in the following sections.

	Distance offshore [km]	Measurement type	Measurement type		Height [m]
			Duration	Time step	
Meteomast Ijmuiden	82	Mast	3.8 years	10 min	27, 58, 92 m MSL
		LIDAR	-	-	90 - 315 m MSL (+25m)
LiDAR Buoy	75	Buoy	3.8 years	-	0 m MSL

Table 6 Measuring instruments of the Meteomast IJmuiden

8.2.1 Data selection

The filtered raw data of the wind measurements from the *Meteomast IJmuiden* website (during the period 01.11.2011 - 31.08.2015) for the wind resource study was provided by my tutor Javier Rodriguez Ruiz at *ScottishPower Renewable Offshore (Iberdrola Group)*. The dataset covers a period of four year.

	Position of the anemometers			
	NW 0°	SE 120°	SW 240°	Height m
Wind data (MMIJ)	x			92 m MSL
		x	x	58 m MSL
		x	x	27 m MSL

Table 7 Position of cup anemometers for Wind Mean Speed Measurements

The wind speed measurements at the mast are extrapolated to hub height (at 100m). Further detail, included the vertical wind shear profile, is provided in the following section.

8.2.2 Uncertainty in wind speed measurement

The uncertainty in measurement accuracy have not been assessed for each of the met mast datasets, in terms of instrument accuracy and mounting, as well as data quality and processing. The following *Table 8* shows us the results of the uncertainties relating to wind speed measurement. More detailed descriptions of the uncertainties are provided in the reference [**5**].

	Instrument accuracy	Instrument mounting	Data quality	Data processing	Total
Meteomast Ijmuiden	2.0%	1.5%	0.5%	1.0%	**2.7%**

Table 8 Uncertainty in wind speed measurement at Meteomast Ijmuiden

Normally, Met masts may experience flow distortions due to the platforms and equipment, and the effects of tower shadow at the mast have been quantified, as shows in an ECN report [**25**]. There is a relatively large uncertainty due to instrument mounting, since the booms are relatively short for the size of the mast, although effects are mitigated by the use of a filtering protocol to select the least disturbed instruments. The uncertainty is lower for *Meteomast IJmuiden*, since the top measurement height is taken as the primary reference, which is relatively unaffected by tower effects, and is newly calibrated.

An inter-comparison of the wind speed measurements of the LiDAR at 90 m and the met mast at 92 m will be one way to verify the data quality and the uncertainly due to mast and instruments. In this study, it was not carried out. As the reference *Site Studies Wind Farm Zone Borssele*, the comparison of these measurements determined an excellent correlation and a linear relationship and indicated that overall there is no significant issue with either the LiDAR and the top anemometry. This study showed us some data points with significant deviation between the LiDAR and met mast anemometry, but this affects only an estimated 0.1% of available data [**5**].

A comparison of cup anemometers (SE & SW) for wind measurements at 27m and 58m verified the data quality and showed a tight uncertainly due to mast and instruments. The comparison of the measurement at 27-58m determined a good correlation (R = 0.97-0.98) and a linear relation (1.004-0.999) respectively (see *Figure 47*).

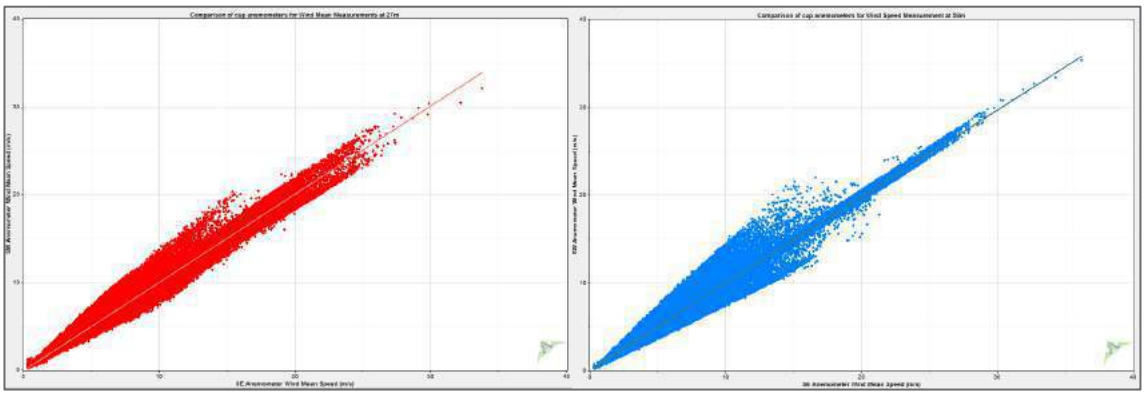


Figure 47 Comparison of cup anemometers for Mean Wind Speed Measurements

The following combined Figure 48 shows us the monthly mean wind speeds at 27m and at 58m.

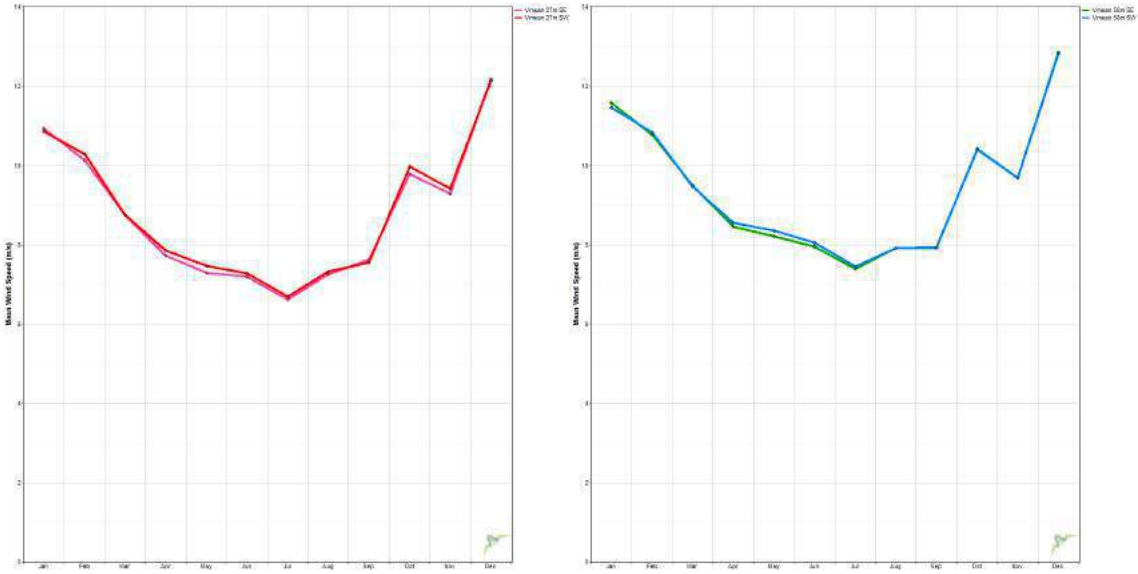


Figure 48 Monthly Mean Wind Speeds at 27m and at 58m, Comparison of cup anemometers (SE & SW)

The monthly differences between the two anemometers (SE & SW) *up 0.5%* are shown in bold.

	Vmean at 27m SE (m/s)	Vmean at 27m SW (m/s)	Delta (m/s)
Jan	10.946	10.873	0.073
Feb	**10.132**	**10.292**	**-0.160**
Mar	8.762	8.766	-0.004
April	**7.728**	**7.859**	**-0.132**
May	7.306	7.473	-0.167
June	**7.206**	**7.292**	**-0.086**
July	**6.623**	**6.705**	**-0.082**
Aug	7.265	7.330	-0.065
Sep	7.629	7.559	0.070
Oct	**9.799**	**9.981**	**-0.182**
Nov	**9.287**	**9.428**	**-0.142**
Dec	12.182	12.144	0.038

	Vmean at 58m SE (m/s)	Vmean at 58m SW (m/s)	Delta (m/s)
Jan	**11.589**	**11.467**	**0.122**
Feb	10.778	10.835	-0.057
Mar	9.504	9.486	0.018
April	8.465	8.559	-0.094
May	**8.225**	**8.356**	**-0.131**
June	**7.963**	**8.054**	**-0.092**
July	**7.396**	**7.460**	**-0.064**
Aug	7.923	7.919	0.003
Sep	7.928	7.941	-0.013
Oct	10.420	10.415	0.005
Nov	9.694	9.697	-0.003
Dec	12.856	12.826	0.030

Table 9 Monthly Mean Wind Speeds at 27m and at 58m, Comparison of cup anemometers (SE & SW)

The wind measurements at 27m between the two anemometers position show us a bigger

deviation than wind measurements at 58m. The difference of annual mean wind measurement at 27m during the four years was 0.07m/s, while the difference of annual mean wind measurement at 58m during the same period was only 0.02m/s. Furthermore, it should particularly be emphasized that the highest values are registered at the anemometer placed at 240° (SW) normally. Since south-southwest SSW and southwest SW winds are predominant in this zone, as shown the *Figure 49*. It can mean a shielding effect is minimised on the anemometer placed at 240° (SW).

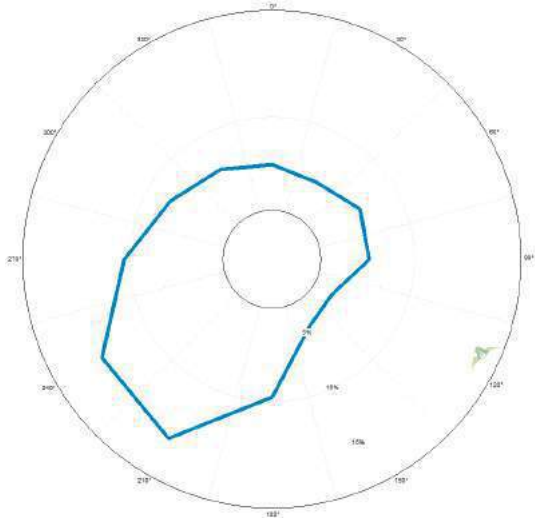


Figure 49 Wind direction vs. Frequency at 58m (SW)

To sum up, the accuracy of the dataset for this study is considered high. The data provided by ECN for *Meteomast IJmuiden* merits a lower uncertainty, since the mast are well documented and accuracy calibrated. For the next sections the combined anemometer results, which are calculated with *Windographer*, for the wind measurement at 27m and 58m are not used. The measurement of anemometers at 27m and 58m located at 240° (SW) were used.

In addition, the short-term measurements are corrected to the long-term data and extrapolated horizontally to the study zone, based on *Mesocale reference data (VORTEX)*, which were sustained by my tutor. Also a second *Mesocale reference data* (*MERRA*) was downloaded from *Windnavigator* and provided by *AWS Truepower* has been downloaded, which will contain hourly time series values of the wind speed at 50m MSL, during 37 years. This source is used as like a long-term reference data. The measurements source is located 18.1km NNW from the *Meteomast IJmuiden*, coordinates N53°00' E3°20'. These procedure are described below and being aware of the limitations of this kind of data.

8.3 Mesocale model

In addition to the measured wind data, a *mesocale model* is used in the calculation of the wind climate. A model was validated by modelled time series with wind measurements from the offshore mast (*Meteomast IJmuiden*). Based on this validation exercise, a MERRA data mesocale model and a VORTEX data were selected for use in this wind resource assessment. The characteristics of the mesocale model time series are presented in *Table 10.*

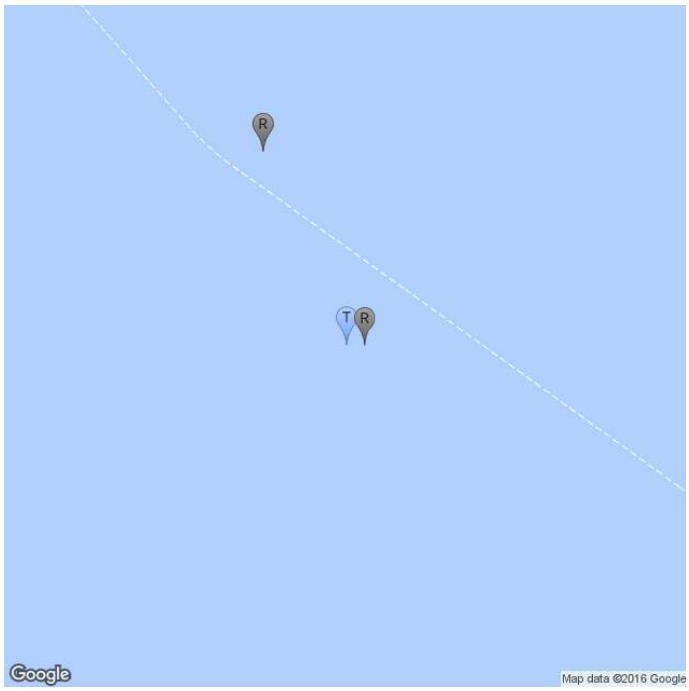


Figure 50 Position of Meteomast Ijmuiden and the two references

	Coordinates	Distance to Meteomast IJmuiden	Measurement type			
			Method	Duration	Time step	Height m
MERRA data	N53°00' E3°20'	18.0 km	Mesocale model	37 years	60 min	50 m MSL
VORTEX data	N52°50'50'' E3°26'08''	0.35 km	Mesocale model	33 years	60 min	100m MSL
Meteomast Ijmuiden	N52°50'09'' E3°26'14''	-	Measurements data	3 years	10 min	100 m MSL*

* Vertical extrapolation calculates from measurement data at 92m, 58m, and 27m (*Windographer*)

Table 10 Mesocale model characteristics

8.4 Wind climate calculation

The preceding analysis of offshore measurement datasets identified that the *Meteomast IJmuiden* offshore met mast represents the most suitable primary source for this wind resource assessment. In order to quantify the on-site wind climate at the study zone, several calculations are necessary:

- Extrapolation from measurement height to hub-height
- Long-term correction from 4 years to a 33-year period and 4 hours time step monthly
- Extrapolation from the measurement location to the study area

These calculations are described in the next sections, with the results and estimated uncertainty of each step.

8.4.1 Hub-height wind speed

First, the measured wind speeds are extrapolated to a height of 100m (hub height). The idea is to propose a present turbine model for the study zone.

The quality-controlled measurement dataset from the top measurement height of 92 m are extrapolated using *Windographer*, based on a matrix of average power law exponents per hour and wind direction bin. The calculated mean wind speed at 100 m height, and the associated uncertainty in vertical extrapolation is shown in the *Table 11*. The following values are not yet correlated with a long-term database, in next *Section,* and the selection of a wind turbine is described in *Section 8.6*.

	Meteomast Ijmuiden			
Selected measurement period	01.11.2011 - 31.08.2015			
Measurement height	92m	58m	27m	**27m,58m,92m**
Data availability	95.89%	96.76%	96.67%	**95.89%**
Measurement mean wind speed at measurement height	9.930m/s	9.433	8.819m/s	**9.930m/s**
Resulting mean wind speed at 100m	10.047m/s	10.180 m/s	10.593 m/s	**9.997m/s**
Estimated uncertainty in vertical extrapolation (at 100m)	-	1.3%	5.2%	-0.5%

Table 11 Extrapolation of wind speed measurement to height of 100m w/o MCP method

The uncertainty in vertical extrapolation is estimated based on sensitivity tests of different shear profiles.

8.4.2 Long-term mean wind speed

The dataset covers a period of four year. In order to represent the long-term wind climate, the data is compared to a long-term reference, by means of a Measure-Correlate-Predict (MCP)

procedure [**29**], which lets us analyse the relationship between wind speed and direction data measured concurrently at a target site and a nearby reference site.

The MCP method analyses the relationship between the short-term measured wind speed and direction data as well as concurrent data from a nearby reference, in this case mesocale data from three sources, see *Table 5*. This statistical relationship is used to predict and synthesise site data from the reference data. *Windographer* presents the option to correlate wind speed and wind direction to perform an MCP analysis with one reference data. Different algorithms for correlating wind speeds can use to set the works best in each situation, e.g. *linear least squares, total least squares, Weibull fit, matrix time series*, …)

At first, a linear least squares algorithm was originally considered the first option, which is simplest and the results suggest a good approximation. This algorithm is method of correlating target and reference speed data based on a straightforward application of the linear least squares (LLS) procedure to the scatter plot of target speed versus reference speed. The resulting linear curve fit is described with a slope and intercept value, i.e. as a relationship of the linear form $y = mx + b$.

A statistical relationship is typically established between a monitoring site and one or more reference stations using a technique known as MCP. Two different wind model datasets are validated against wind measurement at *Meteomast IJmuiden*.

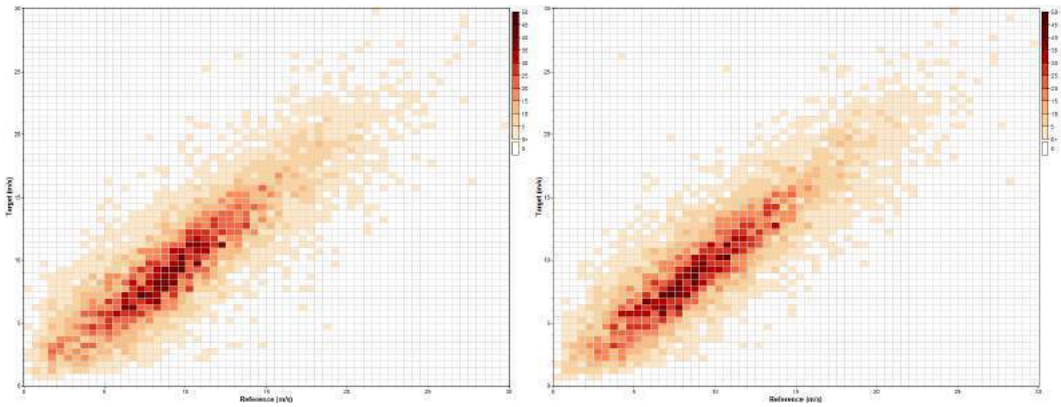


Figure 51 Occurrences of joint probability speed distribution VORTEX data (left) & MERRA data (right)

Firstly, the VORTEX data have chosen to carry out this wind study. Mainly, there are two reasons for selecting the VORTEX data for the long-term study: the first is the nearness to the *meteomast IJmuiden (MMJI)*, and the second thing, while the MERRA data are related to a height of 50m, the VORTEX data refer to a height of 100m. The deviation of the annual mean wind speed between the two sources compared with the measurement data means two-fold increase, see *Table 12*.

	Meteomast Ijmuiden data	VORTEX data @100m	MERRA data @50m
Mean wind speed @ 100 m	9.997m/s	10.012m/s	10.025m/s
Mean wind dir. @ 100 m	233.9°	240.4°	245.7°
Estimated mean wind deviation	-	0.15%	**0.28%**
Estimated wind direction deviation	-	6.5°	**11.8°**

Table 12 Comparison of Mean Wind Speed and Wind direction between three data sources

The second question on this matter should be to consider should be to find the best algorithm for correlating speeds and directions. The linear least square algorithm has been compared with other algorithms. An extension of the 4-year dataset to cover 33-year period would results in a minor increase in mean wind of 0.15% and increase in wind direction of around 2.5%. Since this comparison shows that there is minimal variation between the short-term measurements and long-term mean, it is decided to use the 4-year dataset with long-term correction to expend the

duration of the study, see *Table 13.*

		Algorithm			
	Target data MM Ijmuiden	Linear least squares (LLS)	Total least squares (TLS)	Matrix Time Series (MTS)	Weilbull Fit (WBL)
Mean speed @ 100 m	**9.997m/s**	**10.012m/s**	**10.013m/s**	**10.011m/s**	**10.021m/s**
MoMM spd. @ 100 m	9.971m/s	10.018m/s	10.020m/s	10.016m/s	10.028m/s
Weibull k @ 100 m	2.182	2.584	2.253	2.299	2.269
Weibull c @ 100 m	11.287m/s	11.280m/s	11.302m/s	11.307m/s	11.310m/s
Mean wind dir. @ 100 m	**233.9°**	**240.4°**	**239.9°**	**240.5°**	**239.9°**
Estimated mean wind deviation	-	**0.15%**	**0.16%**	**0.14%**	**0.24%**
Estimated wind direction deviation	-	**6.5°**	**6.0°**	**6.6°**	**6.0°**

Table 13 Comparison of Mean Wind Speed and Wind direction between four algorithms using the MCP

Sensitivity analysis of was performed by data reduction. The cut-off wind speed of 4m/s represented an increase of the mean wind speed and wind direction@100m of over 1%. Then, completed data of the VORTEX source was used.

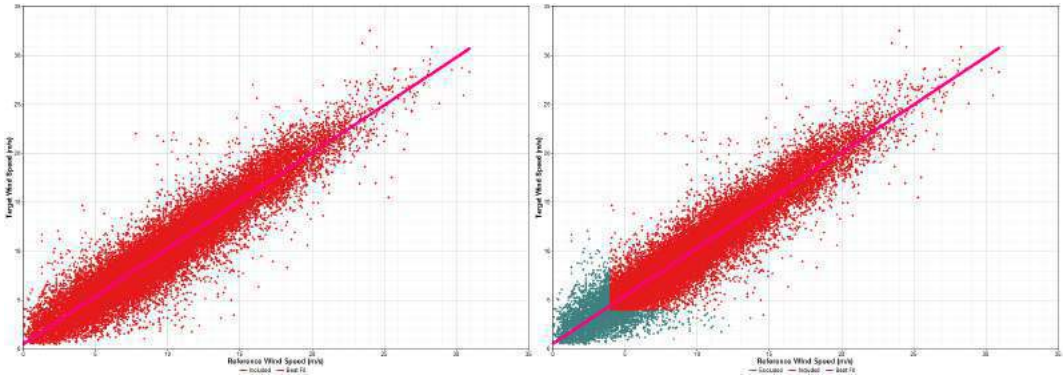


Figure 52 Scatter plot Target Wind Speed vs. Reference Wind Speed. Cut-off wind speed (right)

Windographer was used to calculate the slope of 0.895 and intercept of 1.335m/s for the LLS algorithm and a correlation of 0.757. It is important a slope high to get a better correlation between data sources. The wind direction was correlated too. The south-southwest SSW and southwest SW winds are predominant in this zone with the scaled MMJI Data, as shown the *Figure 53 and Table 13*. It can mean a small variation between the short-term and long-term measurements.

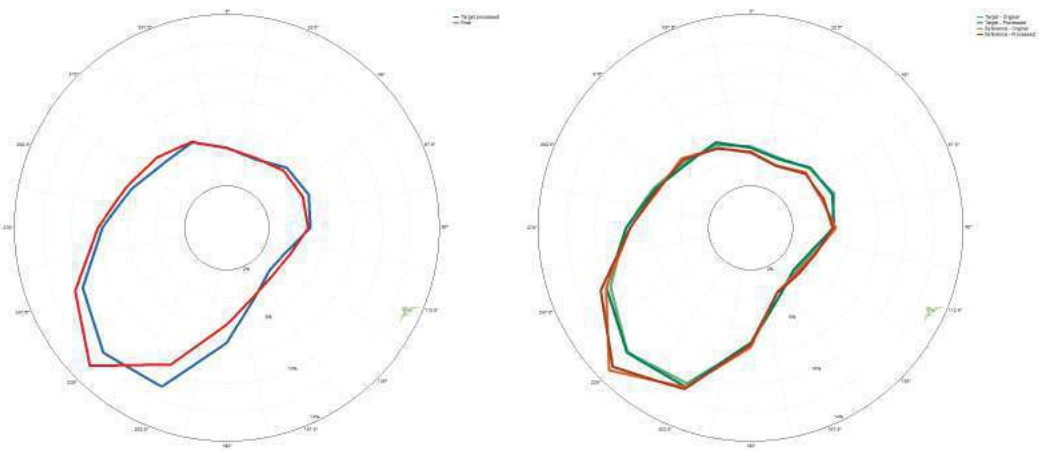


Figure 53 Wind Frequency Rose - Scaled the MMJI Data to long-term average wind conditions (left) & Comparison of the wind direction measured at a target site and a nearby reference site (right)

A correction factor is derived between the relative difference in mean wind speed between the short-term and long-term measurements.

	Meteomast Ijmuiden
Long-term mean wind speed at measurement location	9.689m/s
Relative difference of mean wind speeds between mesocale model and the measurement data	-3.18%
Calculated long-term hub-height mean wind speed at site	**10.01m/s**
Estimated uncertainty in horizontal extrapolation	0.15%

Table 14 Calculated long-term hub-height mean wind speed at site

Considering the data quality uncertainty of ±0.5m/s, the calculated mean wind speed at MMJI zone at 100m is **10.01 m/s**.

For symmetrical distributions, the mean wind speed can be expressed as the P50 value (the value that will be exceeded with a probability of 50%). It is also common to use the P90 value (the value that will be exceeded with a probability of 90%), or other exceedance probabilities (Pxx). Assuming a *Normal distribution* of the results, the different exceedance *probabilities* can be calculated as a function of the uncertainty, as shown in *Table 13* and *Figure54*.

Percentile	Mean wind speed @100m (m/s)	Mean wind speed @70m (m/s)
P95	9.4	9.1
P90	**9.6**	**9.3**
P75	9.8	9.5
P50	**9.9**	**9.6**
P25	10.2	9.9
P10	10.5	10.2
P5	10.6	10.3

Table 15 Different probability levels of mean wind speeds at site

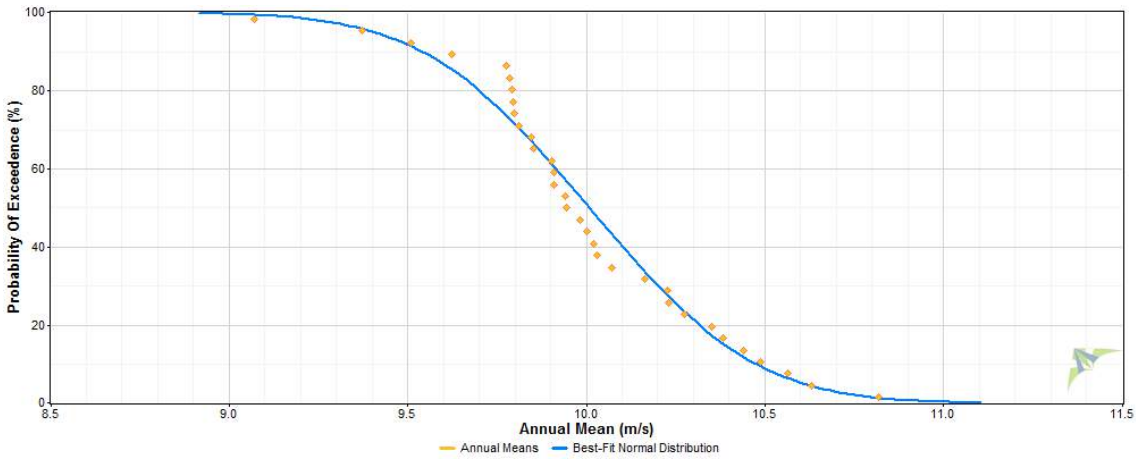


Figure 54 Probability of Exceedence: Annual Mean Wind Speed at 100m (Windographer)

8.4.3 Offshore wind atlases

The calculated wind climate at the study zone could be compared to several other sources. A number of independent scientific studies of the Dutch offshore wind climate has previously been performed. In *Figure 27 (Chapter 5)* the *Atlas CORMA (DNV GL)* was introduced and lets us to justify the assumption of uniformity of the wind resources in the study zone.

Other offshore wind atlases can be found in the following websites:
www.renewables-atlas.info, www.marinedataexchange.co.uk, www.norsewind.eu

For instance, the annual mean wind speed at 90m is estimated on **9.91m/s**, and according to the ECN wind atlases, the average wind speed at 90 m at the study zone is about 10.0 - 10.2 m/s (2004 version), as shown in *Figure 55*. The ECN wind atlas estimates differ by only around 0.2 m/s, but it is difficult to compare directly with the calculated wind climate.

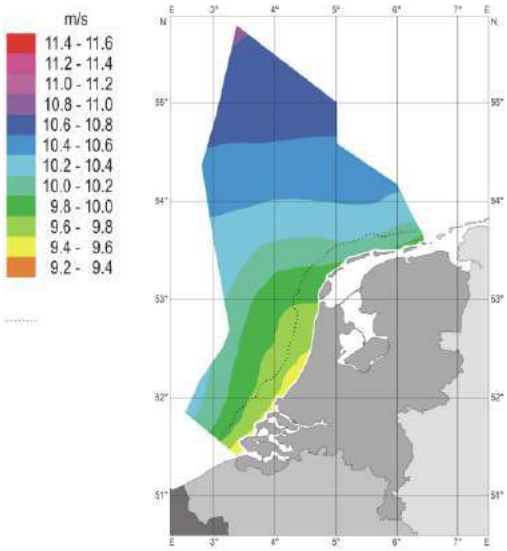


Figure 55 The average wind speed above the Netherlands' Exclusive Economic Zone (NEEZ) in the 1997-2002 period. Height: above average sea level

8.5 Wind resource assessment

The analysis in these sections below is based on the calculated long-term time series of wind speed. The short-term time series is used to compare with the long-term time series. Some date checks were performed with Know-how of my tutor and *Windographer* software, in order to validate general trends and identify outliers.

8.5.1 Mean wind speed

The annual average wind speed at the height of the hub is defined by the norm IEC-61400-1. We consider the average measurement (over 10 minutes) of wind speed at the height of the hub [**1**]. The mean wind speed at 100 m at the *Meteomast IJmuiden* point is 10.01 m/s, within ± 0.5m/s [**5**]. The wind speeds are also extrapolated from MMJI long-term correlated dataset, between 10 and 150 m MSL, based on the measured shear profile at *Meteomast IJmuiden*, not using mast and LiDAR measurements. The mean wind speeds are shown in *Table 16*.

	Mean wind speed (m/s)		
Height (m)	MMJI measurement data 10min	MMJI long-term calculated data 4hours	Delta (m/s)
150		**10.42**	
100	10.00	**10.01**	-0.01
92	9.93	**9.93**	0.000
80		**9.80**	
70		**9.67**	
58	9.43	**9.49**	-0.06
27	8.81	**8.81**	-0.01
10		**8.01**	

Table 16 Comparison of annual mean wind speeds at various heights from Meteomast IJmuiden wind measurements and MMJI Data to long-term average

The calculated variation in wind speed across the site is only 0.06 m/s; the greatest difference wind speed is due to the extrapolation between at 58m. The annual mean wind speeds profile is shown in Figure 56. So it is logical that the MMJI measurement data and the MMJI long-term calculated data are closely related; the shear also aligns very well with the measured wind shear.

8.5.2 Wind shear

Figure 56 shown the wind profiles of wind speed with height. The wind shear described in *Unit 4*.

It can be characterised by the α power law exponent in the power law equation:

$$\frac{v_2}{v_1} = \left(\frac{h_2}{h_1}\right)^{\alpha}$$

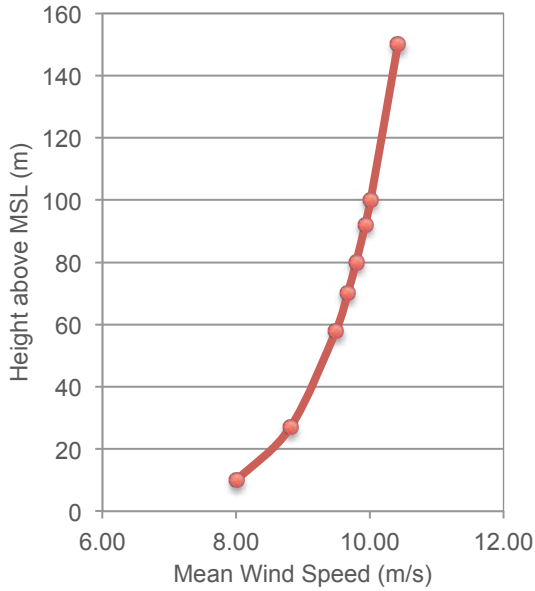


Figure 56 Mean wind speeds profile at site from MMJI long-term calculated data (4-hours period)

Windographer uses linear least squares regression to find the best-fit value of the power law exponent. The power law exponent is calculated to represent the best-fit value of the vertical wind speed profile using all measurement heights, up to 150 m.

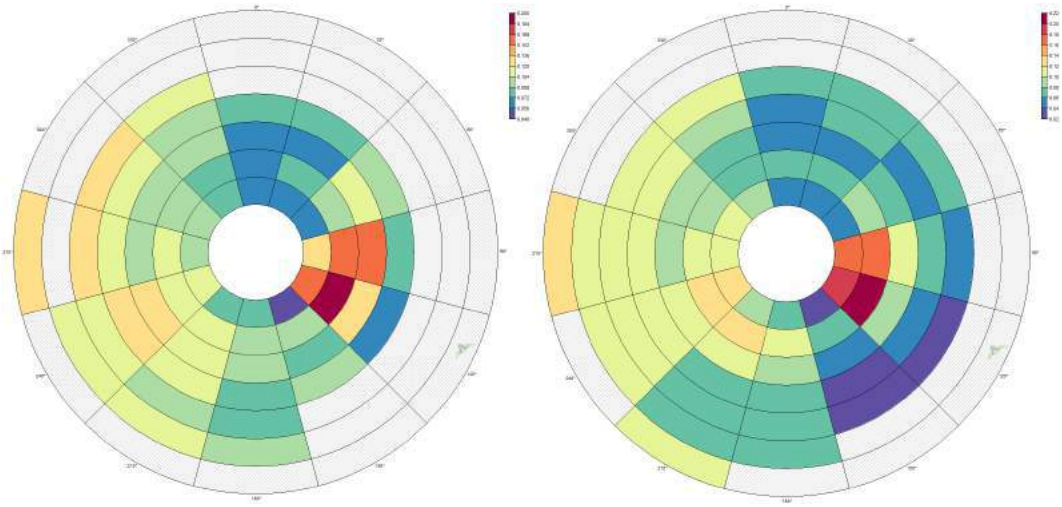


Figure 57 Mean Power law exponents by wind direction at 92m and at 27m from MMIJ measurement data

The calculated power law exponents for 12 wind direction sectors are shown in a radar plot in *Figure 57 and Figure 58* from the *Meteomast IJmuiden* measurement data source. The extrapolation to other heights, e.g. at 100m, is based on the *Meteomast IJmuiden long-term correlated data*.

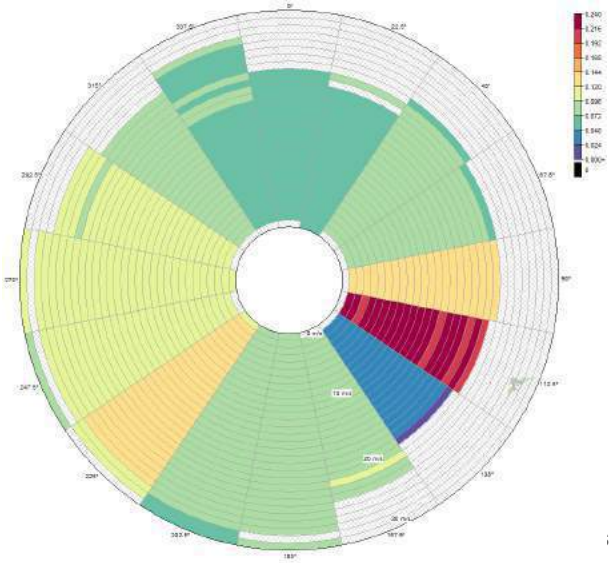


Figure 58 Power law exponents by wind direction at 100m from MMIJ long-term correlated data

In addition, *Meteomast IJmuiden*, at 80 km from shore has lower shear to the southeast.

8.5.3 Diurnal variation

Figure 59 shows the diurnal variations in the calculated mean wind speed for the study zone, based on the *Meteomast IJmuiden* measurement data. The patterns are quite similar at all heights with generally little variation throughout the day.

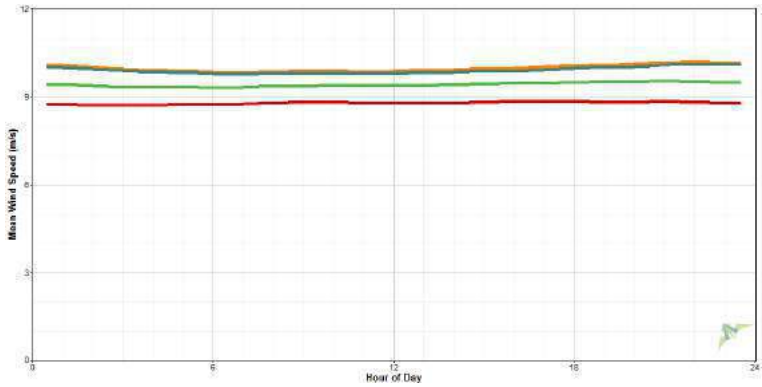


Figure 59 Mean diurnal profile of wind speed at 100m, 92m, 58m and 27m at Meteomast IJmuiden

At the upper heights, there is a slight increase of about 0.3 m/s in wind speeds in the evening, while at lower heights the slight peak is earlier in the day.

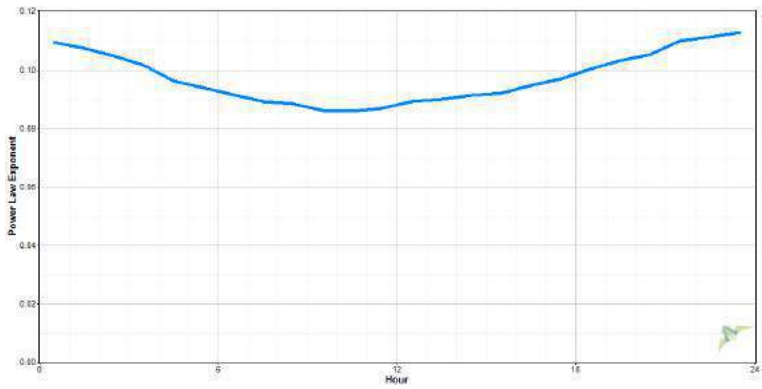


Figure 60 Power law exponent variation (daily) based on Meteomast IJmuiden 10min measurements

The power law exponent also exhibits a diurnal pattern, with lower value during the day, about 0.087, than at night, about 0.112, although the change is relatively small, as seen in *Figure 60*.

8.5.4 Monthly variation

The trends in mean monthly wind speed are similar at all heights, as seen in *Figure 61*. There is a significant difference between the high mean wind speeds in winter compared to summer, which is typical for the offshore wind climate in Northern Europe and is also seen in the mast measurements.

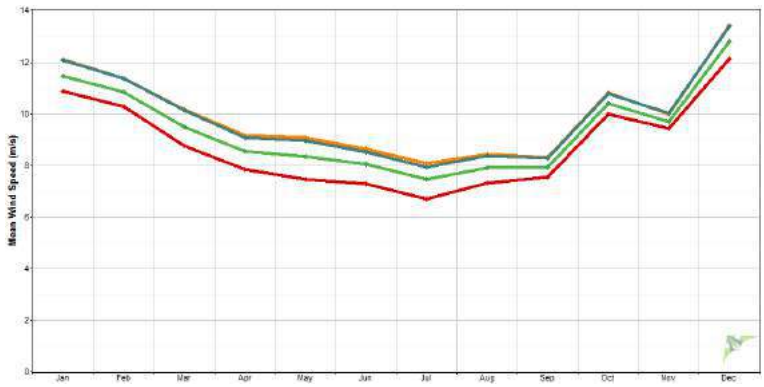


Figure 61 Monthly mean wind speeds at 100m, 92m, 58m 27m based on Meteomast IJmuiden measurement data

The comparison of mean monthly wind speed at 100m based on *Meteomast* IJmuiden measurement data and the calculated MMJI Long-term mean wind speed is showed in the *Figure 62 and Table 17*. There is a significant difference between the last months of the year,

principally September, November and December. It is particularly remarkable in December 2011 the value of the diurnal mean wind speed was 14.87m/s.

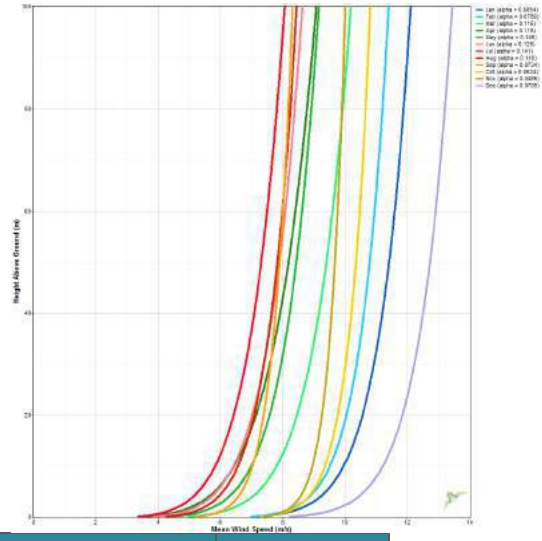


Figure 62 Monthly vertical wind shear profile based on Meteomast IJmuiden measurement data

	MMJI measurement data extrapolated @100m (m/s)	MMJI Long-term mean wind speed @100m (m/s)	Delta (m/s)
Jan	12.11	12.03	0.07
Feb	11.38	11.26	0.12
Mar	10.18	10.52	-0.34
Apr	9.17	9.42	-0.26
May	9.08	9.33	-0.25
June	8.64	8.54	0.11
July	8.05	8.48	-0.43
Aug	8.45	8.49	-0.04
Sep	8.31	9.22	**-0.91**
Oct	10.82	10.76	0.06
Nov	10.01	10.72	-0.71
Dec	13.44	11.49	**1.96**

Table 17 Monthly mean wind speeds at 100m based on Meteomast IJmuiden measurement data (blue) and MMJI Long-term mean wind speed (red)

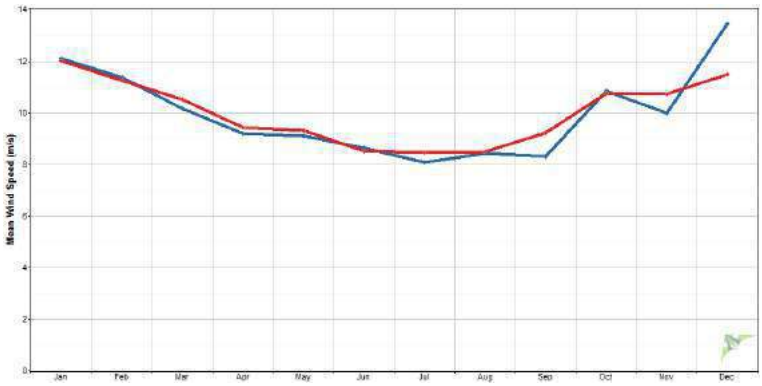


Figure 63 Monthly mean wind speeds at 100m based on Meteomast IJmuiden measurement data (blue) and MMJI Long-term mean wind speed (red)

Normally, it indicates an overall consistent regional wind climate [**34**].

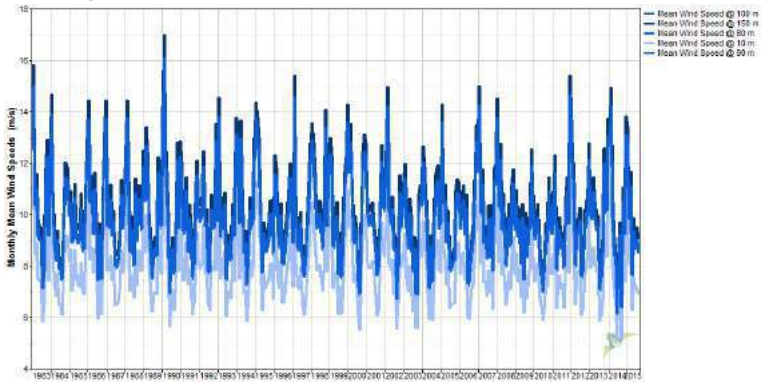


Figure 64 Monthly mean wind speeds at 150m, 100m, 90m, 80m and 10m based on the calculated MMJI 33-years long-term mean wind speed (blue colour scale)

8.5.5 Inter-annual variation

The wind climate at the point zone is based on the 4-years of wind measurements from *Meteomast IJmuiden*, without long-term correction at 100m (pink colour), from 2011 to 2015. Thus, the graph of annual mean wind speeds in *Figure 65* shows a comparison of those four years and the scaled MMJI long-term mean at different heights (blue colour scale). The mean wind speeds from the scaled model, at 100 m for the *Meteomast IJmuiden*, are also shown to further confirm that the 4-year dataset is generally representative of the long-term mean.

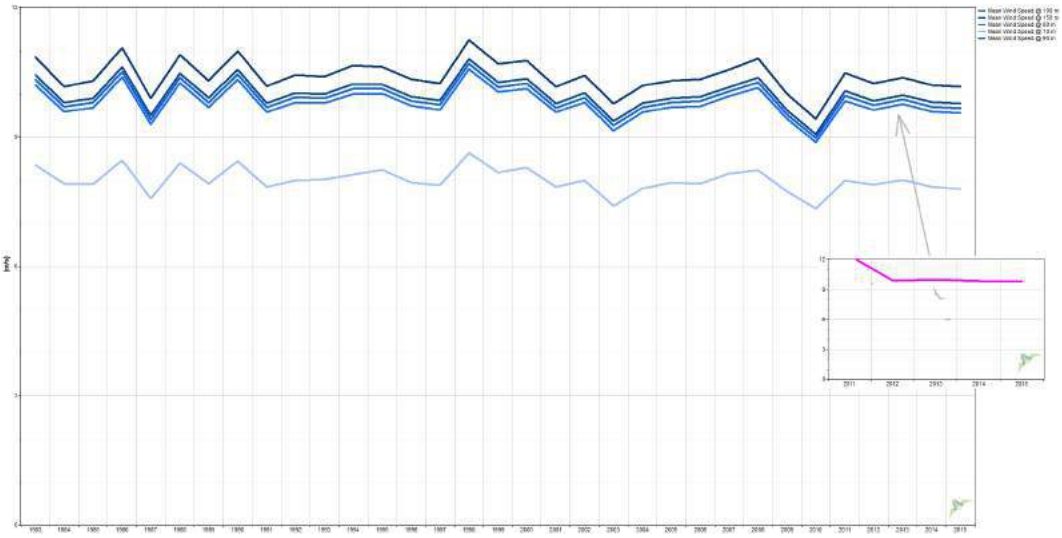


Figure 65 Annual mean wind speeds showing that 4-years period is generally representative of the long-term mean

8.5.6 Frequency distribution

The *Weibull* distribution is, at best, an approximation of the true wind speed frequency distribution. The observed distribution in this case is modal, that is, it has only peaks, and does follow the Weibull curve very well. At most sites, the *Weibull function* fits the observed distribution better than this.

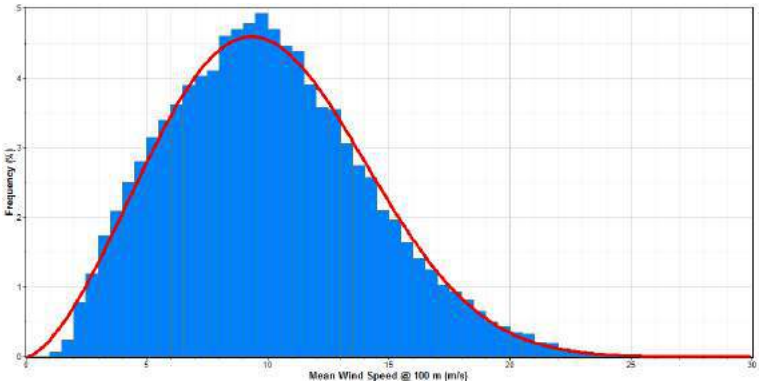


Figure 66 Frequency distribution plotted against a fitted Weibull curve (k = 2.58, c = 11.28 m/s)

8.5.7 Wind rose

In the next *Figure 67 (left)* the wind rose indicates the relative frequency of occurrence for each wind direction sector at 100m.

A second comparison is made in terms of energy content of the wind in each sector, based on wind speed and air density, as shown in *Figure 67 (right).* The most frequent and strongest winds are from the *SW*. These wind roses show close agreement with the mast measurements, although there are indications that the fit is not perfect. The calculated data is based on wind measurements from *Meteomast IJmuiden*, and it experiences most frequently *SSW* winds, while *WSW* winds dominate slightly more at the other sites

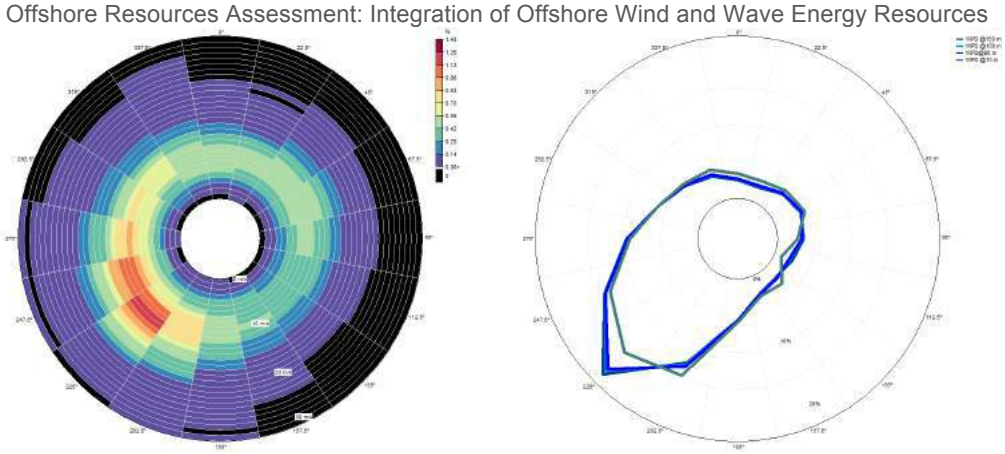
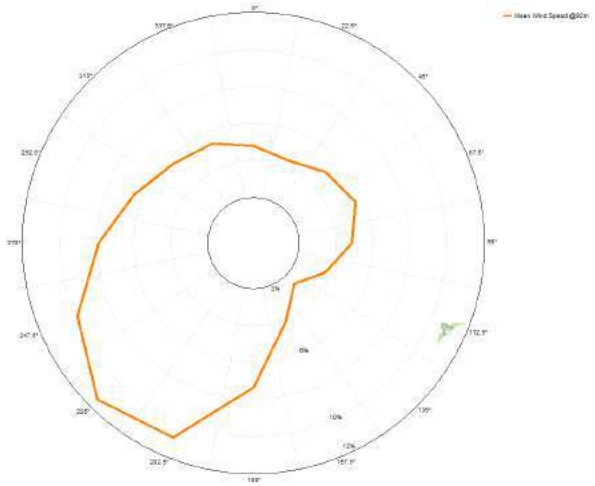


*Figure 67 Wind Frequency Rose – **Mean Wind Speed vs. Wind Direction at 100m** (left) & Comparison of the Wind direction measured at different heights (right): **Energy density per sector**. Source: scaled MMJI Data to long-term average wind conditions*

*Figure 68 Wind Frequency Rose – **Mean Wind Speed vs. Wind Direction at 92m** Source: MMJI Measurement Data to short-term average wind conditions*



8.5.8 Turbulence intensity

The ambient turbulence intensity should be considered as function of the wind speed. The ambient turbulence intensity levels are not modelled. Considering the measured turbulence intensity levels at *Meteomast IJmuiden* @92m gave by the reference [**5**], the representative turbulence intensity values are around 10%. At wind speeds of 15m/s, the representative turbulence intensity level is 5.7% (see in *Figure 69)*.

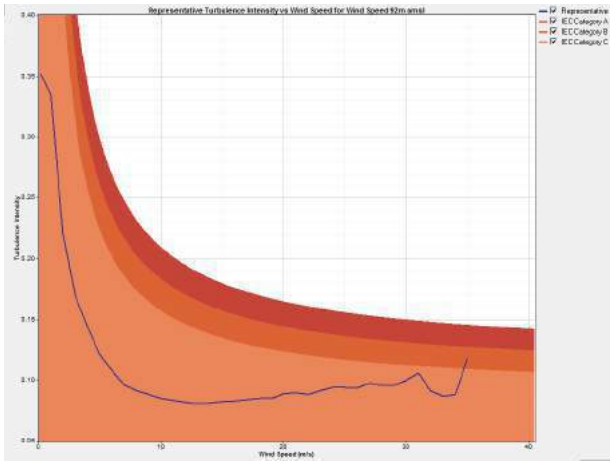


Figure 69 Representative curve of the Turbulence intensity vs. Wind speed @92m (blue) and the IEC turbulence categories in graph, Windographer

In any case, the atmospheric turbulence was determined previously. Therefore it will be necessary to calculate the turbulence with the influence of the wind turbines due to the wakes of the WTGs. The *Frandsen method* is normally used for this purpose. The turbulence intensity levels may be below the *IEC limits* for Category B wind turbines and generally independent of wind direction.

8.5.9 Extreme wind speeds

V50 is defined by the norm IEC-61400-1 as the extreme wind speed (over 3 seconds) with a recurrence interval of once every 50 years, to which the turbine might be subjected. The

extreme wind speeds are defined based on a long-term on-site measurement. For this zone, two alternating sources are considered:

1. VORTEX mesoscale model hourly dataset (33 years, long-term time series)

2. *Meteomast IJmuiden* 10min wind speed measurements (4 years, short-term time series)

As recommended by the reference [**5**] to calculate the 50-year extreme wind speed on the study area, a *Gumbel analysis* was chosen using two methods within the *Windographer Extreme Wind Analysis* tool [**30**]. For both methods, the settings are based on best practice to provide a suitable number of data points for the extrapolation.

- Period maxima method fits the *Gumbel distribution* to the peak wind speeds measured in each foxed time period.
- Independent storm method fits a *Gumbel distribution* to the peak wind speed occurring in each independent storm event.

	Period Maxima method	Independent storm method
Return period (years)	Extreme 10min wind speed (m/s)	Extreme 10min wind speed (m/s)
10	40.0	39.0
25	43.2	42.6
50	45.4	45.1
75	46.7	46.5
100	47.5	47.4

Table 18 Extreme wind speeds at 100 m MSL at the study zone, using calculated wind climate based on Meteomast IJmuiden 10min wind speed measurements

The extreme wind speeds at 100 m MSL for different return periods are shown in *Table 18*. The two extreme wind calculation methods yield similar results, of about 45m/s for the 50-year return period.

The extreme wind speed estimates based on the 33-year VORTEX mesoscale model data are shown in *Table 19*. The VORTEX model dataset is hourly; the estimated extreme winds for a 10-minute averaging period are based on the conversion factors given in the *GL Guidelines for Certification of Wind Turbines* [**31**]. The two extreme wind calculation methods yield similar results, with an extreme wind speed of about 37m/s for the 50-year return period.

The 50-year extreme 10-minute wind speed can be directly compared to the IEC design class of the wind turbine (IEC61400-1) [**33**].

The estimates based on the VORTEX mesoscale model hourly dataset, around 37m/s, are slightly below the extreme wind speed limit for *Class III wind turbines*, V_{ref} = 37.5 m/s, whereas the estimates derived from the *Meteomast IJmuiden* 10min measurements, about 45m/s, are below the extreme wind speed threshold for a *Class I wind turbine*, V_{ref} = 50 m/s.

It should be noted that, in order to obtain a the best WTG for specific project, it has to be shown that the rotor nacelle assembly loads due to wind & waves do not exceed type approved loads.

	Period Maxima method	Period Maxima method
Return period (years)	Extreme 10min wind speed (m/s)	Extreme 10min wind speed (m/s)
10	34.4	33.8
25	36.2	35.6
50	37.5	36.9

75	38.3	37.6
100	38.7	38.1

Table 19 Extreme wind speeds at 100 m MSL at the study zone, using calculated wind climate based on VORTEX model hourly dataset wind speed measurements

8.5.10 Air temperature

It will not be considered.

8.5.10 Air pressure

It will not be considered.

8.5.11 Relative humidity

It will not be considered.

8.5.12 Air density

The annual average air density is 1.22 kg/m at 100 m at the MMJI, comparing well with similar values at the measurement mast. The monthly mean air density varies by ±3% throughout the year (1.15-1.35kg/m3).

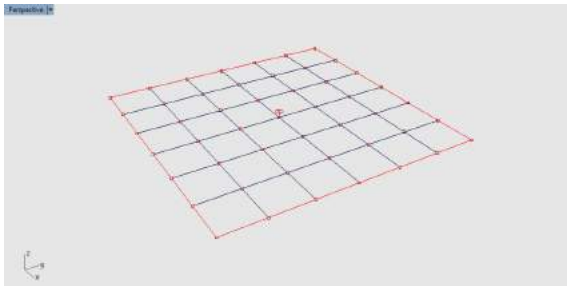
8.6 Estimation of wind energy

In the Chapter 6, the most important points for the energy production estimation in offshore wind farm were introduced. The optimisation of the wind turbines layout is a very important aspect in the design of an OWF, because it has an impact on the power production of wind farm. In this case study, two interactions of simple layout are considered in this case study, see *Figure 42.*

It is planned to develop an offshore wind farm with a total of 400 MW. The wind farm sites are located at least 85 km off the coast, see previous Section 8.1, and the centre of the OWF is considered the meteorological mast. The main characteristics of the wind resource assessment were previously presented.

8.6.1 Array of the offshore wind farm

The proposed development site is located between the IJmuiden a port city in Velsen province of North Holland and the coast of England.



An area of around 85km^2 is allocated to the development and shall include the wind turbines and all wind farm infrastructures excluding the offshore transformer platform. Considering other OWF projects, the allocated area shall be located within the larger project area of 140km2.

It was considered a typical offshore wind turbine array of *8D x 8D* for both layouts, see *Table 20*, where *D* is the rotor diameter.

Array	Vestas V164 8.0MW km	Siemens SWT-3.6-120 km
8D x 8D	1.312	0.96

Table 20 Distance between wind turbines

8.6.2 Selecting a turbine

The maximum rated capacity of the wind farm is limited to 400MW. The farm will feature from 50 to 112 turbines (or Wind Turbine Generators, abbreviated WTGs) depending on the rated energy of the selected turbines corresponding to the range of 3.6 to 8.0MW. There is a possibility that more than one model of the turbine will be installed due to the rapid development of the wind turbine industry and a multi-year construction programme. Three commercial WTGs were compared, as show in the following *Table 21*.

	Vestas V164 8.0MW	Gamesa G132 5 MW	Siemens SWT-3.6MW-120
IEC-Class	S	IB	IA
Nominal power (MW)	8	5	3.6
Diameter (m)	164	128	120
Swept area (m2)	21124	12868	11300
Blade length (m)	80	62.5	58.5
Hub Height above MSL (m)	105	95	90
Air Gap at MSL (m)	59	33	30
Nominal speed at (m/s)	13	13	12
Cut-in wind speed (m/s)	4	1.5	4
Cut-out wind speed (m/s)	25	27	25
Maximum tip height (m)	182	160	150

Table 21 Commercial offshore wind turbines

The wind turbine will comprise tubular towers and three blades attached to a nacelle housing containing the generator, gearbox and other operating equipment. Blades will turn clockwise, when viewed from the windward direction.

Generally, the wind turbines will begin generating power when the wind speed at hub height is between 2 and 5m/s. The turbine power output increases with increasing wind speed and the wind turbines typically achieve their rated output at wind speeds between 12 and 15m/s at hub height. The design of the turbines ensures fail a safe operation, such that if the average wind speed exceeds 25 to 30m/s for extended periods, the turbines shut down automatically. Preliminary dimensions of the turbines are not expected to exceed a maximum tip height of 182m above mean sea level for the largest turbine size (8.0MW). The air gap between mean sea level (MSL) and the lower tip of the blades will be determined based on the actual project; however a minimum of approximately about 23 m above MSL is expected (typically the elevation of the platform is determined by the extreme wave height).

Following is a summary of the most important points of the Choosing a large or smaller turbines:

Reasons for Choosing Larger Turbines

1. There are economies of scale in wind turbines, i.e. larger machines are usually able to deliver electricity at a lower cost than smaller machines. The reason is that the cost of foundations, road building, electrical grid connection, and a number of components in the turbine (the electronic control system etc.), are somewhat independent of the size of the machine.
2. Larger machines are particularly well suited for offshore wind power. The cost of foundations does not rise in proportion to the size of the machine, and maintenance costs are largely independent of the size of the machine.
3. In areas where it is difficult to find sites for more than a single turbine, a large turbine with a tall tower uses the existing wind resource more efficiently.

Reasons for Choosing Smaller Turbines

1. The local electrical grid may be too weak to handle the electricity output from a large machine. This may be the case in remote parts of the electrical grid with low population density and little electricity consumption in the area.

2. There is less fluctuation in the electricity output from a wind park consisting of a number of smaller machines, since wind fluctuations occur randomly, and therefore tend to cancel out. Again, smaller machines may be an advantage in a weak electrical grid.
3. The cost of using large cranes, and building a road strong enough to carry the turbine components may make smaller machines more economic in some areas.
4. Several smaller machines spread the risk in case of temporary machine failure, e.g. due to lightning strikes.
5. Aesthetical landscape considerations may sometimes dictate the use of smaller machines. Large machines, however, will usually have a much lower rotational speed, which means that one large machine really does not attract as much attention as many small, fast moving rotors

At first, three different offshore WTGs were analysed for this case study, see *Table 22*, but the *Vestas V164* and *Siemens SWT-3.6-120* were selected. This decision was made to check a greater deviation, e.g. area, number of WTGs, AEP, wake losses, …

For example, if we consider the number of WTGs to reach the 400MW, it will be necessary only approx. 50 WTGs of 8MW, see the comparison in the next *Table 22*.

	VESTAS V164 8.0MW	GAMESA G132 5 MW	Siemens SWT-3.6MW-120
Number of units	50	80	112
Installed power (MW)	392	405	399.6

Table 22 Number of WTGs

8.6.3 Analysis of wake effect

Wind turbines extract energy from the wind and downstream there is a wake from the wind turbine where wind speed is reduced, see *Wake effect losses in Chapter 6*. As the flow proceeds downstream there is a spreading of the wake and the wake recovers towards free stream conditions. The wake effect is the aggregated influence on the energy production of the project, which results from the changes in wind speed caused by the impact of the turbines on each other.

Find on this section, the wake effects are calculated using the linearized CFD Wake model, known as *Fuga*, which is developed by *DTU Wind Energy*. *Fuga* is a simplified CFD model that solves the linearized RANS equations (momentum + continuity) considering the turbine rotors as actuator disks.

Due to a problem with the software license, my tutor simulated and sent me the following results of the wake losses of the two proposed layouts.

The layout optimisation is typically based on simple grid spacing, but and new industry best practice regarding wake losses and turbulence intensity levels may dictate different spacing. The following *Figure 70* shows a scheme of the wake resolution with *Fuga*.

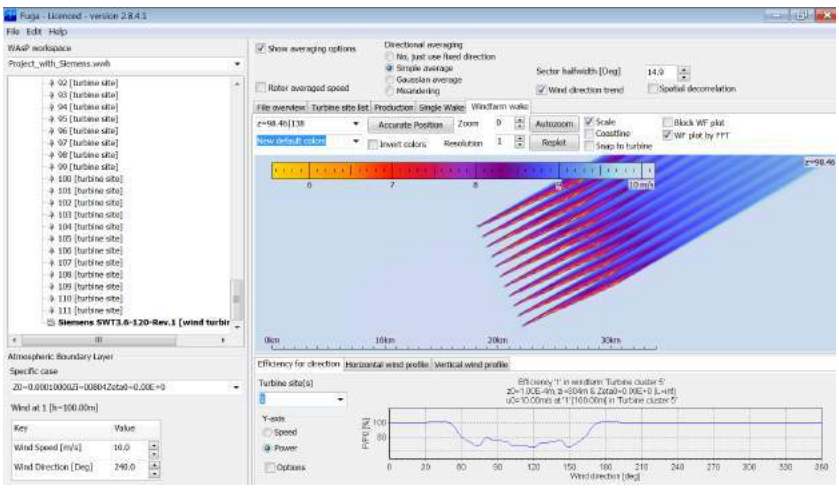


Figure 70 Screenshot of the wake effect analysis carried out with the Fuga software

The primary purpose was to determinate the wake losses and the power loss. Only two simulations were conducted with this software to calculate the wake losses of the proposed wind

turbines layouts with a wind speed of 10m/s and a wind direction of 240° (SW) at 100m height.

The following wake loss are calculated:

	Mean wake loss (%)
Vestas V164 - 8MW	6.92
Siemens SWT3.6-120	8.15

It was expected a higher value of the wake loss in the proposal with the higher number of turbines, considering some parameters, which were mentioned in *Chapter 6*, like layout spacing,

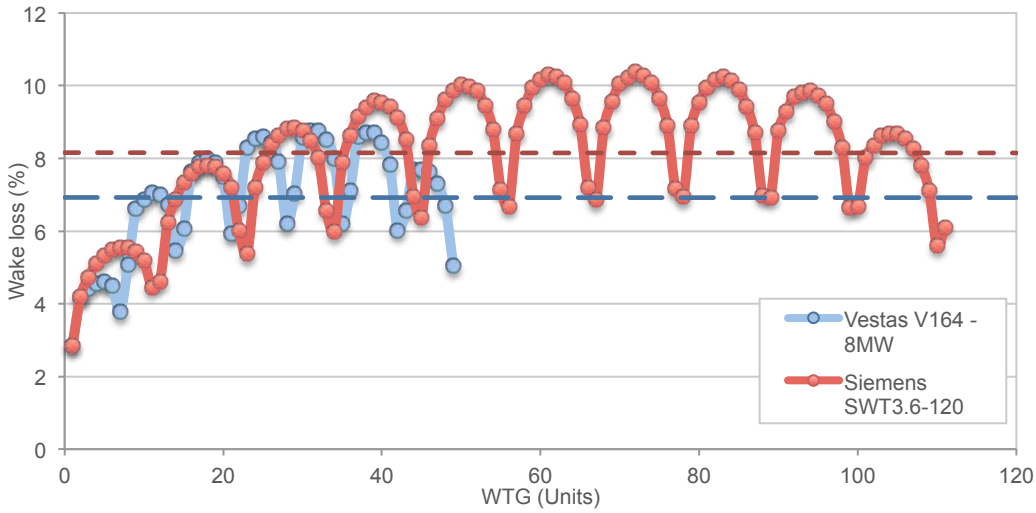


Figure 71 Representation of the wake losses (%) of the two proposed layouts per turbine

The following *Figure 73* represents the power loss (%) vs. wind direction with a mean speed of 10m/s of the turbine located northernmost.

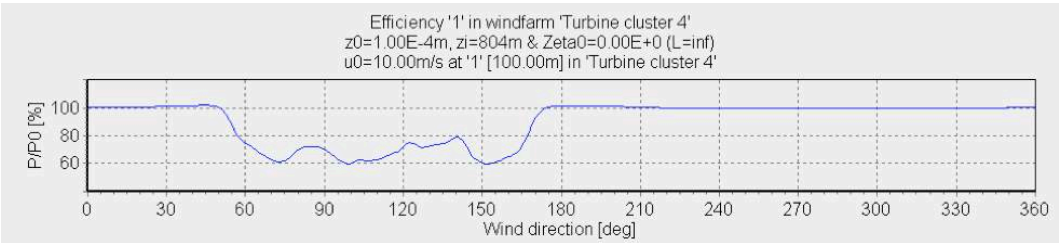


Figure 72 Power losses vs. wind direction

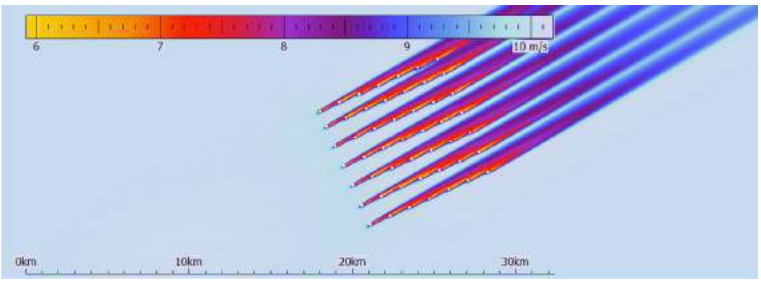


Figure 73 Reduction of the wind speed due to the wake effect in the proposal with 8MW turbines

8.6.4 Gross and Net energy production (or AEP)

Following the Sections in Chapter 6, the next step is to calculate the annual gross and net energy production, and give us a performance of the combining losses too.

Considering the availability, *DNV GL* recommended an availability assumption for projects in mature operation of 94%, which corresponds a *Site Type B: Distant project site in moderate wave climate.*

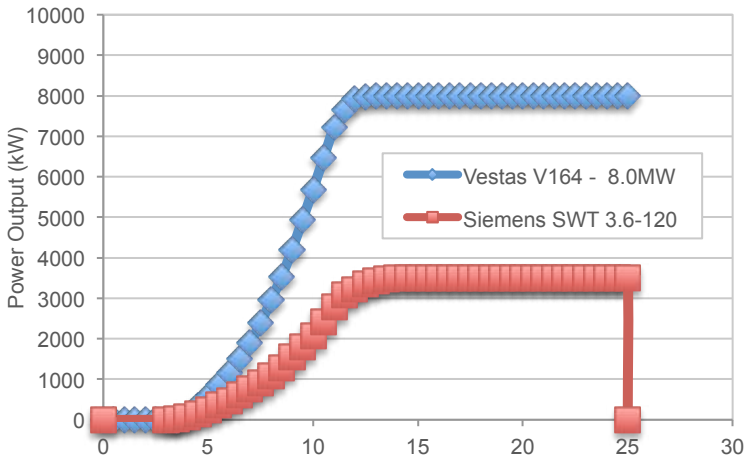


Figure 74 Power output of the two turbines studied

On the other hand, it has been assumed that the electrical efficiency is 97.5% representing an Offshore Substation with medium voltage.

	50 Units 8MW	112 Units 3.6MW
Wake effects %	7	8.15
Electrical %	2.5	2.5
Turbine performance %	1	1
Environmental %	1	1
Curtailments %	1	1
Availability %	6	6
TOTAL losses %	**18.5**	**19.7**

Table 23 Operation losses and availability of Vestas V164 and Siemens SWT 3.6MW

In summary, the previous *Table 23* provides a breakdown of the main loss categories, as a percentage of energy production, encountered in practical application. This factor included to the expected energy availability of the project Balance of Plant (cables, offshore substations, export cables, …). It is considered the same losses, with the exception of the wake effect losses. For a more realistic study the availability depends on the wind turbine model.

Considering both types of turbine, *Windographer software* calculated directly the Gross energy production per turbine, in *Figure 74.*

	Vestas V164 8.0MW		SWT-3.6MW-120	
	Before Losses	After Losses	Before Losses	After Losses
Mean power output (MW)	4.835	3.999	1.966	1.606
Annual energy output (MWh/yr)	42356.7	35030.2	17223.0	14067.8
WTG units	50 units		112 units	
Total mean power output (MW)	241.8	199.9	220.2	179.9
Total annual energy output (MWh/yr)	2117833	1751510	861150	703390
Gross equivalent hours (hr)	5294.6	4378.8	2135.8	1744.5
Capacity factor	60.4%	50.0%	54.6%	44.6%

Table 24 Breakdown of the Gross and Net energy production per turbine

The following *Table 25* shows us the annual net energy production of the offshore wind farm with two types of turbines. The annual net energy production of an offshore wind farm with 112 units of *Siemens SWT 3.6MW* is around 10% less than the same offshore wind farm with 50 units of *Vestas V164*.

	Vestas V164 8.0MW	SWT-3.6MW 120
Annual Gross Energy Production (MWh/yr)	2117833	1928976
Annual Net Energy Production (MWh/yr)	1751510	1575593

Table 25 Breakdown of the annual gross and net energy production

8.6.5 Analysis of uncertainties

Some of the most important sources of uncertainty associated with long-term energy predictions are listed below [**56**]:

- Measurement accuracy
- Data correlation uncertainty
- Historical wind variability
- Future wind variability
- Wind Flow modelling

The uncertainties analysis and the exceedance probabilities were done according to the standards of *Iberdrola Renovables*. The estimation was calculated according to it for the proposed offshore wind farm (50 WTGs of 8MW) at IJmuiden.

Source of Uncertainty	Wind speed		Energy
	(%)	Sensitivity	(%) (GWh)
1. Basic wind speed measurement	2		
2. Anemometer calibration	0.5		
3. Anemometer type	0.5		
4. Mast induced distortion	0.5		
5. Period representative of long-term	1.9		
6. Horizontal wind flow model	3.0		
Total wind speed uncertainty	**4.2**	**1.50**	**6.25**
7. External wakes			
8. Internal wakes			4.0
9. Power curve			5.0
10. Energy availability/accessibility losses			5.0
Long Term Energy uncertainty			**10.2**
Wind Speed Variability (1 year)	**6.0**	**1.50**	**9.0**
Wind Speed Variability (10 years)	**1.9**	**1.50**	**2.8**
Wind Speed Variability (20 years)	**1.3**	**1.50**	**2.0**
Total energy uncertainty (1 year) - Sigma			**13.6**
Total energy uncertainty (10 years) - Sigma			**10.6**
Total energy uncertainty (20 years) - Sigma			**10.4**

Table 26 Source of uncertainty (Iberdrola Renewables model)

Assuming a *Normal distribution* of the results for the annual energy production. The normal distribution provides the cumulative probability that the forecast average level is exceeded (exceedance probability). The different *exceedance probabilities* can be calculated as a function of the uncertainties, as shown in next *Table 26* and *Figure 75*. For symmetrical distributions, the estimation of offshore wind energy can be expressed as the *Pxx* value (the value that will be exceeded with a probability of XX%).

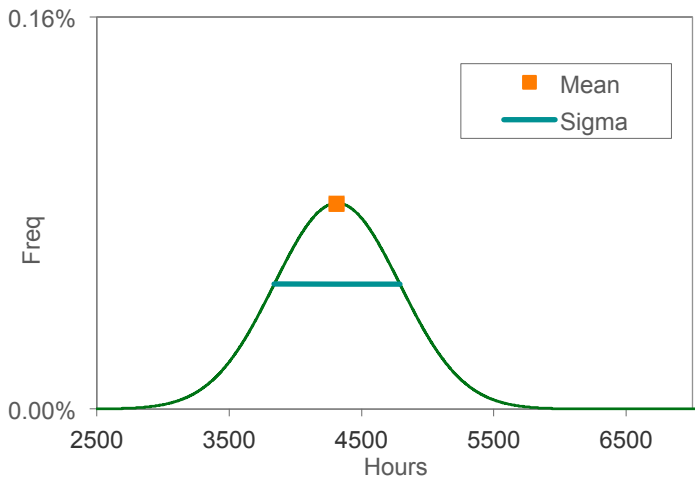


Figure 75 Normal distribution of the exceedance probabilities

Hence, P95 means: There is a chance of 5% that the P95 level will not be reached. P50 of course is the base case, as there is a 50:50 chance it will be exceeded. If the No. of equivalent hours (NEH) is 4315 and overall uncertainty is *σ=11%,* then the P95 is at 3534hr while the P75 is at 3995hr. With more uncertainty, those values decrease!

The *Pxx* values are widely used by potential investors and banks as basis for financing decisions. These will typically apply the P90 or P95 level for their revenue forecast in order to determine if the interest cover is sufficient. More uncertainty results in lower *Pxx* levels and therefore higher cost of capital, as investors have to assume more risk. The *Table 27* shows us the *Pxx* values calculated in this case study.

P	Deviation (%)	NEH	NCF
5	18.1%	5096	58.1%
10	14.1%	4923	56.2%
23	8.1%	4666	53.2%
25	7.4%	4635	52.9%
30	5.8%	4564	52.1%
35	4.2%	4498	51.3%
40	2.8%	4435	50.6%
45	1.4%	4375	49.9%
50	0.0%	4315	49.2%
53	-0.8%	4279	48.8%
60	-2.8%	4195	47.9%
70	-5.8%	4066	46.4%
75	-7.4%	3995	45.6%
80	-9.3%	3916	44.7%
85	-11.4%	3823	43.6%
90	-14.1%	3707	42.3%
95	-18.1%	3534	40.3%

Table 27 Values of Pxx and No. of equivalent hours (NEH)

PART 2 – Wave Resource

9. Background & theory

Ocean energy is the one of these promising energy resources, Given that oceans cover roughly two thirds of the surface of the planet. This enormous energy resource is contained in the oceans' waters, the air above the oceans, and the land beneath them. The renewable energy source of ocean mainly is contained in waves, ocean current and offshore solar energy.

Ocean waves are a huge, largely untapped energy resource, and the potential for extracting energy from waves is important. Using waves as a source of renewable energy offers significant advantages over other methods of energy generation including the following [**10**]:

- Sea waves offer the highest energy density among renewable energy sources. Waves are generated by winds, which in turn are generated by solar energy. Solar energy intensity of typically 0.1–0.3kW/m^2 horizontal surface is converted to an average power flow intensity of 2–3 kW/m^2 of a vertical plane perpendicular to the direction of wave propagation just below the water surface.
- Limited negative environmental impact in use. Thorpe [**12**] details the potential impact and presents an estimation of the life cycle emissions of a typical near-shore device. In general, offshore devices have the lowest potential impact.
- Wave power devices can generate power up to 90 per cent of the time, compared to approx. 30 per cent for wind and solar power devices.
- Natural seasonal variability of wave energy, which follows the electricity demand in temperate cli- mates.
- Waves can travel large distances with little energy loss. Storms on the western side of the Atlantic Ocean will travel to the western coast of Europe, supported by prevailing westerly winds.

Long before scientist ever put forward a reasoned description of the phenomenon based on fluid mechanics equation, men were aware of the power of ocean waves through observing natural phenomena. The first record of the desire to exploit this energy is a French patent dated 1799, entitled *Pour divers moyen d'employer les vagues de la mer comme moteurs* by Messrs. Paul Grasset, a French Engineer, conceived of a wave energy conversion process, called *bélier-siphon barométrique*. Until the first oil crisis of 1973 for academics to begin to review again the question of harnessing wave energy. The article *Wave Power*, which published in the journal *Nature* in 1974 by Professor S.Salter of the University of Edinburgh marked a turning point and reawakened scientific interest. At the time, the research into wave energy converters will be concentrated mainly in European countries and Japan. Today, the number of projects with innovative ideas has increased considerably, and the research effort has spread to other continents.

Some efforts have been made to harness this reliable and predictable resource. Harnessing of ocean energy needs detail knowledge of underlying mathematical governing equation and their analysis. With the actual computational resources it is now possible to predict the wave climatology in lab simulation. Several techniques have been developed mostly stem from numerical analysis of Navier-Stokes equations. Models of 1st, 2nd and 3rd generations have been developed to estimate the wave characteristics to assess the power potential.

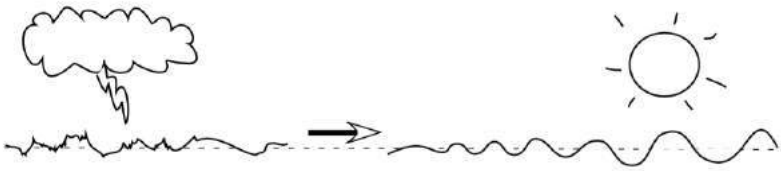
The Theoretical aspects of wave energy conversion related with the sea environment and the interaction with offshore devices is a very important issue. In early stages of the development of WEC this is critical and numerical modelling validation and calibration through tank tests is the most important duty to obtain a successful design.

Knowing the sea environment where an offshore project is deployed is very important. To do this, it is necessary a database that includes meteorological, oceanographic and wave data to establish the *Metocean* conditions for the structural design and operation of offshore structures,

and related infrastructure. Therefore, hydrodynamic, structural, mechanical, installation, wind turbine, WECs and electrical engineers, which are involved in a WEC project, should be familiar with the sea environment.

9.1 Sea waves formation

The ocean is a place of inspiration and mystery; 95% of the ocean is still a mystery. I have spent many amazing moments at sea, mainly surfing wave or enjoying Stand Up Paddle. For without any special effort, a surfer who spends many years waiting for, and riding, different waves in different part of the world. The surfers end up acquiring a large amount of information. Waiting, watching, discussing, waiting and thinking gives surfers an insatiable thirst for knowledge. Therefore, there are some concepts and that surfers know intimately, in a qualitative way that are barely recognised by the scientific community.

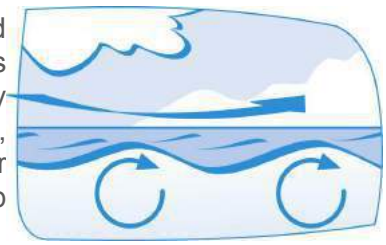


As *Tony Butt & Paul Russell* consider in *SURF SCIENCE: An introduction to Waves for Surfing* [**2**], the first part of their book is about a journey in which the traveller is not an individual wave, it is energy package. The journey starts with the Sun's energy heating our atmosphere. The Equator gets hotter than poles, but air moves to compensate for temperature difference between them. This motion produce the formation of large, swirling vortices of surface air, called depressions.

The transferring the energy from one fluid to the other is produced by the rubbing of the air along the surface of the water, and generating the recognizable from of wind-generated waves.

9.1.1 Ocean Waves

The waves themselves are generated by the action of the wind blowing over the surface of the sea. Exactly how the air transmits its energy to the water to produce waves is still not completely understood. The energy is transferred from one medium to the other, the air transmits its energy to the water to make waves, the air motion converting a smooth and static sea surface into a ruffled-up and dynamic wave-field.



The water particles excited by the wind have in each location of the ocean circular trajectories with highest diameter at the surface and diminishing exponentially with depth. The rise and fall of the waves are due to this progressive orbital motion of the wave particles.

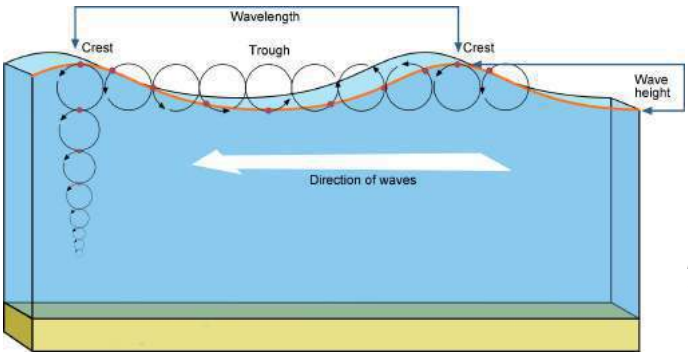


Figure 76 Propagation of the sea waves, www.aquaret.com

The size of the waves generated will depend upon the wind speed, the duration, and the distance of water over which it blows (the fetch), bathymetry of the seafloor and currents. The

resultant movement of water carries kinetic energy that can be harnessed by wave energy devices. Emerging from a fetch is a collection of waves known as the wave train.

The best wave resources occur in areas where strong winds have travelled over long distances. For this reason, for example the best wave resources in Europe takes place along the western coasts.

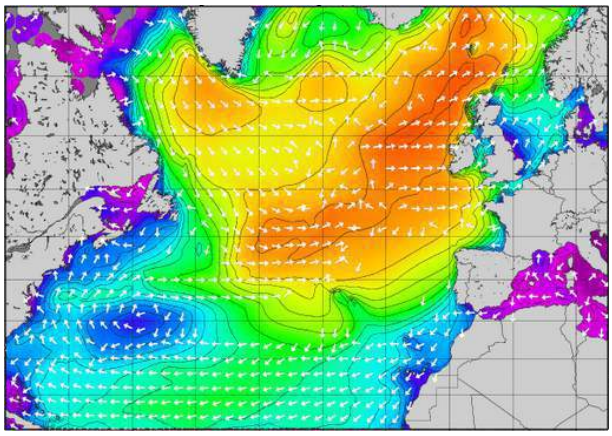


Figure 77 Numerically significant wave height calculated in North Atlantic. Source: Wave Watch III Data courtesy

9.2 Wave theory and description

Ocean waves are generated by the effect of the wind on the surface of the ocean. After the wind has passed they keep propagating as swell [**17**]. The stronger, longer and further the wind blows, the higher the waves are. When looking at a stormy sea the typical appearance is often rather chaotic. A variety of waves with different heights, lengths and directions appear and disappear in what seems to be a completely random manner. However, it is an unstructured chaos of waves is in fact much more structured [**18,19**].

An ocean wave propagates with no loss of energy. Therefore, it can travel long distances before dissipating on the shores. Contrary to what it looks like when waves are rolling towards the shore, only very little mass is actually being transported but merely energy.

Larger waves in a random wave series may be given the form of a regular wave that may be described by a deterministic theory. Numerous water wave theories have been developed and are applicable to different environments dependent upon the specific environmental parameters, such as water height, wave period or wave depth. Generally, these three parameters are used in the design of an offshore structure.

In order to represent the flow associated with the propagation of ocean waves. Considering an ideal fluid, the effect of viscosity will be ignored. The flow will be considered incompressible and irrotational. Therefore the velocity of the fluid *V* derives from a la potential function Φ.

$$\vec{V} = \overrightarrow{grad}\phi$$

By combining this equation with the hypothesis of incompressibility, the potential across of the whole domain is governed by Laplace's equation:

$$\Delta\phi = 0$$

In order to solve the problem, it is also useful to introduce boundary conditions at the fluid domain. For instance, on the assumption that the floor of the ocean is flat, in other words, the vertical velocity component is zero at the bottom. The most interesting are those which are written at the free surface, defined by

$$z = \xi(x, y, z)$$

The free surface of the wave is governed by a kinematic boundary condition, which expresses that the free surface is a physical barrier, and a dynamic boundary condition, which expresses the continuity of pressure on the interface (without surface tension on the scale of waves ocean). These conditions at the limits are [**1**]:

$$\begin{cases} \frac{\partial \phi}{\partial z} = \frac{\partial \phi}{\partial t} + \frac{\partial \phi}{\partial x}\frac{\partial \xi}{\partial x} + \frac{\partial \phi}{\partial y}\frac{\partial \xi}{\partial y} \\ \xi = -\frac{1}{g}\left(\frac{\partial \phi}{\partial t} + \frac{1}{2}\left|\overrightarrow{grad}\phi\right|^2\right) \end{cases}$$

, where *g* is the acceleration of gravity and ρ is the density of the water.

These two conditions on the free surface are nonlinear, and it is difficult to go much further analytically using these formulations:

$$\xi = \varepsilon\xi_1 + \varepsilon^2\xi_2 + \cdots + \varepsilon^{n-1}\xi_{n-1} + \theta(\varepsilon^n)$$

$$\phi = \varepsilon\phi_1 + \varepsilon^2\phi_2 + \cdots + \varepsilon^{n-1}\phi_{n-1} + \theta(\varepsilon^n)$$

, where a non-dimensional perturbation parameter ε, which is defined in terms of the wave slope (wave height *h* /wavelength λ),

$$\varepsilon = \frac{h}{\lambda}$$

This development is called the Stokes development, in which the wave associated with the order of truncation of order *n Strokes Waves*.

Wind-induced seaways can be approximated by the superposition of regular waves of small wave height (elementary waves, Airy waves). Each elementary wave has a sinusoidal profile, of pulsation ω, amplitude *A*, with an infinite number of wave troughs and wave crests [**20**], which are perpendicular to the direction of wave propagation.

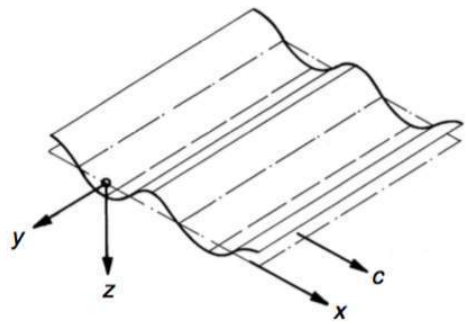


Figure 78 Elementary waves

The simplest and most useful of all wave theories is the small amplitude wave theory [**20**]. Such elementary waves are an important building block for some computational methods, such as for linear seakeeping problems. This wave theory is known as **1st order Stokes wave or Airy wave theory**. It is based on the assumption that the wave height is small compared to the wavelength or water depth.

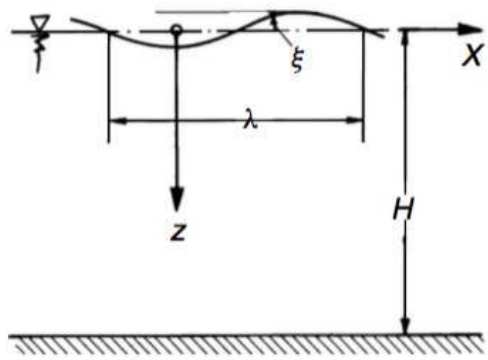


Figure 79 Definition of the main parameter of a regular wave

Generally, regular waves are described by a length parameter (wavelength λ or wave number *k*) and a time parameter (wave period *T* or frequency *ω*). *k* and *ω* are defined as follows:

$$k = \frac{2\pi}{\lambda}; \omega = \frac{2\pi}{T}$$

The celerity *c* denotes the speed of wave propagation, i.e. the speed of an individual wave crest or wave trough, which is proportional to the wavelength (typically 100 meters).

$$c = \frac{\lambda}{T} = \frac{\omega}{k}$$

As the speed of waves is proportional to the wavelength, longer waves will travel at a higher speed than short ones. When the wind has passed by and lost its generating influence on propagating waves, the initial chaotic state will gradually be rearranged into a more arranged and periodic appearance. After a long stretch of water the waves will be arranged after wavelength. When approaching the coastal line in absence of winds, wide, periodic waves will be rolling in, hitting the shore [**5**].

Note that the free surface conditions imply a relationship between the wave number *k* and the pulsation ω, known as the dispersion relation.

$$k = \frac{\omega^2}{g} \quad on\ deep\ water \qquad\qquad ktan(kH) = \frac{\omega^2}{g} \quad on\ finite\ depth$$

The potential ϕ of the wave travelling in the direction of the positive x's in an environment of deep water is:

$$\phi(x,y,z,t) = Re\left(-icAe^{-kz}e^{i(\omega t - kx)}\right)$$

And for finite depth *H* is:

$$\phi(x,y,z,t) = Re\left(\frac{-icA}{\sinh(kH)}cosh(k(z-H))e^{i(\omega t - kx)}\right)$$

In deep water, we can therefore see that the wave number, and therefore the wavelength, depends only of the pulsation of the wave and the acceleration of the gravity *g=9.81m/s* [**1**]. In oceanic engineering, the waves in question have periods of 2-20s, heights of 0.5-15m, and wavelengths of 6-600m.

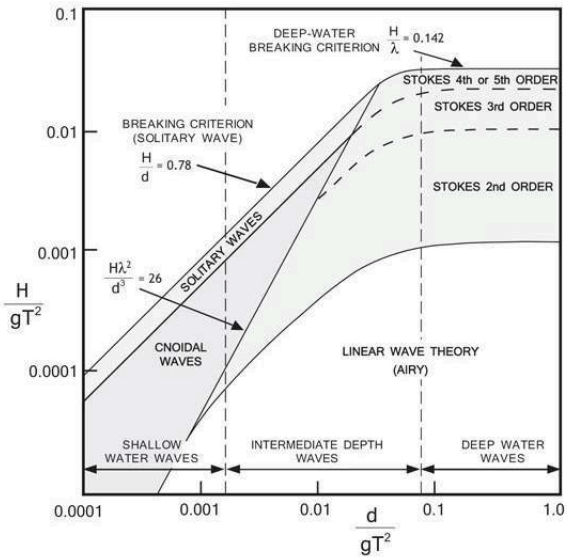


Figure 80 Applicability ranges of various waves (Le Méhauté, 1976, Sorensen, 2005 and USACE, 2008)

The several waves theories that are normally applied to offshore structures are described in the *Figure 80*. Unfortunately, in using the superposition principle for elementary waves, all properties of the seaway that are nonlinear with wave steepness (wave height/wavelength) are lost.

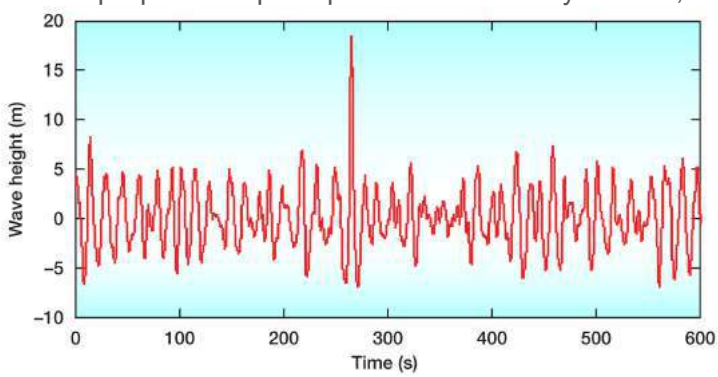


Figure 81 Wave height measurement as a function of recording in the North Sea (January 1995)

In practice, ocean waves are irregular, and time recordings measured at sea may suggest that they are a perfectly arbitrary phenomenon. Today, we are dealing with the spectral content of irregular waves.

There are two basic approaches considered in choosing the design wave environment for an offshore structure. One of these uses a single wave method in which the design wave is represented by a wave period and wave height. The other approach to selecting the design wave environment considers the *wave spectrum* [**21**].

9.2.1 Sea spectrum

The random ocean wave, by the other hand, is described by an energy density spectrum. The wave energy spectrum describes the energy content of an ocean wave and its distribution over a frequency range of the random wave. The random wave is generally described by its statistical parameters [**14**].

A suitable wave spectrum model is chosen an appropriate density distribution of the sea wave at the site under consideration. The most suitable spectrum is a measured design wave spectrum at the site, or one chooses one of the theoretical spectrum models available based on wind speed, duration and the fetch, wind and other meteorological conditions of the site. Attempts have been made by various researchers to formulate the spectrum of a wind-wave or swell, for purposes of forecasting. The spectrum method and significant wave method are two classes of forecasting procedure. However, improved forecasting methods are becoming available which are based on more reliable and accurate field data. This is the existing of theoretical wave spectrum models.

The mathematical spectrum models are generally based in one or more parameters, e.g. significant wave height, wave period, shape factors, etc. The most common single parameter spectrum is the *Pierson-Moskowitz model (1964)* based on the significant wave height or wind speed, which is given by

$$S(\omega) = \frac{A}{\omega^5} e^{-\frac{B}{\omega^4}} \;\; ; \;\; A = 173\frac{H_s^2}{T_{01}^4} = \frac{5}{6}H_s^2\omega_p^4 \;\; ; \;\; B = \frac{5}{4}\omega_p^4$$

where H_s is significant wave height, ω_p is the peak frequency and T_{01} is mean period.

Bretschneider (1969), Scott (1965), *ISSC (1966)* and *ITTC (1966)* are two-parameter spectra available. *JONSWAP spectrum* is a five-parameter, but usually three of the parameters are held constant, was developed by *Hasselman (1973)* during a joint North Sea wave project, and is given by

$$S(\omega) = \alpha H_s^2 \frac{\omega_p^4}{\omega^5} e^{\left[-1.25\left(\frac{\omega_p}{\omega}\right)^4\right]\gamma^a} \;\; ; \;\; a = e^{-\left[\frac{(\omega-\omega_p)}{2\tau^2\omega_p^2}\right]}$$

, where H_s is significant wave height, ω_p is the peak frequency, γ peakedness parameter, and τ shape parameter (τ_1=0.01 for $\omega_p \geq \omega$ τ_2=0.09 for $\omega_p < \omega$).

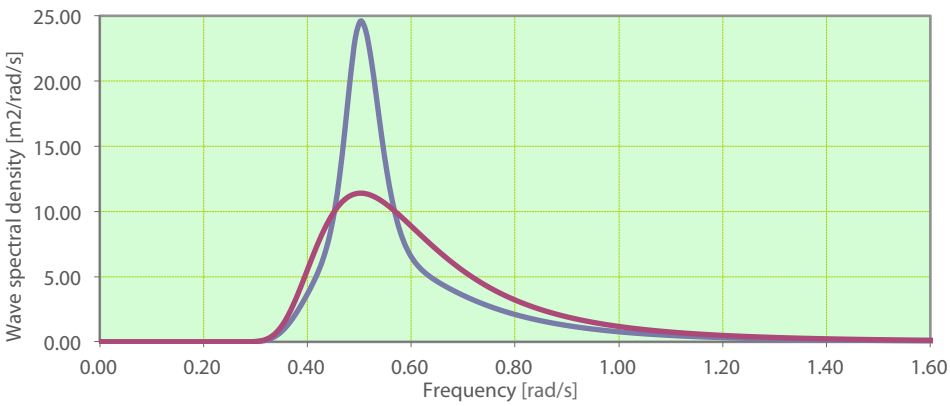


Figure 82 Comparison of the JONSWAP and Pierson-Moskowitz spectra (H_s=8m; T_p=12.5s)

A more complex spectral model was presented by *Ochi and Hubble (1976)*, which a six-parameter spectrum and it describes two peaks in a wind-generated sea mixed with swell,

During the design life of an offshore structure, two types of storms will be encountered by the structure. One is generated in form of wind and associated wind-waves, which waves depend on variable wind and moving fetches, and are random in nature. The other is a long distant storm is called swell. The swells are more stable and regular in nature.

Wave measurements are necessary in site. Generally, the wave theories are used to determinate the response of an offshore structure, which are based on the assumption that are regular and their properties remain unchangeable from one cycle to the next. However, ocean waves are random in nature, therefore, the statistics will define their properties.

For instance, the wave record is considered stationary and its properties are considered to be invariant over a few hour period (for a period of 2-3 hours). Short-term ocean waves are treated as random processes, which are functions of the time variable governed by a probability law. In this case, the wave record is assumed to follow the short-term statistics.

While the probability distributions are applicable for short-term waves. For extreme value statistics, a long-term distribution (for a period of 20-100 years) of the environmental parameters (e.g. wind speed, wave height, etc.) is required.

Tp \ Hs	0.6	1.2	1.8	2.4	3.0	4.6	>4.6	
0-4	10.3	2.6	0.7	0.4	0.1	0	0	14.1
5-6	10.5	2.7	0.6	0.2	0	0	0	14
7-8	13.7	5.7	2.7	1.6	1	0.9	0.3	25.9
9-10	3.8	0.8	0.2	0	0	0	0	4.8
11-12	11.8	7	2.7	1.5	1.1	2	1.8	27.9
13-14	2.9	1.5	0.6	0.3	0.2	0.1	0	5.6
15-16	1.7	0.9	0.4	0.2	0.1	0.1	0	3.4
>17	3	0.9	0.2	0.1	0.1	0	0	4.3
Total	57.7	22.1	8.1	4.3	2.6	3.1	2.1	100

Figure 83 Average percentage frequency of occurrence of significant wave height H_s
Master in Marine Renewable Energies, Bureau Veritas Business School

The prediction of the characteristics of long-term extreme values deals with the occurrence of rare events as opposed to the short-term statistical method is preferable in the design of an offshore structure. Therefore, a long-term statistical method is necessary in the design of an offshore structure, and a design life for the structure should be first established with the long-term statistical criteria [**21**]. The choice of long-distribution functions for wave height modelling (e.g. Draper, Gumbel, Fréchet, Weibüll…) is governed by the principle of the best fitting to the data. Analysing the parameters, significant wave height and peak period and possibly even including the wave direction, μ, *wave scatter diagram* will group the characteristic wave data values and arranged in a table. We can find more detailed information about this topic on the reference book [**5,21**].

10. Wave Energy Resource

Wave energy is a variable resource in the sense that the energy output at any instant is uncontrolled by humans. Instead it is governed by something beyond our control, in this case wind or more strictly the solar irradiation through the atmosphere.

Nearer the coastline, wave energy decreases due to friction with the seabed, therefore waves in deeper, well exposed waters offshore will have the greatest energy. The wave energy flux is stored in waves as both kinetic energy and potential energy, or in the movement of the seawater particles and in the amount of mass of seawater displaced from the mean sea level (or deformation of the wave profile). Then, Ocean waves transport mechanical energy.

When the waves reach shallow water, they will loose speed, increase in amplitude and change in shape. Due to lost stability, the waves will finally break, as illustrated above. Waves also break at larger depths in strong winds.

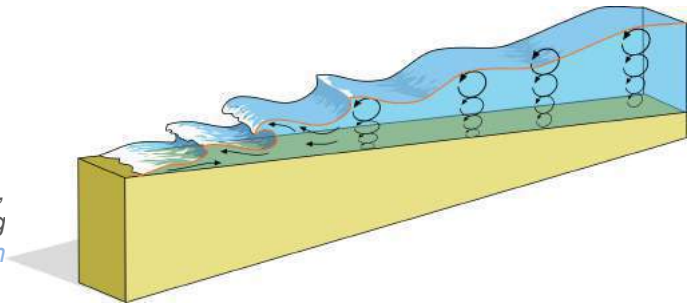


Figure 84 An illustration of ocean wind waves, and how their shape changes when propagating into more shallow waters, www.aquaret.com

When waves propagate across the oceans, energy is transported with very low losses. In a natural state, this energy will transform into heat due to friction and breaking of the waves at the coast.

Wave power is defined by the flow of energy through a vertical surface perpendicular to the direction of its propagation, W/m^2. Generally, the energy flow of ocean waves is expressed as the average the average energy per second passing under one meter of wave crest from the surface to the seabed, W/m [**5**].

The energy is not evenly spread over the depth water, it is found to decrease exponentially with depth. Normally, 95% of the energy is captured in a depth equal to half its wavelength. The wavelength is typically between 60 and 160m.

Taking the spectral nature of the swell into account, the power per meter front of the swell associated with the significant wave height H_s, and the energy period T is given by, in W/m,

$$P_w = \frac{\rho g^2}{64\pi} {H_s}^2 T_e \left[1 + \frac{2k_e H}{sinh 2k_e H}\right] tank_e H$$

, which reduces to the following analytical expression applicable in deep water $(k_e H > \pi)$ [**40**],

$$P_w = \frac{\rho g^2}{64\pi} {H_s}^2 T_e = 490 {H_s}^2 T_e$$

where ρ is the seawater density and g is the gravity acceleration [**1**]. In deep water, i.e. where water depth h ≥ λ , in swell dominated areas such as the North Atlantic for h≥100m where waves are not modified by the sea bottom, wave power can be computed by

$$P_w \approx 0.5 {H_s}^2 T_e$$

that is given in kW/m. It is generally assumed that the relation between H_s, T_e, T_p, and P_w are well fitted, as shown in [**26**].

Note that, at this simplified level the seawater density is about 850 times the air density. So the kinetic energy is proportional to mass, a lot of energy is in motion when water waves propagate, which is the reason why the energy density of waves is high compared to other renewable energies, such as wind and solar.

10.1 World resource of the wave power

Today about 87 per cent of the world's energy consumption depend on non-renewable energy sources such as oil, natural gas and coal.

Wave energy is a variable resource in the sense that the energy output at any instant is

uncontrolled. Other advantageous features of the wave energy is its persistence, meaning that the most likely output from a wave energy converter the next hour is the same as during the previous hour, and is often possible to forecast the power of waves and their occurrence in a specified area a certain period in advance.

The wave energy resource can be defined in different ways. The theoretical resource i.e. the hydrodynamic power captured in the ocean waves or the global wave energy flux was already in the seventies estimated to be in the order of 1-10 TW [**22**]. While the share of this energy hitting the coasts has been estimated to be around 1 TW, the global estimates for wave energy potential are still relatively uncertain. It is estimated, based in different sources, a ranging from 2000 TWh to even 80000 TWh, which means a potential worldwide wave power resource is 2TW. For instance, the Intergovernmental Panel on Climate Change (IPCC), in 2012, reported a theoretical potential of around 29500 TWh/yr considering all areas with wave energy densities higher than 5 kW/m (Lewis, et al., 2011) [**23**].

Wave energy is unevenly distributed over the globe. The *Figure 85* below presents the worldwide distribution of average annual wave power in kW/m. The best wave resources occur in areas where strong winds have travelled over long distances. Nearer the coastline, the wave energy decreases due to friction with the seabed. Therefore, waves in deeper well-exposed waters offshore will have the greatest energy. The north and south temperature zones have the best sites for capturing wave power. Increased wave activity is found between the latitudes of 30° and 60° on both hemispheres, induced by the prevailing western winds blowing in these regions, which are strongest in winter.

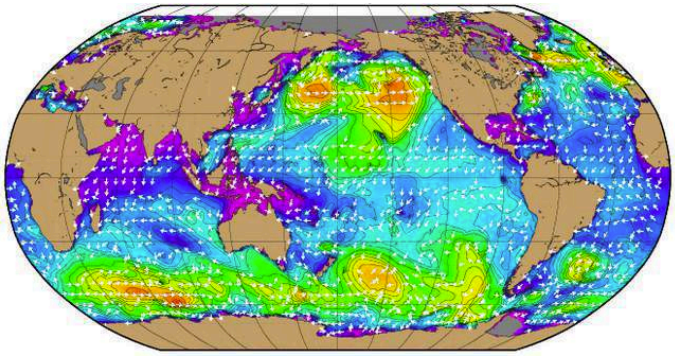


Figure 85 Global Numerically calculated significant wave height, 03.03.2016 at 00:00hr, Wave Watch III Data courtesy.

The maximum, around 100kW/m, is found around Cape Horn. The average energy flux hitting the coasts at the southern tip of South America, the Falkland Islands or parts of New Zeeland might be as high as 100 kW/m [**23**]. Huge areas of the globe have densities of wave power greater than 20kW/m, which is generally accepted as the profitability threshold for wave energy converters. Some of the most exposed areas are found in the North Atlantic, southern Australia and on the Pacific coast of South America. The seasonal variance of the wave energy resource is, however, far more marked in the northern hemisphere. For example, countries like Australia, Chile, Ireland, New Zealand, South Africa, the UK and the US have excellent wave resources with average power densities of 40-60 kW/m.

In Europa the first wave energy converters were developed. The best wave resources in Europe occur along the western coasts, which lie at the end of a long fetch, the Atlantic Ocean. UK, along with Ireland, has the greatest exposure to the wave energy resource in that area of the globe. The average annual wave power on the Atlantic coast of Europa varies between 15 and 80 kW/m, with the UK´s realistic potential being 7-10GW [**25**], and in France it is around 40kW/m in the Bay of Biscay.

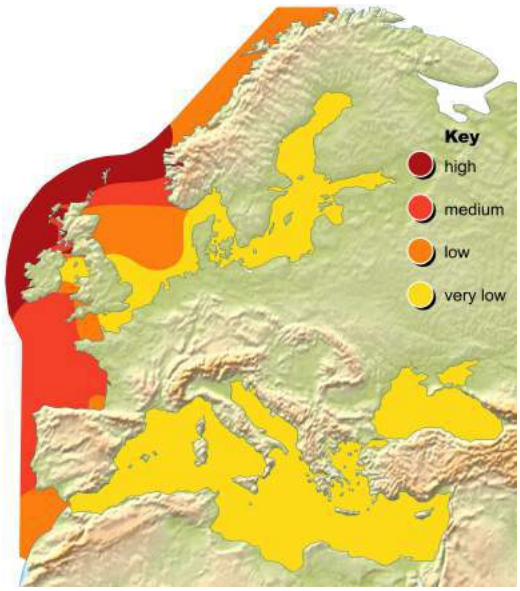


Figure 86 Wave resource distribution in Europe, www.aquaret.com

10.2 Wave power technology and mechanisms

Wave energy sector is currently under development and in continuous evolutions.

The Wave Energy Converters (WECs) are complex devices whose main target is to obtain the energy of the waves. Since the power in ocean waves grows with the square of the wave height, the wave energy converters occasionally have to handle huge power and forces, with out the ability to stop as for example wind power plants. In occurrence of smaller waves and calmer wave climates the WECs are still supposed to absorb energy in an efficient way.

Intensive research into wave energy began in the 1970s, when the oil crisis promoted an increased the interest in renewable energy, but as the energy crisis came to an end, interest in wave power diminished and the early 1980s saw many of the trials discontinued. In the 1990s wave energy began its march toward commercialization in the 1990. The evolution of the technologies remained slow until early in the 21st century following a new drive for renewable energy, the offshore devices capable of exploiting the more powerful waves out at sea. Today there is a large number of devices for wave energy conversion that are nearing commercialization and a large variety of technology concepts are under development. In 2008, the first experimental wave farm was opened in Portugal at *Aguçadoura Wave Farm*, which was designed to use three *Pelamis Wave Energy Converters*.

Considering at the design stage, an important parameter is what kind of sea state the converter will be subject to. In large ocean swells the device will experience large, slow forces, and in order to cope with this a large mass and more inertia to produce power is required. On the other hand, in more shielded oceans where the waves are smaller it is beneficial to have a small and light device, which can utilize the higher velocity of the motion. Therefore, there are also concepts utilizing a turbine to generate electricity.

Full-scale WECs are structures with large pneumatic, hydraulic and electrical devices, whose mechanisms internally or externally fitted on it for the energy conversion and transmission. There are deployed with a rigid structure and station keeping system that aim to survive for years in the harsh ocean climate.

The Main components of Wave Energy Converters are:

- Power Take Off System (PTO) – It is the main mechanisms that can be implemented in a WEC to convert wave energy into mechanical and/or electrical energy. Whatever the control strategy is, a wave energy converter must have power take-off machinery that can be ordered to do things according to the changing values of digital and analogue commands e.g. switch on, switch off, increase, decrease, connect, disconnect [**27**].

- Structure – While the PTO and hydrodynamics are the main parts of a wave energy converter, the *skeleton* is the structure.

This structure has to accomplish with several elemental requirements:

- Stability and hydrodynamics requirements
- Design to withstand the stresses that it is subjected to the ocean during its lifecycle
- Accommodation of the PTO and the ancillary system
- Space for surveys, accessibility, and ballast system

Prototype devices are mostly built from steel today, but some studies indicated that concrete, and composites (GRP) could all provide economically viable solutions, although the last of these would need to be in sandwich form.

- ***Ancillary System*** – It is the support of the rest of components. Typical ancillary systems are ballast, anti-corrosion and fire fighting systems, as well as control, communications and SCADA,

and electrical equipment and heat exchangers.

- Mooring and Station Keeping Systems – These floating devices keep their position using mooring systems. The mooring and station keeping systems are very important systems within the overall project, and will be considered economically and technically since early stages.

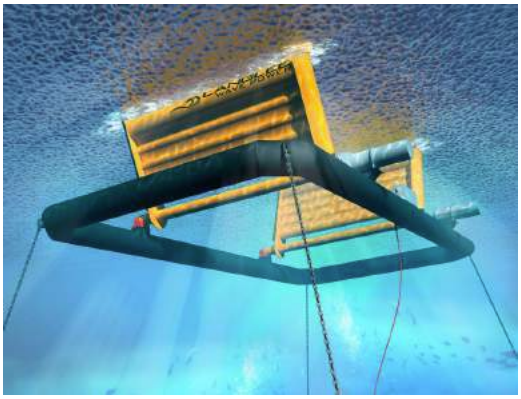


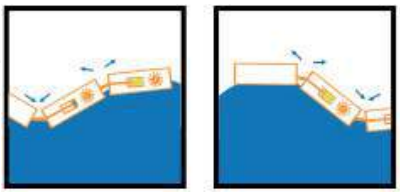
Figure 87 Mooring concept of Langlee Wave Power converter, http://www.langleewavepower.com/

10.2.2 Wave energy converters classification

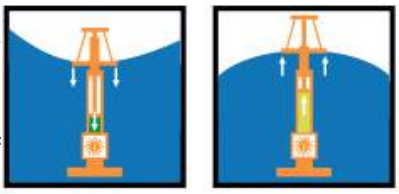
Numerous criteria can be used to draw up classifications. Generally, WECs can distinguish by the shape of the device, the location and the type of hydro-mechanical conversion.

Despite the large variation in design and concepts WECs can be classified into eight main types [**13**], each with its own method of harnessing energy from ocean waves:

A. *Attenuators* are horizontally large floating devices that function in line with the direction of wave propagation of the waves and operate parallel to the wave direction and effectively rides the waves. The wave energy is captured from the relative motion of the arms. Good examples of attenuators are: *Pelamis, Oceantec…*

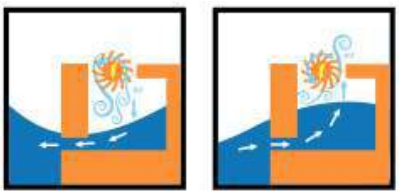


B. *Point absorbers* are axisymmetric devices that are small in relation to the wavelength. These floating structures absorb energy from all directions through its movement at/near the water surface. The motion of the buoyant top is converted into electrical power. The power take-off system depends on the configuration of displacers/reactors. Typical examples of point absorbers are: *OPT, AQUA buoy, APC PSYS, Wedge…*



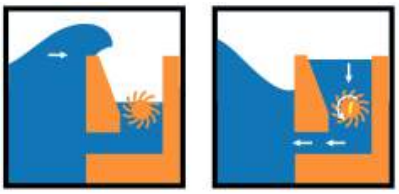
C. *Oscillating wave converters (OWC)* are an elongated floating body that are subject to the swell, they undergo movement according to its 6 degrees of freedom: translation (surge, heave and sway) and rotation (pitch, roll and yaw). They can be classified according to the degrees of freedom used. For instance, an oscillating wave surge convert extract energy from wave surge, in this case, an arm oscillates as a pendulum mounted on a pivoted joint in response to the movement of water in the waves.

D. *Oscillating water column converters (OWSC)* utilise changes in pressure in the air contained in a hollow structure with a first opening below the water line. The movements of the wave influence the column of water to rise and fall. In the top, a second opening allows to flow to/from the atmosphere via a turbine, which is designed so that their direction of rotation is regardless of the direction of the airflow. This rotation is used to generate electricity.

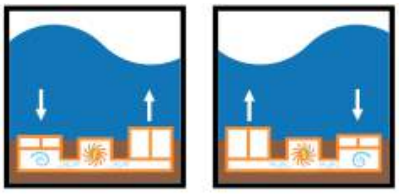


E. *Overtooping/Terminator devices* have the main axis perpendicular to propagation direction of the waves. Those are designed to capture and concentrate the waves in order to increase their height, and then waves break into a storage basin. Then, the water runs back into the sea

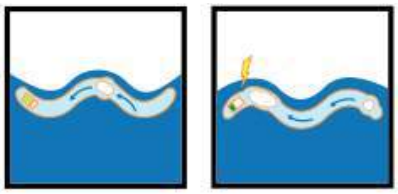
passing through low-head turbines to generate power. These devices may concentrate the wave energy (gravitational potential energy storage). Typical examples of point absorbers are: *Wave Dragon, Wave Roller, Oyster…*



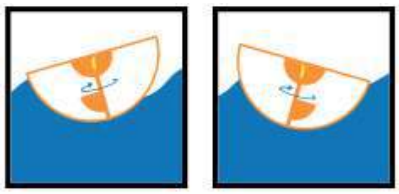
F. *Submerged pressure differential devices* are located in shallow waters and attached to the seabed. The waves cause the sea level to rise and fall above the mechanism, inducing a pressure differential in the device. The vertical motion caused by the differential drives a fluid pump that creates mechanical energy.



G. *Bulge wave devices* are used to capture energy by the movement of the device heaving and swaying in the wave. The water enters through the stern and the passing wave causes pressure variations along the length of the tube, creating a bulge, which travel through the device, gathering energy to be used to drive a low-heat turbine at the bow, and the water returns to the sea.



H. *Rotating mass devices* capture energy by the movement of the heaving and swaying of the device in the waves, causing a gyroscope or weight to rotate within. This movement is attached to a generator inside the device that produces energy.



I. *Other devices*, like the Wave Rotor, is a form of turbine turned directly by the waves. Flexible structures have also been suggested, whereby a structure that changes shape/volume is part of the power take-off system.

10.2.2 Description of a wave energy converter

The *Archimedes Waveswing* device is a submerged wave power buoy, which designed to provide reliable and affordable power for maritime communities and offshore applications. This device reacts to changes in subsea water pressure caused by passing waves and converts the resulting motion to electricity via a direct-drive generator. The system is suitable for deployment in water depths in excess of 25m and can be configured for ratings between 25kW and 250kW by selecting the appropriate scale [**29**].

As a result of the horizontally travelling wave energy, water molecules will undergo a circular motion, which decreases exponentially with depth. This technology was tested offshore Portugal in 2004 and narrowly missed a world first for delivery of offshore wave power to a national electricity grid, being beaten by *Pelamis* by some 6 weeks. Since that time, the *Waveswing* has been refined and developed to focus on customer needs in an emerging market and a pre-commercial basis will be offered from mid-2017.

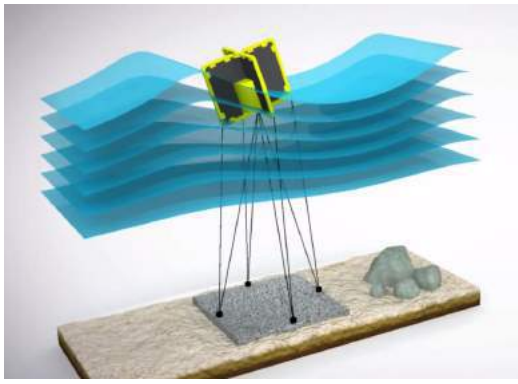


Figure 88 1/4 scale model with real sea testing (Phase 2) (left) and Figure 89 Archimedes Waveswing, Wave Energy Converter of Laminaria (right)

10.3 Wave Energy Market

Wave energy devices have not yet reached the same stage of development, if we compare with other sources, such as offshore wind and tidal. Some concepts have undergone large-scale testing, and the sector presents a vast number of different concepts, with no clear convergence in design. This is partly due to the different characteristics of the wave resources available at various water depths, which will ultimately require different technical solutions for power capture. These technical solutions could be slightly more complicated and expensive. This might mean a delay in final stage of the enlargement development, and the abandonment of a design and later the demise of some companies.

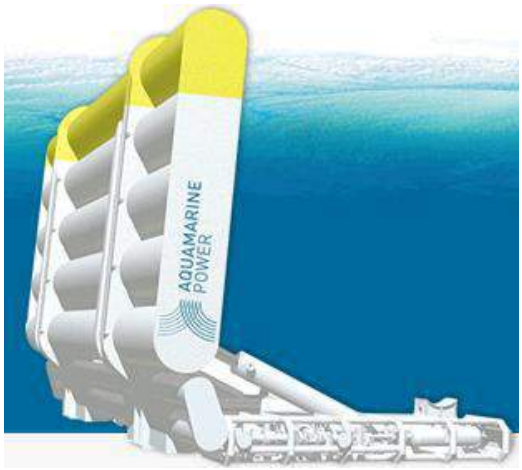


Figure 90 Oyster 800 wave energy machine

From one point of view 2015 has been a terrible year for renewable energies, especially for wave energy technology. Since no offers were made for Aquamarine Power as going concern, the company dues to cease to trade their business and its Oyster concept [**28**].

Also the revolutionary concept *Pelamis Wave Energy converter* was the world's first commercial scale machine to generate electrical energy into the grid and the first used in commercial wave park projects. *Pelamis Wave Power* called in an administration process the year before after failing to secure development funding for their market-leading wave energy converter [**30**]. These examples just show the difficultness of developing technology in a non-mature market and with possibly the hardest conditions to design in among all of the marine renewable energies, despite the apparently huge potential available for wave power. If these technologies cannot be proven it will be hard for investors and projects to attract the necessary funding to keep on developing these technologies as perceived risks will remain high and not within reasonable levels that could be mitigated.

Despite this situation, there are a large variety of new technology concepts and on-going projects, and the wave power industry is experimenting a second youth. But this time, this industry is developing with more experience on full scale testing, a more developed supply chain and also more support from the Governments, especially in regions like Scotland which is about to lead the deployment of marine renewable energy in all of its forms.

The fact that wave power companies are struggling to perform their deployment plans and up-scale their technologies is at the same time motivating new stakeholders to propose new and innovative ideas on how to harness wave energy from the sea, e.g. *Laminaria*, a Flemish wave energy developer, has done signing up for a wave berth in EMEC where the plan to deploy their wave energy converter (*Archimedes Waveswing*) design by 2017, or *NEMOS* is an innovative system, which has developed by the German company *NEMOS GmbH*, to generate electricity from ocean waves. It is represent at low costs and particularly suitable system for combination with offshore wind turbines.

Regarding tidal energy, the feasibility of deploying tidal farms in the coming years is far more realistic that for the wave energy scenario. For instance, Atlantis Resource Limited, together with Siemens, is making big steps towards the deployment of the first pre-commercial scale tidal energy project called *Meygen*, again in the North part of Scotland where marine renewable energies are finding the most stable and convincing support. Finally, the wave and tidal energy technologies have certain advantages over other energy sources in Europe, especially in Scotland. These technologies provide an opportunity to generate energy at a wide range of locations throughout Europe. Additionally, wave and tidal power produce energy at different times, and more consistently, than other renewable energy sources, such as wind and solar. It means a better stability of Europe's energy networks, and they present a minor visual impact of

electricity generation sources in areas where this is a concern. In addition, wave and tidal energy technologies can achieve an extra value for the exploration of synergies with, such as, offshore wind. It will provide considerable costs reductions.

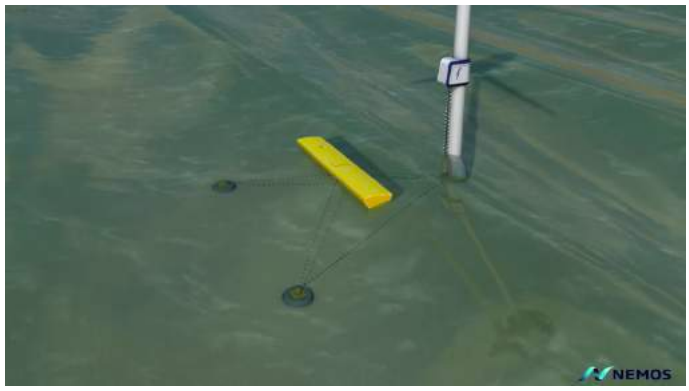


*Figure 91 NEMOS-system consists of an elongated floating body, which is connected to the seabed by three ropes [**30**]*

10.3.1 Cost of Energy Estimate

LCOE is a term that DOE uses to determine the "break even" cost for a technology assuming a minimum rate of return. The LCOE provides an estimate of the cost of electricity from a farm of devices over a 20 years period, and it is reflective of capital cost, operating cost, decommissioning cost, and of the expected annual energy production. The simplified LCOE is required four inputs and can be represented using [**35**]:

$$LCOE = \frac{(FCR \cdot CapEx) + OpEx}{AEP}$$

, where *Capital costs (CapEx)* of equipment, including installation and project management, are incurred mostly at the beginning of a project while *Operating costs (OpEx)* such as maintenance, rent and insurance are spread over the lifetime of the array. Decommissioning costs are also considered. The other two inputs include the *Annual Energy Production (AEP)* provided to the grid, which will depend on the resource available, the characteristics of the devices in converting the resource into electrical energy and the availability of the device, and the *fixed charge rate (FCR)*, e.g. a typical discount rate of 12% for a lifetime of 20 years. CapEx costs are broken down even further depending on the specific design.

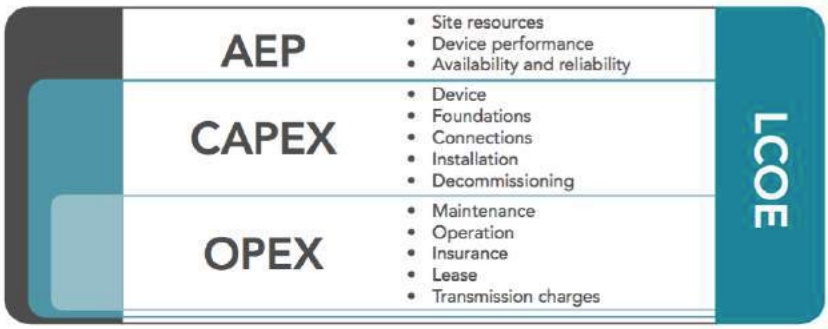


*Figure 92 Components of the levelised cost of the energy [**34**]*

CapEx	OpEx
Development	Insurance
Infrastructure	Marine operations
Foundation & Mooring	Shore-side operations
Support structures	Replacement parts
PTO system	Consumables
Subsystem integration & profit margin	Post installation environmental
Installation	
Contingency	

Figure 93 Cost categories for CapEx and OpEx

An analysis was performed in the reference model project to determine the cost of electricity production for a single-unit, 10-unit, 50-unit and100-unit array sizes. Considering a floating oscillating surge device [**35**], the chart below shows the LCOE breakdown of single-unit

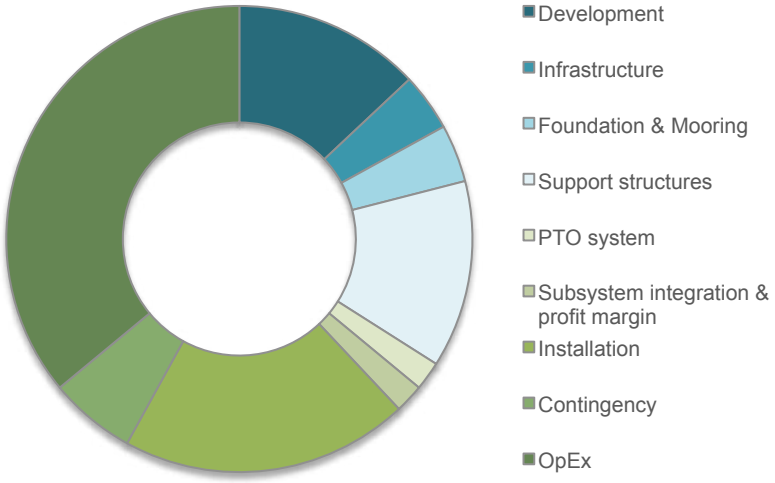


Figure 94 LCOE breakdown of a single unit

For instance, the WECs have a large percentage attributed to annual OpEx costs, and another significant cost for a single-unit is the development cost, particularly the environmental cost for permitting and leasing. A this point, WECs have today a higher development cost that is associated to the variety of sizes and working principles [**36**]. The next *Figure 95* shows the LCOE breakdown of 100-units array in this occasion.

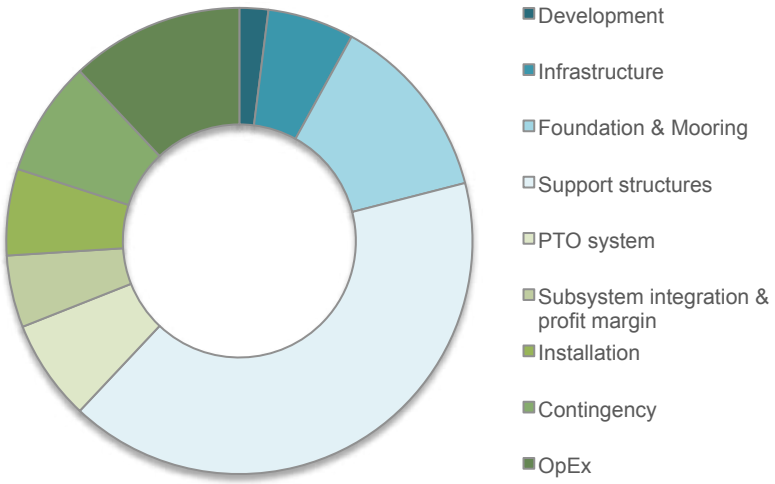


Figure 95 LCOE breakdown of 100-units array

Comparing the LCOE breakdowns for this kind of device at single-unit and 100-unit array deployments. The largest deviation is the contribution of structural costs with 41% of the LCOE at 100-units array. Much of this deviation can be associated with overdesign. Also high mooring costs are expected in the WECs, which has a bigger impact than the PTO system cost. Therefore, a more structurally optimized design will reduce considerably these costs.

As shown in the following *Figure 96*, based on the reference [**37**], the baseline commercial LCOE estimate for one WEC is around 3.25 EUR/kWh (or 360 cents/kWh) and for an array of 100 units, this value drops to approx. 0.64 EUR/kWh (or 70 cents/kWh). These values are based on an AEP value of around 900MWh, CapEx, OpEx, and prescribed fixed charge rate. Based on these values, it is critical that the device cost must decrease and/or the device performance must increase for this technology to become economically viable.

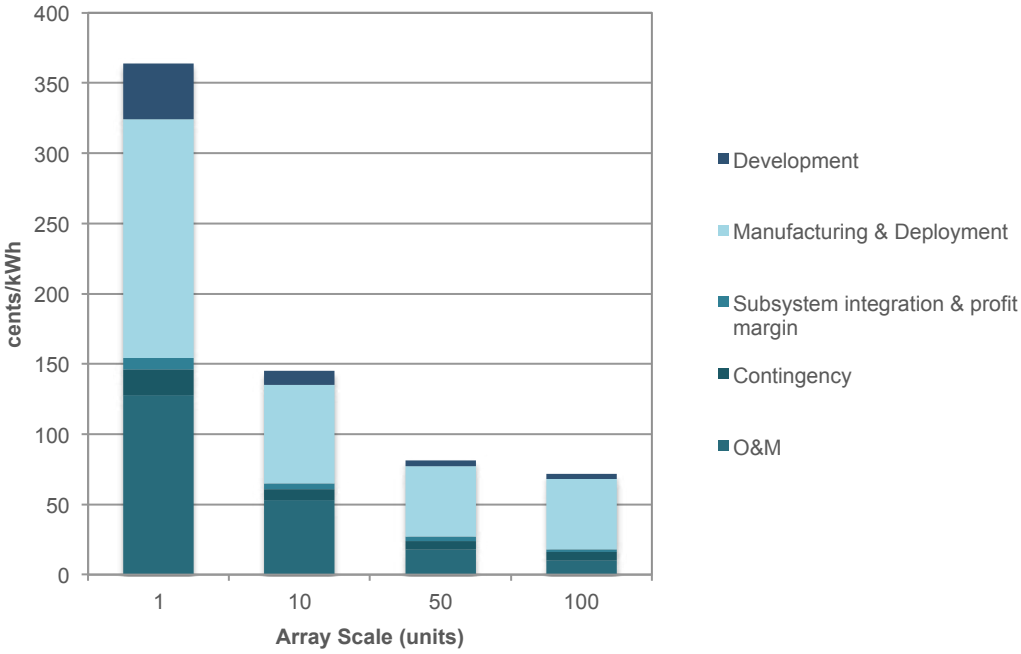


Figure 96 LCOE (cents/kWh) breakdown per deployment scale

The cost of manufacturing and deployment contributes 47% of the total LCOE for a WEC, and O&M contributes another 35%. As expected, these two categories are the dominant cost drivers for the WEC design. To make this WEC design cost competitive with other renewable energy technologies, it is essential to conduct research and development that is geared toward reducing the cost of manufacturing, deployment, and O&M.

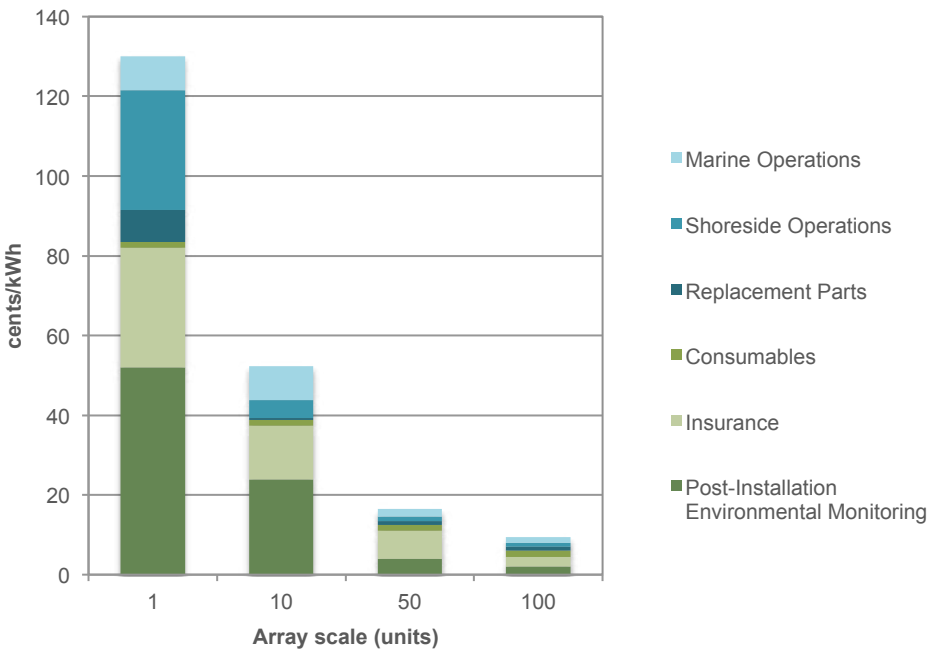


Figure 97 OpEx contributions to LCOE (cents/kWh) per deployment scale

The *Figure 97* shows how OpEx costs contribute to the LCOE, and a breakdown of the OpEx contributions to LCOE is provided for a single-unit, 10-unit, 50-unit and100-unit array sizes.

It is true that there are some cost savings associated with volume production in components such as the power conversion chain, and installation and permitting in mooring systems;

however, large cost reductions to the device structure will still need to be made even as installation and permitting costs diminish.

A preliminary design, like all other previously presented reference model designs, has inherent uncertainties surrounding the device's performance, design, and economics. The economic uncertainties of each cost breakdown category, for both CapEx and OpEx, will be consulted as shown in the reference [**37**]. To capture the CapEx and OpEx uncertainties, a qualitative assessment of the design uncertainties was performed.

Figure 98 Wave Energy Converters Farm Concept, Langlee Wave Power

10.3.2 Current Costs and Cost Projections

With the estimated levelised cost for 10MW-wave energy farms between EUR 330-630/MWh (SI Ocean, 2013). Given the early stage of technological development and where arrays of 10 MW of total installed capacity still need to be tested, it is not surprising. This directly affects the *economies of scale*, including the assumptions considered to estimate the current and projected costs. However the potential for wave energy is significantly greater than the tidal resources across the same geographic area and is also site specific, so the expectations are that wave energy costs will fall to levels similar to those of tidal current technologies [**33**].

10.3.3 Cost distribution for a wave power plant

Wave and tidal stream energy technologies have made significant progress in recent years. A number of full-scale prototypes are now in operation and generating to the electricity grid and the plans for first arrays are well advanced. It is important for policy makers and those who might invest in ocean energy generation to have a picture of the current costs for ocean energy generation and how these are likely to reduce over time.

Construction of waves parks should be made in areas of moderated or lower environment sensibility, using safe technologies. The relative costs distribution of the wave plant is shown displayed in the following graph.

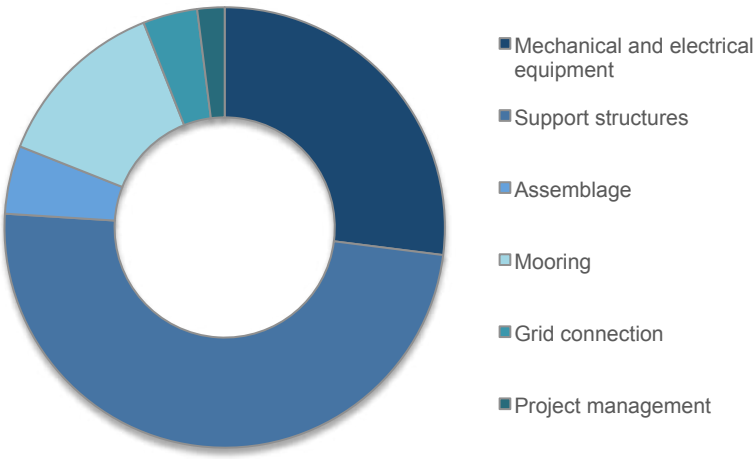


Figure 99 Cost distribution for a wave power plant

It can be seen from the diagram that he cost of the devices makes up only part of the total cost of array projects. Balance of plant items such as cabling, connectors and switchgear as well as

installation, operation and maintenance make up a significant proportion of total lifetime costs. The MW cost installed to be competitive is situated in 500-600 EUR/MW and the economic competitive is attainable when is installed at least a power of 6.8 GW. Presently, the world wave energy market is situated in 750 MEUR. The European Union estimates a wave energy cost of about 5 EUR/kWh [**9**].

Wave energy is not expensive to operate and maintain, no fuel is necessary and no waste is produced. However, this energy source depends on the intensity of the waves and their consistency.

10.3.4 How cost can reduce over time

Cost of energy for any energy technology, including marine technology, can reduce with Scale and volume, with learning by doing, and through innovation. Scale and volume production offer significant opportunities for cost reduction. Increasing scale can bring down costs in several different ways, e.g. upscaling of device, number of devices, scale of production, scale of engineering support [**33**].

Also the know-how, and improved efficiency it brings, is the mechanism by which learning by doing can offer cost reduction. The track record of marine energy devices will have an important influence on the perception of risk and therefore on project cost assessment. Learning through Research & Development also contributes to the overall increase in sector experience

In practice the boundaries between innovation and learning by doing will be not clear. The third mechanism for cost reduction is innovation. Innovative cost reduction involves radical changes in design or process to remove costs. It can include everything from fundamental new energy capture concepts through to innovative deployment and operation processes.

11. Wave Resource Assessment

Offshore wind farms present benefits that include higher wind resource (larger wind velocities) with lower turbulence levels than adjacent land sites.

The two basic sources of wind and wave information are data obtained from direct and indirect measurement techniques, i.e. using remote detection both ground-based sensors and satellite systems, and results of numerical models.

In the *Chapter 11*, some concept of wave resource assessment will introduce and a small analysis of the wave conditions in a study zone, around the *meteomast IJmuiden* in the *Dutch North Sea*, is presented. Therefore, a short-time time data and the JONSWAP spectrum (long-term distribution), which is introduced in *Chapter 9*, were used to assess the wave conditions. The short-time dataset was achieved of the *meteomast IJmuiden* with typical characteristics of wave assessment (peak period, significant wave height, …)

11.1 Metocean data

Metocean data has become an accepted industry abbreviation for meteorological and oceanographic data, terminologies that embrace the subject of marine climatology. Simply stated, *metocean data* refers to the combined effect of the meteorology and oceanography. These effects include [**14**]:

- Local surface wind
- Wind-generated local waves
- Swell (long-period waves) generated by distant storms
- Surface current also generated from the local storms
- Energetic deep water currents

- Non-storm-related currents (e.g. loop current in the Gulf of Mexico or in the Norwegian northern North Sea)

The *metocean* conditions are generated from two independent physical effects: *astronomical forcing* and *meteorological forcing*. In this context, *metocean data* used as a group name for following types of data; water movements, offshore wind conditions, and sea state (wave conditions).

The variation of these processes over time and for different locations is directly relevant to engineering design, construction and operational planning, and to determine issues such as trends, cycles and extremes. Then, an exercise that determines the importance of each process to overall conditions at a project site is commonly referred to as a *metocean study* [**38**].

11.1.1 Sea state

Sea state describes the wave field resulting from combined the *wind-sea* and *swell* components. Wind-sea waves are generated by local winds blowing over the surface of the ocean. Swell represents wind waves that have travelled out of the area in which they were generated [**38**].

Principally, the individual waves within a given sea state can be defined according to:

Wave conditions observed at a given point of the sea over a few hours (sea states) are described by its *energy density distribution* $S(f,\theta)$, where θ is the mean direction of the spectrum, that is usually summarized by significant wave height H_s (vertical difference in elevation between wave crest and trough, in m), mean wave energy period T_e , which is commonly used in the calculation of the wave power, or peak period T_p , and mean direction θ (direction that waves arrive from).

During the period in which wave observations are made, the sea state will comprise groups of waves with varying height, period and direction. Statistical analysis of these variations provides standard summary parameters, which are commonly related for design purposes to properties of a particular wave spectrum and statistical distribution.

When discussing statistical wave distributions it is necessary to distinguish between distributions used to represent the variability of individual waves in a sea state persisting over the order of hours (*short-term distribution*) and distributions used to represent the variability of a statistical parameter, such as significant wave height over a period of years (*long-term distribution*).

11.2 Data Sources

The first global source of wave data is visual observations carried out for meteorological purposes on board of commercial ships, which started to be archived by 1850 e.g. by British Meteorological Office e.g. Sherman in 1984.

In the following sub-sections, a review of the main sources of wave data for wave energy resource mapping is given quickly. Although there are many different types of measurement principle and various types of numerical wave models, only the most important for wave energy applications are described [**44**].

11.2.1 Buoys

Since the early 1960s the buoys have been used for measuring waves. *Datawell Waverider* and *OCEANOR WAVESCAN* buoys are particularly successful ad continues to collect directional wave data today. Wave buoys, both scalar or directional can give good estimates of the sea state from a 20 or 30 minute sample, but may be too expensive to maintain for long term wave climate estimation, except at a small number of key reference sites.

In offshore waters around the world, long-term buoy wave measurement networks are still relatively few and far between. Networks with directional measurements (directional information is essential for coastal prediction) are even scarcer, e.g. National Networks in Spain, Greece, France and Italy and long-term measurements carried out in Norwegian waters and the North Sea for the offshore industry. For a study of the wave resource, the raw data of the wave measurement (during the period 01.11.2011 - 01.01.2016) for the wave resource study was provided by *Meteomast IJmuiden*, in *Chapter 5*, the *Waverider buoy* was introduced. AWAC

11.2.2 Satellite altimeters

Remote sensed wave data are obtained by radar altimeters that provide accurate significant wave height data, and SAR (synthetic aperture radar) from which estimates of directional spectra $S(f,\theta)$ are obtained for long waves with period larger than 8-9s. The use of SAR spectra can be directly useful for wave energy resource assessment in areas dominated by long swells. Each satellite altimeter has to be validated and calibrated by comparing with long-term offshore buoy data generally.

11.2.3 Global wave models

Today, Altimeter H_s data are assimilated into global and regional wave models. Many meteorological centres today run wave models regionally and a few run global models. Dedicated long-term hind-casts have also been performed. The wave models simulate the growth, decay and propagation of ocean waves based on input winds over the area in question, e.g the wave data from DHI's global hind-cast database, which were established through numerical modelling using *MIKE 21 Spectral Wave model*.

The most update and generally used are third generation models with an explicit representation for the physical processes relevant for wave evolution, namely the non- linear interactions. The *WAM Model* by The *WAMDI Group (1988)* was firstly implemented at ECMWF being described by *Komen et al.*, in 1994. It was further implemented in many centres. *WAVEWATCH III (or WW3)* developed by *NOAA/NCEP* differs in governing equations, numerical methods and physical parameter-rizations. Verifications and inter-comparisons of wave models has shown that model accuracy has continuously increasing namely due to data assimilation and statistical forecasts, its accuracy being presently limited by the quality of input wind fields, *Jansen* in 2006 [**44**].

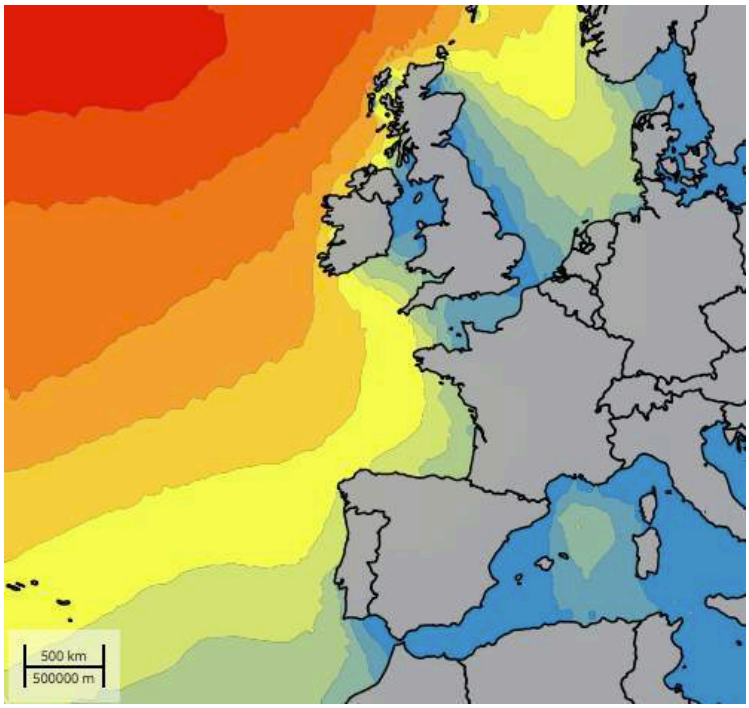


Figure 100 Overall Wave power map, from 0 (blue) to 80 (red), in kW/m, http://si-ocean.dhigroup.com/map

The satellite altimeter data represent a high quality, independent and globally covering wave and wind data set which can be used for the direct validation of the wave model data, although this is limited to significant wave height and wind speed.

11.3 Case Study: Estimation of Wave Energy Source

The North Sea is an inland sea of the Atlantic Ocean in Northwestern Europe with an average water depth of 94m. The North Sea is located between the European Continent (Denmark, Germany, The Netherlands, Belgium & France), the Scandinavian Peninsula and the UK, and is 575000km^2 large.

11.3.1 Wave data

Not all sites in the North Sea are suitable for wave energy exploitation. The average annual available wave power and the dominant mean wave direction need to be calculated to predict the suitability of a specific location. In this study case, it will only be determined the average annual wave power.

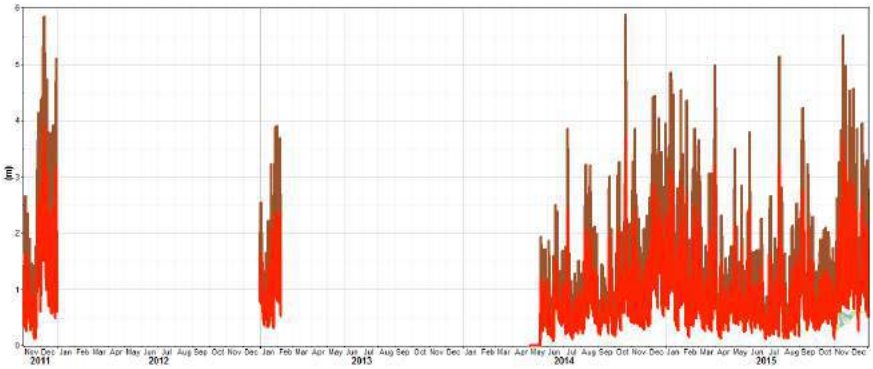


Figure 101 Measurement data of the offshore meteorological station, Meteomast Ijmuiden, coordinates N52°50.89' E3°26.14'

The wave data used in this study is from the *Energy research Centre of the Netherlands (ECN)*. The *meteomast IJmuiden* provided the measurement data, given as energy period, peak period and significant wave height. Data from the years 2011 to 2015 has been used. The data corresponding to the years 2012 and partially 2013 are missing.

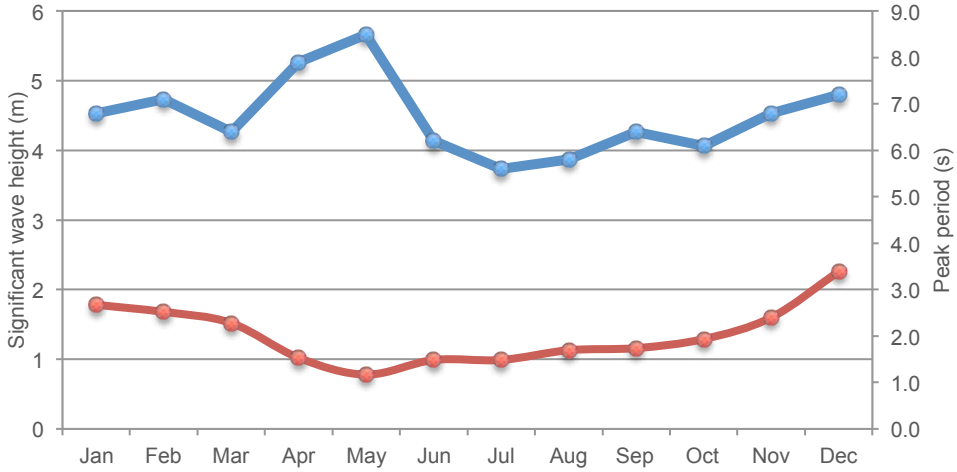


Figure 102 Monthly average of significant wave height and peak period at Meteomast IJmuiden during the study period

To avoid the results being applicable to one and half year only, an average annual energy has been estimated based on during this period, and then the wave pattern in every time step has been taken from year November 2011. A regression was used to get the rest of the data.

Figure 103 Representation of the wave power in the Dutch North Sea, DHI's wave model

A second wave data in the study zone based on *DHI's wave model*, corresponding to the Dutch

North Sea was used and compared with the measurement data. In the *Figure 103*, parts of England, The Netherlands and, generally, in Europa can be seen, and the grid of dots illustrates points where long-term wave data is available. Then, it is possible to compare a short-term wave data (*meteomast IJmuiden*) and a long-term wave data.

If we compare the measurement wave data and the long-term wave data, the most probable wave height is between 1 and 1.25m.

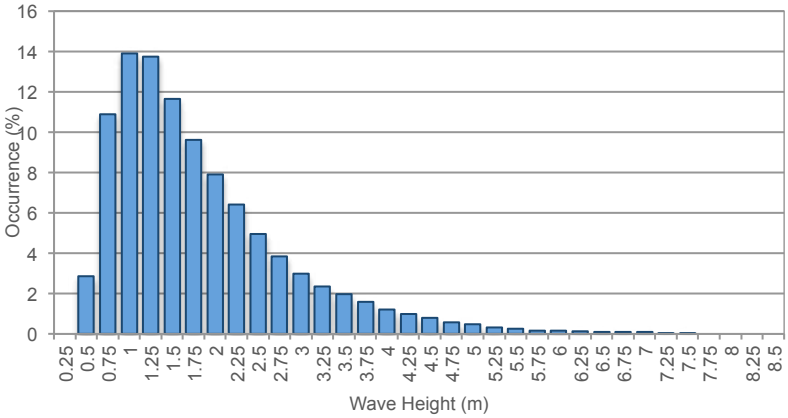


Figure 104 Frequency of occurrence (wave height) in the wave point, DHI's wave model

11.3.2 Scatter diagram

A scatter diagram can be based on wave measurements or numerical calculations. A scatter diagram shows the average occurrence frequency, in %, of different sea states for one year (or one specific month) and a given wave direction. A sea state is defined by a combination of significant wave height and energy period or peak period. Corresponding to representation of the wave power with DHI's model, the scatter diagram shows us the average percentage frequency occurrence of the significant wave height.

AVERAGE PERCENTAGE FREQUENCY OCCURRENCE OF Hs																		
Hs[m]/Tp[s]	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3	3.25	3.5	3.75	4	4.25	>4.5	
0-4	1.2	2.3	0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4
5	1.1	4.6	5.1	3.5	1.2	0.2	0	0	0	0	0	0	0	0	0	0	0	15.7
6	0.4	2.4	4.2	4.5	4.7	4	2.4	1	0.2	0.2	0	0	0	0	0	0	0	24
7	0.2	1	2.7	3.2	2.8	2.7	2.8	2.8	2.6	1.9	1.1	0.5	0.2	0	0	0	0	24.5
8	0	0.3	0.9	1.7	1.5	1.5	1.3	1.2	1.1	1.3	1.1	1.2	1.1	0.9	0.6	0.5	0.6	16.8
9	0	0.1	0.4	0.6	0.9	0.9	0.7	0.5	0.4	0.4	0.5	0.6	0.4	0.4	0.4	0.5	1.5	9.2
10	0	0	0.2	0.2	0.4	0.4	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1	4.5
11	0	0	0	0	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.1	0	0	0	0	0	1.3
12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total	2.9	10.7	14	13.7	11.7	9.9	7.7	6	4.8	4.1	3	2.6	1.9	1.5	1.2	1.2	3.1	100

Table 28 Average percentage frequency of significant wave, IJmuiden zone (see Figure 104)

For each sea state in a scatter diagram the corresponding wave power (kW/m) is calculated. For instance, the spectral distribution of the energy in the North Sea can be described by a parametric JONSWAP spectrum, which is described in *Chapter 9*, with peak enhancement factor $\gamma = 3.3$ and scaling parameter $\alpha = 0.2044$.

11.3.3 Annual average available wave power

The average available wave power was calculated and compared with the wave power value of 8.68kW/m presented in the study by *Beels et al. 2007* [**41**] in the region *IJmuiden Munitie Stortplaats.*

	Meteomast Ijmuiden data	DHI wave model
Significant wave height (m)	1.35	1.5
Peak period (s)	6.76	6.00
Average annual wave power (kW/m)	**7.63**	**7.84**

Table 29 Annual average wave power in kW/m

The values are also in the same order of magnitude. Then, it proved that the wave data of the *meteomast IJmuiden* are the good quality.

11.3.4 Extreme significant wave height

The significant design wave height with a return period of 50 years can be an important parameter when evaluating suitable location for deploying energy devices. There is a possibility that it is more viable to design a device for an extreme wave height of 10m. The DHI´s wave model shows us a representation of the parameter.

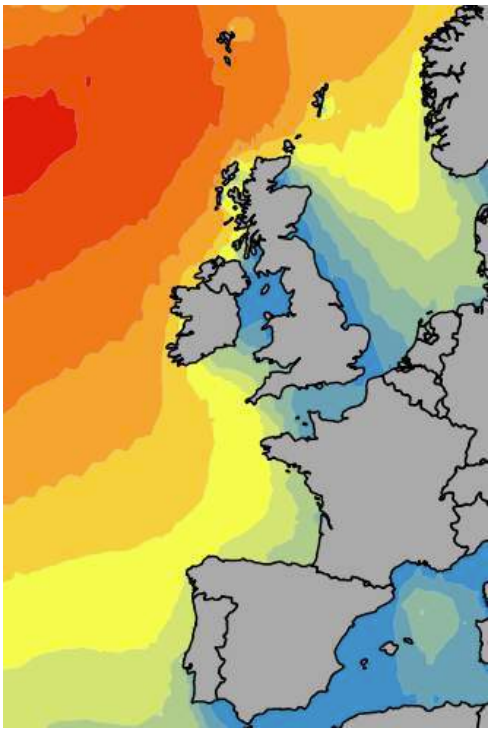


Figure 105 Overall map of the wave power, DHI's wave model

11.3.5 Conclusion

In general the wave power resource in the North Sea is rather small compared with the resource in front of the West European coast (40–50 kW/m) [**41**]. Near shore (< than 30 km), less than 11 kW/m is available, not only on the Belgian Continental Shelf, but also on the Dutch, German, Danish & UK Continental Shelf. Only the resource in the northern part of the North Sea is comparable with the one of the West European coast. The wave climate is less aggressive and this feature makes the North Sea attractive for wave energy conversion. When planning wave power extraction in the North Sea, a detailed study on the planned location is needed. The characteristic sea states, presented in this paper, should only be used for preliminary design.

PART 3 – Integration of Offshore Wind and Wave Energy Resources

12. Correlation between the Wind and Wave Power

In the oceans of the northern hemisphere, there is also a strong correlation between availability and demand over seasons, since most of the wave energy generally is available during winter in these oceans. However, sites exposed to open oceans will primarily experience swell waves, with longer duration and less correlation with the local wind conditions. If wave power is to be combined with solar power, one advantage is that the output from solar is likely to have the opposite pattern to wave, both with respect to weather and season.

When comparing the full year correlation with separate months it was found that the correlation is higher in December than for the whole year, and slightly lower but still high in July. Short-term measurement data of significant wave height and wind speed at *Met mast IJmuiden* are shown in the *Figure 106*.

The correlation between the wave power generation and its complementary source will be very important when quantifying the gains. An Irish study *Fusco et al. 2010* compares combining wind power generation with wave power generation on the west and east coast of Ireland, and finds that due to the big difference in correlation the outcome is also very different with the gains being much higher on the coast with uncorrelated wave and wind power generation.

On the other hand, waves are correlated with wind but offset in time in confined waters such as the North Sea. A high correlation between wave and wind power implies that it is predominantly wind waves present in the area. A high correlation reduces the chances of having benefits from combining the two resources from a smoothing output perspective.

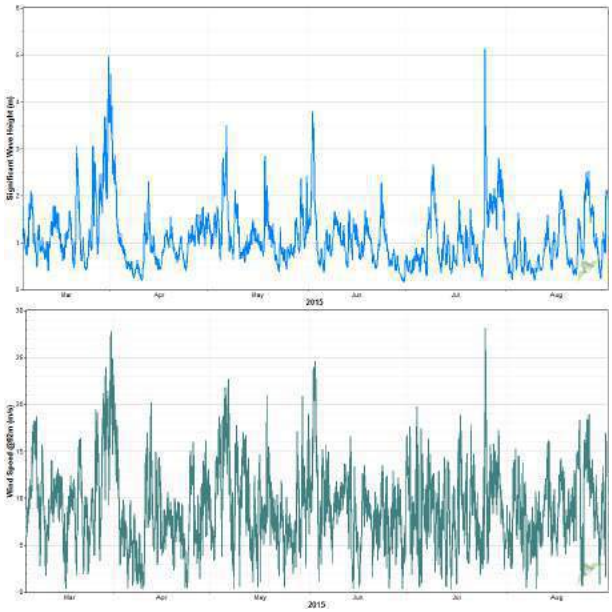


Figure 106 Significant wave height (blue) and wind speed (red) measurement data (1h) at meteomast IJmuiden during a period from March to December 2015

High wave and offshore wind potentials are present along the West and North coasts of Europe. Eventually these renewable sources will become large-scale contributors to the electricity mix. The idea of combining marine renewables has gained lot of interest in the last years and so is the number of designs merging wind and wave conversion mechanisms. However, there is no real deployment combining both resources and research on the area is based on deskwork. Three studies assess the correlation between winds and waves, and the advantages of harnessing them together to improve the reliability and decrease the variability of the power production.

Wave Star and Dong Energy are collaborating towards the deployment of a commercial 600 kW converter at *Horns Rev 2 offshore wind farm*, off the West coast of *Denmark,* a location with comparable *metocean* conditions as those investigated here. This analysis is the first approach towards the study of real power productions of full-scale wave and wind converters. It also focuses on the environmental characteristics of *Hanstholm,* located in an area with continuous interest on renewables development.

After some years of testing, the main conclusions derive from the analyses at *Hanstholm, Fusco et al. 2014* [**1**]:

- There is high correlation between waves and winds.
- Waves have 2 to 3 hours average delay to winds, which can go up to 9 hours.
- Waves are less variable than winds.

Figure 107 Wave Star machine is in place in Hanstholm (2009) (below)

The wave power would benefit from a finer time resolution since wind values, too quick to affect the waves would be visible, increasing the fluctuation of wind power generation compared to with three hourly data. Earlier studies have shown that a temporal resolution of three hours is sufficient to identify the system effects of wind power in western Denmark [**1**]. The temporal resolution of the weather data is an important factor when investigating the correlation between wave and wind power.

Another important aspect of this thesis is that wind power is completely land based. Offshore wind has a higher capacity factor but is also much more costly. Co-locating wave and offshore wind can have benefits such as better utilisation of the expensive cable connecting to main land. Again, since there is a high correlation between waves at the west coast and winds in southern Sweden, this effect would be small in this case.

A dispatch model with a temporal resolution of three hours has been applied to study the effects of implementing wave power into the electricity system of Dutch North Sea. The wave power availability and the correlation to wind energy in some areas in the North Sea has been investigated, and also potential benefits of combining the two sources.

It is clear from the results that there are benefits of implementing a combination of wave and wind power to the electricity system compared to only increasing the wind power penetration equally much, revealed as reduced system operational costs.

13. Potential gains of combining these two sources

In the upcoming years a large number of offshore wind farms will be build. At the designated water areas some considerable energy potential exists through ocean waves. Combining wave power with offshore wind power has been gaining interest in the scientific literature, as it offers several advantageous synergies such as sharing infrastructure [**3**] and reducing the cost of energy by more the 50% compared to stand-alone wave farms, while simultaneously smoothing production due to only partial correlation in power output [**4**].

The sustainable development of both offshore wind and wave industries requires a reasonable and responsible use of the natural resources, optimizing their exploitation where present. The joint exploitation of offshore wind and wave energy resources enjoy a number of advantages [**4,5,6,7**]:

- Higher availability of produced power when swells continue after the wind
- Higher quality of power from wind and wave energy
- Lower structural and erection costs per MW if the two converters share the same structure
- Lower electric cable cost per MW by sharing the same transmission cable
- Lower operation and maintenance costs
- Less area and environmental impact for combined farms

13.1 Synergies of combining offshore wind turbines and wave energy converters

The combination of marine energies gains momentum as a real alternative. This is supported by

a number of synergies ranging, such as increased energy yield and a reduction on the Operation & Maintenance (O&M) costs.

- An increased energy yield: These combined concepts will increase the global energy yield per unit area of marine space, contributing to better use of the natural resources.
- Better predictability: The wave resource is more predictable and less variable the wind resource (about 23%) and the power output of WECs is 35% [**5**] more predictable than for wind turbines.
- Shared O&M costs: The combination of both energies would lead to an important cost reduction as result of the shared use of these installations and technicians.
- Shadow effects: It is clear that the energy extraction of an array of WECs creates a wake that modifies the local wave climate by reducing the mean wave height - shadow effect, as shown in *Carballo and Iglesias, 2013* [**7**]. Combining WECs and offshore wind parks at the same location, in a way in which this shadow effect can be used to obtain a milder wave climate inside the park, may lead to more weather windows for accessing the wind turbines for O&M, and to reduced loads on the structures.
- Environmental effect: The combined option presents an important advantage in environmental terms in that it is likely to have a reduced impact (e.g. installations), leading to a better utilisation of the natural resources. Moreover, this could result in a transfer of knowledge on the environmental impacts from one sector to another [**4**]**.** Other synergies are *smooth power output, common foundations and grid infrastructure, logistics*…

13.1.1 Risks and challenges

There are some predicted risks and challenges associated with combined wind and wave arrays. Some recommendations are listed as follows [**5**]; Technology readiness of wave energy, uncertainty of mooring lines failure due to lack of experience, impact risk, project insurance, …

13.2 Example: Combination of offshore wind and wave resources

NEMOS is an innovative system to generate electricity from ocean waves, based on research finding since 2010, it achieves a higher efficiency level than comparable systems, can be implemented at low costs and is particularly suitable for combination with offshore wind turbines. The WEC's system is shown in the following *Figure 1* [**8**].

Figure 108 The NEMOS-system consists of an elongated floating body, which is connected to the seabed by three ropes. Excited by the movement of the waves, it transmits mechanical energy to a generator, which is positioned at any suitable offshore structure and protected from seawater [**9**]

In August 2015 the NEMOS 1:5 test device in Denmark started the fully automated. Currently a first large-scale device is in preparation to be installed in the North Sea. Parallel to this endurance-test the first components of the full-scale prototype are investigated. For example, to simulate realistic conditions the ropes were sprinkled with saltwater and have already been moved millions of cycles over pulleys with different coatings and fibres.

Conventional systems with vertical movement only achieve efficiency levels of up to 50%. With innovative trajectories and control algorithms, up to 80% of the incoming wave energy can be used to drive electric generators. With a change in the wave direction the orientation of the body and its trajectory can be adapted by a patented system. To protect the system from extreme wave loads in heavy storms, it can be lowered to calmer waters. One of the strengths is that this innovative application has been designed which can be realised at low costs. Leading energy companies already evinced their interest in a commercial use of this concept.

13.3 Conclusion

In 2010 a study of the Stanford University for an American site with a coupled energy park detected that about 20% higher yields can be derived compared to a pure wind farm with a given line capacity. Shared infrastructure, e.g. subsea cables and substations could be better utilized, whereby the pro rata costs for energy transport would also decrease.

These synergies are strong and make from the combined wave-wind energy a real alternative. From these synergies, the one regarding to the shadow effects should be considered with special detail and further research should be conducted to understand the interaction between the weak of the WECs within the farm array and the find turbines substructures.

A number of risks have been identified for combined wave-wind projects, which are mostly due to the early stage of development of wave energy or for the reduced of experience on full-scale projects.

Today, it has been identified that there are WECs that are susceptible for combining considering their actual development status, and a great example is presented above, but I am agreed that the Tidal energy devices are a step ahead.

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